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The European Bathroom & Kitchen Product Markets

- 2021 Edition -

Turkey

BRG
BUILDING SOLUTIONS

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Overview

BRG Building Solutions is a specialised market research consultancy serving the global building technologies industry. Combining unparalleled industry experience with expert research and analysis, our team delivers lasting results to hundreds of companies worldwide.

Our core expertise lies in our coverage of the bathroom, kitchen, HVAC, and other building products/technologies. Throughout a history spanning over 40 years, we have assisted industry players in identifying and realising growth opportunities through a combination of regular market monitoring and ad-hoc consultancy services.

BRG's unique Enterprise Database is at the centre of our offering. This is a comprehensive tool and very detailed source of historic, current and forecast market trends and market share data (by holding and brand) on a country comparable basis. The database provides an essential tool to our clients as a platform to support their strategic decision making.

BRG Building Solutions is a genuinely global company, staffed with individuals from varied backgrounds and nationalities, working collaboratively from four offices across four countries. Combining a high level of industry experience with expert research and analysis, our team delivers lasting results to hundreds of companies worldwide.

Our market coverage is also global, reflecting the nature of the industries we serve. Currently, 66 individual markets are covered annually in detail across the following key geographical regions (and number of countries):

- Africa (3)
- Australasia (2)
- East & Southeast Asia (6)
- Central America (4)
- Central Asia (1)
- Europe (31)
- South Asia (1)
- Middle East (7)
- North America (3)
- South America (5)
- Sub-Saharan Africa (3)

BRG has its own in-house dedicated modelling and forecasting unit. In cooperation with clients, we have since extended our offering to include market sizing for countries outside the traditional geographic reach of our main research

programme, via global market models

Product Coverage

Since the 1980s, BRG's Bathroom & Kitchen programme has provided industry players with insightful analysis and evidence-based information. In addition to detailed data on trends, values, end use, technical segmentations, market shares and distribution, our programme also includes perceptive text commentary, enabling our clients to easily identify and understand market characteristics and the latest industry developments.

BRG's programme has evolved steadily in terms of both geographic scope and product coverage and continues to do so each year. Each country report follows a uniform structure, to ensure that the data can be readily compared across regions and countries.

The product coverage has also expanded to cover virtually all bathroom, sanitary and kitchen related products, used in both residential and non-residential buildings, including:

Baths and Shower Trays

- Baths: cast iron, steel and synthetic
- Hydromassage baths
- Shower Trays: ceramic, steel/ cast iron and synthetic
- Shower systems (NAM): shower modules, integrated tub showers

Sanitary Ware

- WCs: wall hung, floor standing 1-piece and squat
- Washbasins: hand washbasins and countertops
- Other CSW: bidets, kitchen sinks, pedestals, shower trays, urinals and cisterns
- Plastic Cisterns: concealed and exposed.

Taps and Mixers ('Faucets'), both sanitary and kitchen:

- By application: basin/sink, bath, bidet, shower, kitchen sinks
- By type: pillar, two head, one head, thermostatic, self-closing, electronic
- By type (Americas): 4in centre set, 8in widespread, electronic, hand spray, pressure, single handle/one-head, self-closing, thermostatic, three handle.

Shower Wall

- Bath screens, conventional shower cubicles, hydrotherapy cubicles, bath & shower surrounds (NAM)

Bathroom Furniture

- Base units, mirror / mirror cabinets and separate cabinets.

Bathroom Accessories

- Soap holders and dispensers
- Towel holders, towel rings and towel bars
- Toilet paper holders and toilet brushes
- Towel and bathrobe hooks
- Toothbrush holders and tumblers
- Shelves, wastepaper bins, shower storage units and tissue holders.

Shower Accessories

- Shower heads, flexible hoses, shower supports, shower panels and shower pipes.

Kitchen Sinks

- Cast Iron, ceramic, enamelled steel, stainless steel and synthetic.

Definitions

Please note that all volumes given in the report refer to the number of units sold, where not indicated otherwise. In-depth product definitions are included in a separate document and available upon request.

All prices, both manufacturer and retail, exclude value-added-tax (VAT) and installation costs.

Methodology

The methodology behind the data remains true to our origins in that the industry participates in the research process, in many cases by providing BRG Building Solutions with annualised volume sales by product category per country and contacts for detailed expert interviews in all countries covered.

Our approach follows a multi-pronged strategy, gathering data inputs and information from many different sources. A further USP of BRG is the fact that we live and breathe the markets in which we specialise (bathroom, kitchen, plumbing & HVAC) and are in constant contact with manufacturers, distributors and other key influencers all of the time during our repeat annual research programme. The markets are built via a bottom-up approach. Extensive primary and secondary research complement the picture and provide all necessary insight for many

additional segmentations shown in the reports, like e.g. price segmentations, technical and end-use segmentations.

The reports are constantly evolving and refined based on industry feedback, and always reflect the latest developments in the industries.

All research is conducted in-house with a team of around 30 individuals who speak 20 different languages between them. As a result, we can transfer that knowledge and experience into services designed to assist stakeholders in understanding the opportunities available to them, at a fraction of that charged by the larger and less nimble global consultancies. BRG operates a policy of direct management of the team required to conduct the regular research work and avoids subcontracting its market studies to outside agencies.

The analysis has grown in complexity as the number of drivers affecting the markets grows, like e.g. economic uncertainty, construction cycles, new technologies, legislation. As a result, BRG Building Solutions considers all drivers through a holistic approach, recognising that products, end use segments and countries are all inter-linked.

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TURKEY

BACKGROUND

THE EUROPEAN BATHROOM & KITCHEN PRODUCT MARKETS



Geography and Demographics

Turkey – officially the Republic of Turkey – occupies an area of 783,562 km². Approximately 97% of the country's land area lies in Asia, on an extended peninsula called Anatolia, with the remainder located in Europe. Turkey's geographical location makes it a natural bridge between Europe and Asia. Its location on two continents plays a central part in Turkish history and gives the country a major advantage in serving the markets of Europe, the Middle East, Central Asia and North Africa.

The country is bordered by three seas: the Black Sea to the north, the Aegean to the west, and the Mediterranean to the south. Turkey borders Greece and Bulgaria to the west, with the republics of Georgia, Azerbaijan and Armenia to the north, and Syria, Iran and Iraq to the east. It is divided into 81 provinces. The capital is Ankara and Istanbul is the largest city. The population distribution of Turkey is closely related to topographic conditions and rainfall patterns. Settlements are most heavily concentrated in European Turkey, and in the fertile valleys and lowlands of the Marmara, Aegean and Black Sea coasts. Around half of the population inhabit this area, accounting for about 30% of the country's territory.

As of 2020, the population of Turkey stood at around 83.9 million. The Turkish population is relatively young, with around 34% being under 18 years old. The average age is 28.4 years, so this means that 50% of the total population is less than 29 years' old. According to the statistics released by the government, life expectancy stands at 75.1 years for men and 80.3 years for women. Around 66% of the total population is between 15 and 64 years old. Presently, Turkey's population is much younger than those of the developed European countries where the populations are ageing rapidly; however, the share of young people within the total population will progressively decrease over the coming decades. As the fertility rate declines, the number of newborns joining the population every year will decline. Hence, the size of the adult population will constantly increase throughout the next three decades.

Macroeconomic Overview

Following the number of conflicts and highly debatable political decisions of 2019, Turkey experienced another turbulent year in 2020, with controversial political developments and economic challenges. President Recep Tayyip Erdogan has led Turkey through 16 years of continuous growth, making it globally one of the leading emerging markets, but much of the growth of the last few years has come through unrestrained borrowing.

Economists had been warning of a recession for years, as foreign investment in Turkey dried up over growing concerns about the rule of law as well as its economic policies. Turkey's growth was fuelled in part by the availability of cheap foreign credit, but that credit flow started to slow down in 2018, as in other

emerging markets, after the Federal Reserve in the United States raised interest rates and continued in 2020. Loans have become more expensive for Turkey and other emerging markets because of investors being wary of the risk, while rising interest rates and buoyant stocks have lured money back to the United States. The challenges from 2019 carried on in 2020, due to foreign politics of the country.

Moreover, like in most other countries, the worldwide COVID-19 pandemic has caused various problems in Turkey and it climbed to the top of the list of concerns and affected the whole world negatively. From April 2020 onwards, in Turkey most of the time schools, universities and many commercial establishments were closed. Factories and some construction sites had to stop work or reduce the number of workers. Many people lost their jobs and had to be compensated by the government, which allocated large sums of money. Many establishments got into financial difficulty, and rescue plans had to be put into effect in the form of providing loans and deferring tax and other payments to public institutions. Demand for many goods and services, including real estate, shrank under the pandemic pressures.

In March 2020, Turkish President Recep Tayyip Erdogan announced an economic package worth 100 billion liras (14.3 billion euros) to tackle the impact of the coronavirus outbreak on the Turkish economy. The package included measures for the postponement of tax duties, loans and social insurance payments, as well as several incentives for Turkish businesses and citizens. There was a six-month postponement of tax and social insurance payments for all companies. The government also pledged to give financial support to its export companies. It increased the state credit guarantee fund from 25 billion liras to 50 billion liras to shield private banks. However, following the impact of the pandemic, the Turkish economy's growth rate declined by 5% in 2020 before bouncing back in 2021 to grow by 5%, according to the IMF projections from October 2020.

Moreover, Turkey's foreign policy became even more challenging for its partners and neighbouring countries in 2020. Syrian refugees in Turkey, the oil and gas drilling operations around Cyprus and deployment of armed drones in Northern Cyprus, the reinforcement of an existing base in Qatar, and the actions from the US in response to Turkey's purchase of S-400 missiles from Russia all very negatively affected the economy in 2020.

These situations led to a further weakening of the Turkish currency and high inflation, along with declining purchasing power, in 2020. However, the Monetary Policy Committee (MPC) of Turkey put on the emergency brake by raising its key policy interest rate from 10.25% to 15.0%, reverting back to an orthodox monetary policy position. It followed with a further rate growth to 17.0% in December 2020. Financial markets did not appreciate this policy stance, reflected in rising bond yields and sharp currency devaluation. The Turkish Lira lost 40% in value against the USD from the beginning of 2020 to early November. However, at the same time, nominal exports benefitted from the more weak and international competitive currency. On the whole, Turkey's inflation in 2020 stood very high at 12.0%, with the unemployment rate reaching 14.6%.

However, due the ongoing economic challenges up to the third quarter of 2020, on 7th November 2020 Naci Abal was appointed the governor of the Central Bank of Turkey, replacing Murat Uysal, who was unable to stop the fall of the Turkish Lira, which had lost around 40% against the Dollar since January 2020. At the same time, finance minister Berat Albayrak was replaced with the market-friendly Lufti Elvan. Thanks to these new experts in the key positions and their programmes, the Turkish Lira stopped on its losing path and even started to recover again, which affected the Turkish economy positively.

In mid-March 2021, President Erdoan sacked the head of the country's central bank Naci Abal, days after raising interest rates to tackle soaring inflation. As a direct result, the Turkish Lira declined by almost 15% the next day, while the Istanbul stock exchange shed 10% of its value after the president shocked global investors by replacing Naci Abal with ahap Kavcolu, a former member of parliament for Erdoan's ruling Justice and Development Party (AKP). The next day, The Turkish Lira was stabilised again, but the economy still remains very fragile – although with huge potential to grow again.

Residential Construction

Overview

Turkish residential construction activity has, for many years now, been the biggest in Europe even though housing completions experienced some decline due to economic difficulties in 2020. The country's economic issues have severely impacted construction activity; the depreciation in the Lira and high interest rates have pushed up construction costs and also the cost of borrowing – such that the viability of projects has been eroded, and firms are reportedly bringing projects to a halt.

Although COVID 19 disrupted the various industries, public construction (or in short, construction) has been one of the few industries that has maintained progress to some extent. Overall, more and more residential construction continued to go towards big projects, managed either by very big private contractors or by the government, which builds thousands of new dwellings – driven by the Housing Development Administration of Turkey or the Urban Transformation Projects.

New Housing

The growth of new construction activity over the last few decades, which was the locomotive sector of the ruling Justice and Development Party era, stopped in 2019. The very strong weakening of the Turkish Lira in 2019 against foreign currencies started an economic threat to the country, which increased strongly the inflation and scared investors and weakened Turkish consumption in general. This issue was intensified in 2020, due to the weak economy of the country.

According to industry reports, Turkey's construction industry has been among the worst hit by a severe economic downturn sparked by a currency crisis, which began in 2019 and then continued in 2020. Scores of companies went bust while others were forced to seek bankruptcy protection, repay bank loans or restructure them as the housing market slumped and projects were abandoned before reaching completion.

In a bid to stimulate housing transactions, the government introduced a measure in June 2020, according to which the loan-to-value ratio in residential mortgage loans was increased to 90%. The three state-owned banks were to offer mortgage loans under market interest rates and with a longer repayment period. The stimulus measure influenced the national housing market positively, but the decline of housing completions declined further in 2020 as a whole.

In 2020, according to the Turkish Statistical Institute (TUIK), over 30,000 houses and 670,000 apartments were completed. As to the dwelling units, Istanbul had the highest share, followed by Ankara and Izmir. The provinces that had the lowest number of dwelling units were Hakkari, Mus and Sirnak.

However, the Housing Development Administration of Turkey and the Urban Transformation Projects in a number of metropolitan municipalities remain the main drivers for residential building by the government. The government project TOKI and the Urban Transformation Law aim to rebuild or reinforce 6.5 million housing units that are at risk from earthquake over the next 20 years, at a cost of around USD 400 billion. Thanks to these programmes, construction activity in the residential sector did not experience a bigger decline than it possibly could have done.

Housing RMI

Since the boom in the Turkish economy in the 1980s and the construction of big residential dwellings, repair, maintenance and replacement works remain increasingly necessary. Due to the lower quality of building products in Turkey when compared with Western Europe, the general lifespans of building products such as baths and taps and mixers are shorter. This means that repairs and replacements are needed more frequently than in other countries in Europe.

However, RMI increased strongly in 2020, due to several reasons. With ongoing lockdown periods over the year, Turks started to spend more time at home and renovate their homes, which was not very easy before as people did not spend so much time at home. At the same, many Turks escaped from big cities to their homes in rural areas. Here they invested in their homes, from essential and/or old facilities to new and modern bathrooms. As a result, housing RMI became an important driver for the overall bathroom products market.

Non-Residential Construction

International investments declined strongly due to the economic recession in 2020. As a result of decreasing investments in the commercial and public sectors, total output going into new non-residential buildings declined in the year. On the other hand, existing business owners continued their investments, especially renovations, further. Many restaurants, cafes and hotels took advantage of closing their businesses during lockdowns by renovating them, and thus purchased new bathroom products.

However, despite its political and economic troubles, Turkey is still pushing highly ambitious infrastructure projects. The demand for schools, healthcare facilities and government buildings increased as a result of population growth and urbanisation. The growth of non-housing construction is also supported by the country's Vision 2023, under which the government aims to develop the country into one of the world's largest economies by 2023. Under the 5-Year Development Plan 2019-2023, the country continues to build 13,478km of new roads, 5,748km of new highways, and 10,000km of high-speed rail network lines across the nation.

In 2020, the nuclear energy company ROSATOM continued to build the Akkuyu nuclear power plant in southern Turkey. On 8th January 2020, Russian president Putin joined the Turkish president Erdogan in Istanbul on for the official opening of the TURKSTREAM pipeline. Construction began in May 2017 and cost nearly \$8 billion. TURKSTREAM is set to carry Russian gas under the Black Sea to Turkey and supply several countries in southeastern Europe, with an annual capacity of 15.75 billion cubic metres. In addition, the EMBA ELECTRICITY PRODUCTION company, a joint-venture Chinese and Turkish company, is investing USD 1.7 billion to build the EMBA HUNUTLU power plant by 2021.

Other upcoming projects are:

CANAL ISTANBUL, popularly known as President Erdogan's 'crazy project', is also in the government's project pipeline. The 45-km proposed canal would turn western Istanbul into an island. It is aimed at diverting oil tankers away from the Bosphorus, where they clog traffic and pose environmental risks. Following the tender for geological and engineering works, the construction tender is planned within two years. Environmental experts are concerned that the project, which has an estimated cost of over €12 billion, would damage the delicate ecosystems controlling the sea currents and temperatures.

The 1915 CANAKKALE BRIDGE will have the longest main span of any suspension bridge in the world. It will include an 88-kilometre motorway. Construction started in March 2017, and operations are expected to start by 2023.

II.2 Key Figures

TR - DEMOGRAPHICS AND HOUSEHOLDS

Value	Year
Segmentation	Category
Demographics	Population (% Rural)
	Population (% Urban)
	Population ('000)
Households	Average Size (Persons h/h)
	Total ('000)
Persons per Household (% of Total)	1
	2
	3
	4
	5+

Source: BRG based on Country Statistics.

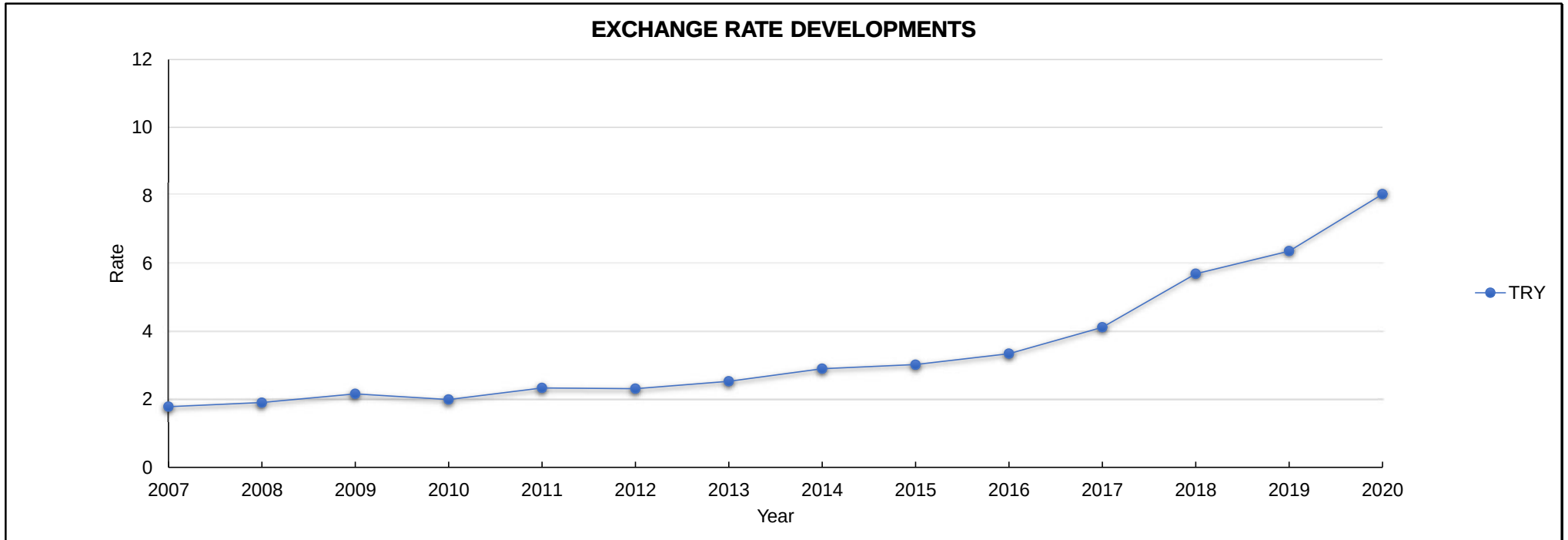
II.3 Macroeconomic Trends and Forecasts

TR - MACROECONOMIC KEY INDICATORS

Value	Year																				
Category	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Consumer Price Index(% change)	7.72	9.65	8.39	10.06	6.53	6.40	10.45	6.16	7.40	8.17	8.81	8.53	11.92	20.30	11.84	14.60	12.50	11.50	11.00	11.00	11.00
GDP (% change)	8.99	6.95	5.04	0.82	- 4.82	8.43	11.20	4.79	8.49	4.94	6.08	3.32	7.50	2.96	0.92	1.79	6.04	3.50	3.50	3.55	3.52
Government Balance (% of GDP)	- 0.75	- 0.69	- 1.93	- 2.64	- 5.84	- 3.40	- 0.68	- 1.82	- 1.46	- 1.42	- 1.26	- 2.34	- 2.18	- 3.65	- 5.61	- 5.38	- 5.71	- 6.07	- 6.11	- 6.04	- 6.00
Private Consumption (% change)	6.31	3.67	5.28	0.28	- 3.66	10.81	12.21	3.14	7.54	3.13	5.31	3.78	5.93	0.53	1.56	3.19	0.00	0.00	0.00	0.00	0.00
Unemployment Rate(%)	9.49	9.03	9.18	10.02	13.05	11.13	9.10	8.43	9.04	9.92	10.28	10.91	10.90	10.96	13.71	13.11	12.36	10.97	9.04	8.82	8.60

Source : IMF and OECD. N.B. Household Saving Rate and Private Consumption not Available for all Forecast Years.

II.4 Historical and Current Exchange Rates
TR - EXCHANGE RATES : HISTORICAL AND CURRENT



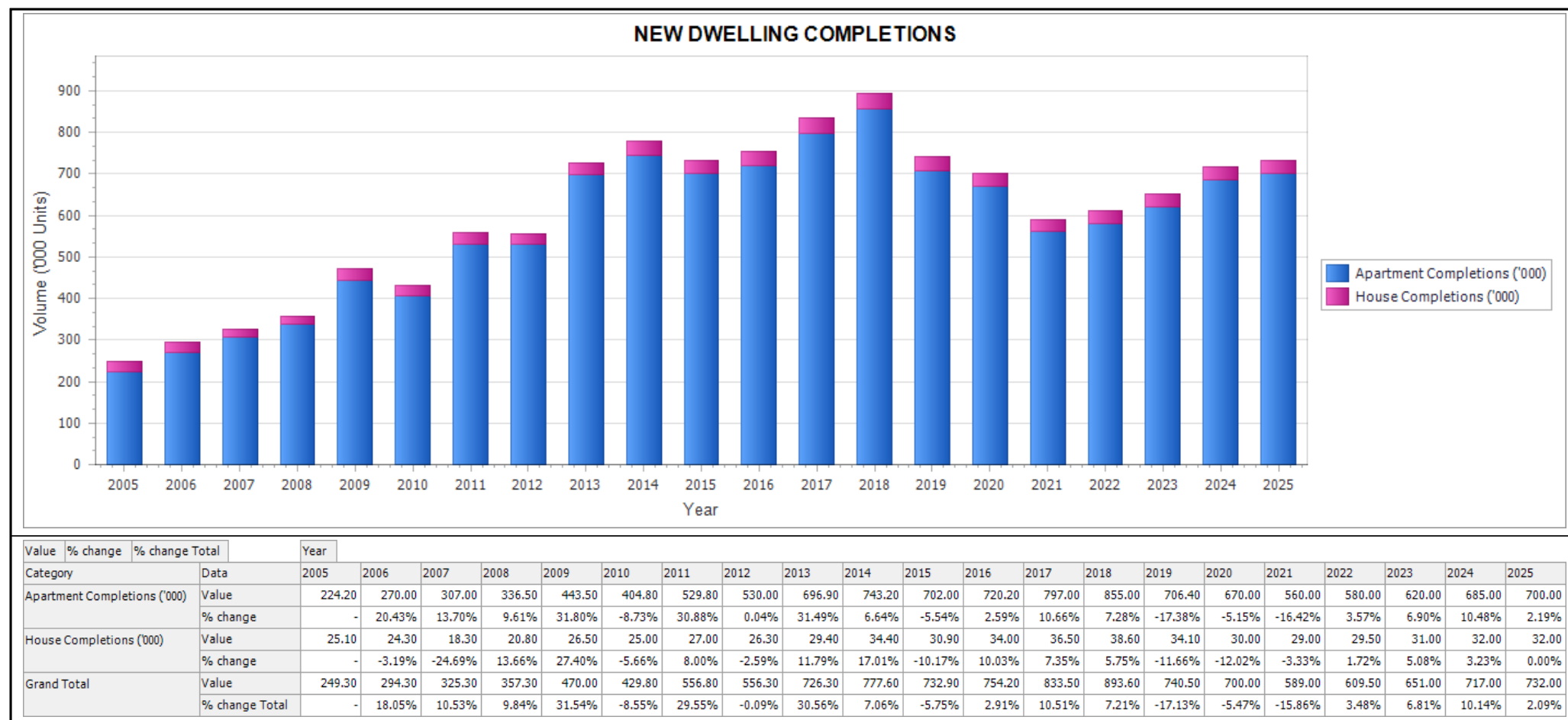
Local Currency	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
EUR / TRY	1.79	1.91	2.16	2.00	2.34	2.32	2.53	2.91	3.02	3.34	4.12	5.70	6.36	8.04
% change		6.71%	13.42%	-7.56%	16.86%	-0.87%	9.34%	14.76%	4.07%	10.60%	23.23%	38.22%	11.67%	26.41%

Source: OANDA

The Turkish Lira (TRY) is the Currency of Turkey. The Exchange Rates Shown are the EUR to the TRY.

II.5 New Dwelling Construction

TR - NEW DWELLING COMPLETIONS ('000 UNITS)



Source: BRG based on Country Statistics.

II.6 Dwelling Stock by Type and Tenure
TR - DWELLING STOCK BY TYPE AND TENURE ('000 UNITS)

Value		Year
Segmentation	Category	2020
Dwelling Stock by Type	Apartments (%)	74.35
	Apartments ('000)	19 210.00
	Houses : Total (%)	25.65
	Houses : Total ('000)	6 626.00
Dwelling Stock by Tenure	Primary Owned (%)	61.08
	Primary Owned ('000)	15 780.00
	Primary Rented : Private (%)	19.00
	Primary Rented : Private ('000)	4 908.00
	Primary Rented : Public (%)	3.10
	Primary Rented : Public ('000)	800.00
	Secondary (%)	16.33
	Secondary ('000)	4 220.00
	Vacant (%)	0.50
	Vacant ('000)	128.00

Source: BRG based on Country Statistics.

TURKEY

DISTRIBUTION

THE EUROPEAN BATHROOM & KITCHEN PRODUCT MARKETS



Overview

In Turkey, the distribution channels cannot be directly compared with any Western European country, as there are no independent national chains of wholesalers or retailers covering the entire territory and the DIY stores that have established a presence there, such as BAUHAUS, TEKZEN, MAXI, and KOCTAS, carry very few bathroom products, if any at all.

Distribution can be divided into two broad groups. Firstly there are the semi-structural distribution channels, over which the large-scale manufacturers belonging to large groups exert a high degree of influence and control. Secondly, there are the unstructured distribution channels, where the small and medium-sized dealers play an essential role.

Turkish business, in general, is characterised by the existence of large multi-specialist companies. In a given sector, these companies often run their manufacturing activities in connection with their own marketing and sales companies. The premium manufacturers in the sector, except for the specialists, sell their products through companies that are directly owned by their holding groups.

Through their regional offices in the main cities, these marketing and sales subsidiaries in turn work with a network of dealers/installers. Headquarters in Istanbul tend to search and follow mass projects, but sales go through exclusive or non-exclusive dealers. The smaller, local companies in the 'economy' or 'standard' segments tend to supply directly to the local dealers/installers, without the benefit of having their own national or multiregional networks.

Due to the COVID-19 pandemic, the shift towards online channels and retailers sped-up, away from other retailers such as specialists. Due to lockdowns, people spent more time at home, which led to more private internet consumption. Moreover, some classic retailers were not allowed to open their shops, which supported the online channels positively.

Wholesale Level

A number of different types of players/distributors are involved in the distribution of bathroom products in Turkey:

Commercial marketing organisations of the large manufacturers: it is common practice for large manufacturers to sell their products to their own commercial marketing departments, which then operate autonomously to commercialise the products, often together with complementary products from outside. These companies are autonomous from their 'mother' or 'sister' companies; they work in different buildings, their accounts are separate, and they even take a margin on the products they sell from the mother/sister companies. The reasons for this strategic

division are believed to be twofold: the marketing/commercial company can freely import to widen its range of products so that it can maintain an exclusive distribution strategy and, due to the political and economic uncertainty, it can freely stop or reduce buying a certain product from the mother company and import it at a more advantageous price. This is often done when there are currency fluctuations.

Merchants/wholesalers are exclusive distributors and generally larger than the semi-wholesalers and dealers. They are usually less-specialised retailers and, in many cases, stock a wide range of building and household products including heating, plumbing and sanitary products. Merchants/wholesalers are more likely to carry larger stocks than semi-wholesalers and dealers, although throughout the distribution system in Turkey very little stock is held, and this is generally the responsibility of the manufacturers. Merchants/wholesalers do not sell directly to the consumer, but rather offer an intermediate distribution service to semi-wholesalers, dealers, sub-dealers and DIY stores.

Semi-wholesalers mostly operate on an exclusive basis (sometimes non-exclusive as well) for manufacturers with little stock holding. The main difference between semi-wholesalers and merchants/wholesalers is that the former not only sell directly to the public/end-users but also offer an intermediate distribution service to dealers and sub-dealers.

Dealers operate on an exclusive or non-exclusive basis for manufacturers. These are mostly relatively small companies, most of which are also installers. In addition to bathroom products, most cover heating, plumbing and air conditioning products. There is little or no stock holding. Dealers also sell directly to the public/end-users and offer an intermediate distribution service to sub-dealers.

Sub-dealers mostly operate on a non-exclusive basis for manufacturers. These are mostly installers and just sell directly to the public/end-users. In general, there is no stock holding.

A great majority of the non-exclusive dealers and sub-dealers are also plumbers who install the products. Usually, small and medium-sized companies sell through these non-exclusive distributors. The range of products can vary enormously. Some even sell white goods and furniture, while others specialise more in bathroom products.

Deliveries are made by the manufacturer, which usually has a limited amount of stock over the territory. Most of the manufacturers contract a transport company to deliver the products to the dealer.

Retail Level

DIY Retailers

According to DIY INTERNATIONAL, hypermarkets accounted for over 31% of total retail sales in Turkey in 2020, with this share expected to rise to almost 35% by

2022. The reason for this increase is investments from chains such as METRO, BAUHAUS, TEKZEN, and KOCTAS to open more and more DIY shops across the country, supplying the market with different bath products such as bath tubs, shower trays and walls, as well as taps and mixers.

Specialist/Other Retailers

There are a large but declining number of independent small retailers in Turkey, which are known in Turkish as *bakkals*. Despite their downward trend, however, these retailers are working together with very successful and popular major supermarket chains, such as MIGROS and discounters BIM, SOK, and A101, which have a very price-driven market. Similarly to DIY retail, some of the discounters have temporarily supplied the Turkish market with bath products such as bathroom furniture and taps and mixers.

Internet Retailers

As is the general trend worldwide, online sales are increasing strongly in Turkey and expected to increase further in the coming years. However, the internet's share is still low when compared to the classic distribution channel, but is increasing very strongly. As bathroom products on average are priced quite highly, Turks do not want to risk their money due to a lack of trust, and use the internet more as a research tool. Cheap bathroom products such as taps and mixers are the most popular items retailed online. The most popular online retailers are www.n11.com and www.hepsiburada.com.

Internet retailers have both a financial and logistical advantage in that there is no necessity to hold stock on their balance sheets. Sales can be done ex-works in many cases, with the internet businesses effectively taking their margin without ever taking physical possession of the product.

III.2 TR - SUMMARY OF LEADING DISTRIBUTORS

Holding	Trading Name	Distributor Type	General Product Orientation Profile	Turnover		Home Market Coverage		Specific Product Orientation													Buying Group Affiliation
				Year End	€ million	No. of Depots	Geographical Coverage	Specialist						Specialist					Other		
								Total	Heat	Bath	Plumbing	Vent/ Aircon	Tiles	Total	Building Materials	Timber	Insulation	Roof	Other	Specify	
ELGINKAN	ELEKS FOREIGN TRADE INC.	Wholesale/Exporter	Heating and Bathroom				Europe, Middle East and CIS countries	xxx	xxx	xxx	xx										
ELGINKAN	ELMOR	Wholesale	Bathroom				Turkey	xxx		xxx											
ECZACIBASI	INTEMA	Wholesale	Bathroom				Turkey	xxx		xxx			xx								
BAUHAUS	BAUHAUS	DIY Retailers	DIY Assortment			10	Istanbul(5), Ankara(3), Bursa(1), Antalya(1)	xx	x	xx			x	x	x	x			xxx	DIY Assortment	
KOCTAS YAPI MARKETLERI	KOCTAS	DIY Retailers	DIY Assortment	2020	180	171	Turkey	xx	x	xx			x						xxx	DIY Assortment	
TEKFEN	TEKZEN	DIY Retailers	DIY Assortment	2020	190	128	Turkey, Romania, Irak	xx	x	xx			x		x	x			xxx	DIY Assortment	

Source: BRG based on company information and industry interviews
x = marginal activity
xx = substantial activity
xxx = core activity

TURKEY

BATHS AND SHOWER TRAYS

THE EUROPEAN BATHROOM & KITCHEN PRODUCT MARKETS



Market Characteristics and Product Mix

The Turkish bath and shower tray market has been characterised as one of the most dynamic in Europe, constantly evolving and often experiencing relatively strong growth rates; the market size has doubled since 2002. However, individual products have not all followed the same development path. Unlike in other European countries, until 2010 the overall bath market did not decline as much as expected despite the increasing popularity of showers (this popularity has, however, probably resulted in slower growth figures for baths). However, from 2011 onwards, similarly to European countries, the shift towards showers away from baths has started and the market volume of baths has been declining since then. Nowadays, Turkey is primarily a showering nation, with a high number of consumers preferring to shower. Moreover, shower tray sales have benefitted from households becoming smaller and, in turn, dwellings becoming smaller. This means smaller bathrooms, which only have space for a shower tray rather than a bath. This can help explain the gradually decreasing growth rate of the bath market in the past few years.

Looking back, the overall Turkish bath market has historically been dominated by cast iron models, which accounted for the majority of sales in the early 1990s. This market has been declining for many years, due to the change in customer taste and growth of other attractive alternatives, such as synthetic and steel baths. However, the retro style has become trendy again and some high-end users and hotels have turned back to cast iron baths. Therefore, the negative trend of cast iron baths is likely to slow down in the coming years.

The synthetic bath market has followed a positive trend from 2002 onwards. The increasing popularity can be attributed to a large variety in design, shapes and sizes available on the market. Synthetic baths are still often installed in new housing, but the share of RMI is gradually growing as well. The share of cast marble baths in the synthetics segment has been gradually increasing, with demand for design becoming a stronger market driver. Turkey remains a very price-driven market, which explains why the very big majority of synthetic baths and shower trays are made of acrylic, followed by very small, but increasing, shares of mineral cast and solid surface.

Steel bath sales have also grown over the same period, but have experienced slower growth rates when compared to synthetic. These baths are often installed in the non-housing sector, particularly in hotels. As the Turkish economy has grown very strongly since 2010 and many hotel chains have opened new branches, sales of steel baths increased for a phase. However, the trend towards showering has affected the bath sales negatively, which is decreasing the market volume again.

Traditionally, and similarly to the bath market, the majority of Turkish shower trays were made of steel, with only a small proportion being made of cast iron. Sales of these products, however, have decreased strongly as consumer preferences have

changed towards synthetic models. Similarly to baths, this is due to the availability of different shapes and sizes as well as lower prices. Ceramic shower trays remain popular in Turkey and are regularly sold as part of an entire suite, often to new build or non-housing.

The availability of different materials, colours, shapes and sizes has increased the penetration of synthetic shower trays – which is the bestseller of all shower trays. Most installers prefer to install these lighter models, which should further increase their share within the market in the coming years.

However, like baths, the shower tray sales are also gradually challenged by different trends in the market. So-called ‘walk-in’ showers are increasingly gaining popularity in Turkey. This type of shower does not contain a shower tray but very often tile or stone floors, and is a cheap and even more aesthetically pleasing alternative to shower trays.

Regarding hydrotherapy products, hydromassage baths, which are almost all produced in Turkey, account for the majority of all sales. Most of these baths are acrylic and only a few have steel shells. Most have air jets rather than air-water or water jets. Sales of hydromassage baths have slowed down over the past five years, as there seems to be a preference for cheaper shower panels and hydrotherapy cubicles, which have both experienced higher growth rates over the years. Additionally, sales of hydrotherapy baths started to decline significantly in 2018 and this continued in 2019.

Recent Market Trends

The decline of the overall bath and shower tray market continued over the course of 2020. Due to the political and economic developments as well as different trends in Turkish bathrooms, sales volumes declined strongly as a whole again in the year. The substitution trend from baths to showers should still be prevalent and is unlikely to slow down given that the factors for the trend are not set to reverse anytime soon, such as smaller living spaces, busier lives and an ageing population. Hence, all bath types continued to decline further in 2020.

On the other hand, the shower tray market has been challenged again by the development of wet rooms, which do not require the use of a shower tray. The popularity of walk-in showers has been growing, due to the modern and stylish look and their ease of accessibility. Moreover, in the price-driven market in Turkey, the possible cheaper costs of tile and stone floors compared to shower trays, and more alternatives regarding design and individual taste possibilities of these walk-in showers, were the main reasons for the former popularity of stone/tile floor showers. Another big reason was the competition of the contractors, as they are always happy to create some differences, especially when they are cheap to be implemented.

However, it is believed that the growing share of walk-in showers had a turnaround in residential houses in 2020, due to several reasons: Turks had bad experiences

with tiled showers when it came to cleaning and hygiene as well as the quality of these products, as very often water started to leak between the tiles after some time. Hence, in new residential buildings walk-in showers remained still popular, as most contractors still prefer cheap solutions. However, many homeowners started to install conventional shower trays after some time, to avoid water leaking between the tiles.

For the duration of the virus restrictions, many Turks escaped from big cities towards rural areas along the seaside and mountains, often to stay in holiday homes designed to be used for only several weeks per year, with limited maintenance work over the years.

However, due to the COVID-19 pandemic many Turks started to renovate their holiday homes. This was supported by the government's stimulus package to help lower the interest rates to push the consumption. This led to an unexpected increasing demand for synthetic and ceramic shower trays in 2020, whereas the steel / cast iron shower tray market remained unchanged, as it operates in niche markets. Bath demand remained the same and could not profit from the growing replacement activity, due to the shift towards showering.

Regarding product mix, no major changes were experienced in 2020. Synthetic options remained the bestsellers within all bath and shower models. Synthetic baths and shower trays have a broader range of shapes and forms, which is one of the limits of the steel material. Steel and cast iron sales declined with the overall market. However, these types of baths and shower trays continued to operate in a niche market, very often in the non-housing segment and the upper residential segment.

Hydromassage bath sales also declined in 2020, for the same reasons as mentioned above. The unexpected decline of new housing activity affected this market negatively. The trend away from baths towards showering is another important reason for the negative trend of hydromassage baths.

Prices and Market Values

The Turkish bath and shower tray market is very competitive, with a high number of domestic manufacturers and low labour costs. Hence, the Turkish bath and shower tray MSPs are of the lowest in Europe. Bath and shower tray pricing can often depend on the locality of the manufacturer and width of their distribution. Local manufacturers are more sensitive to local price shifts and sensitivity, while larger manufacturers can maintain fixed nationwide price strategies. Baths and shower trays usually provide dealers with a margin of approximately 40% of the list price; this figure can increase to 60% if dealers sign one-year contracts with manufacturers in advance.

In comparison to other European countries, where a shift away from the middle segment to the economy and lower segments still continues, the trend is different in Turkey. The share of the economy segment is very high, as the country is still

developing. When the Turkish economy was growing very quickly up to 2017, there was a shift from the economy segment towards the middle and upper segments. However, due to the weakening of the Turkish Lira along with growing inflation and a decreasing purchasing power of Turks, this changed totally in 2019 and then again in 2020, and there has been an overall shift away from the upper and middle segments towards the lower segments.

Regarding the MSP, all products have recorded significant increases in lira. Imported baths and shower trays occupy only a very small share of the Turkish market, as most products are supplied by manufacturing sites within Turkey. These sites are often partly owned by foreign companies. However, importing energy and raw materials is the most costly expenditure for Turkish manufacturing sites, which are usually paid for in expensive US dollars. Hence, producers have been forced to increase their MSPs for all products due to currency fluctuations, but only in a very controlled manner, accepting lower margins as the overall baths market declines strongly and becomes even more competitive. Manufacturers did not want to put their sales in danger in a declining 2020 market. As a result, the MSP in EUR declined in 2020 in comparison to 2019, meaning that manufacturers did not increase their MSP prices in local currencies, in line with the overall economic development with a weakening domestic currency.

New Housing

In Turkey, the bath and shower tray markets heavily depend on sales into new dwellings, the number of which declined steeply in 2020. As a result, bath and shower tray sales, especially those of the synthetic models, into new housing decreased dramatically in the year. The decline of bath sales in this sector was also due to the decreasing average bathroom space in new dwellings, along with the growing trend towards showering. At the same time, sales of shower trays declined as well, due to the growing consumer preference for stone and tile floor showers. Both trends are very often implemented by contractors, as they want to avoid costs. Hence, contractors prefer to avoid baths and so install stone and tile floor showers in new dwellings, as they are more aesthetically pleasing yet also cheaper. The declining number of dwelling completions, as well as the trend away from baths towards showering, negatively affected sales of hydromassage baths, which experienced smaller sales volumes in this segment in 2020, in comparison to previous years.

Housing RMI

Sales of the bath and shower tray market for refurbishment and replacement accounts for the second-largest share of sales volumes in Turkey. As the quality of the products is not as good as in Western Europe, baths and shower trays have shorter lifespans in the country. The same is true for hydrotherapy baths. The quality of these products can be very low, which means they need more frequent replacement.

In 2020 the replacement activity experienced some strong growth, due to unexpected developments. Due to the COVID-19 lockdowns many Turks spent most of their time at home and, as a result, they started to renovate their homes. On top of that, a huge number of Turks escaped from big cities towards rural areas, where they invested in renovations of their holiday homes. Hence, sales of shower trays, especially, in housing RMI increased very strongly in 2020. The overall bath market could not benefit from this development, due to the trend away from baths towards showering.

Non-Housing

Total non-housing sales of baths and shower trays represented the smallest end-use segment in 2020. Steel, cast iron and ceramic models are often preferred in the non-housing sector, for example in hotels and spa centres, as they have a greater durability than their synthetic equivalents. Stand-alone cast iron baths have become especially popular in high-end hotels.

Regarding hydrotherapy baths, even though the non-housing segment has become an increasingly important source of demand for hydrotherapy products in recent years, sales to non-housing are relatively low when compared to other end-use segmentations. Luxury hotels, fitness clubs and spas are the top destination for hydrotherapy products in non-residential installations. The relatively low market share is expected to increase in the coming years, as the tourism industry is becoming increasingly competitive. Hotel owners strive to give their hotels the advantage by providing increasingly luxurious accommodation and facilities, and therefore hydrotherapy baths are extremely relevant here.

Supply Structure

No major changes on the supply side were recorded in 2020. The market remains very fragmented and is supplied mainly by Turkish producers holding small chunks of the market. The market is fiercely competitive, with many small firms operating with unfair business practices, such as working without invoices, to force prices down. Due to the high number of local bath and shower tray producers, it is very difficult to identify all players by their brand.

The hydrotherapy bath market is largely supplied by the established synthetic bath and shower tray suppliers, including SERICO, ARTEMIS, SANICA, SHOWER, SEREL, VITRA, and FORMINA. Steel baths are led by foreign brands, whereas shower trays are dominated by domestic manufacturers.

SERICO, previously named HYDROCABIN, grew very strongly up until a couple of years ago. The company experienced some financial problems in 2017 and sold its shares to TEKFEN, which also owns one of the biggest DIY chains in Turkey, TEKZEN. HYDROCABIN used to primarily sell its products via Turkey's biggest DIY chain, KOCTAS. After the cooperation with KOCTAS's competitor TEKZEN, KOCTAS stopped selling HYDROCABIN products and so its distribution channels decreased significantly in 2017. However, in 2018, KOCTAS added HYDROCABIN

back to its portfolio. Beginning in 2020, HYDROCABIN restructured its business and created the SERICO brand, which could be a very large supplier in the coming years.

KALE developed originally from Turkey's leading tiles supplier holding, the KALE GROUP. The company also follows the trend of growing horizontally – supplying different complementary products. The company is established very well in the bath and shower tray markets, due to aggressive marketing and offering low-cost products. The company imports most of its baths from China and had to increase its MSPs very strongly due to the weakening of the Turkish currency. Hence, sales declined significantly in 2020.

DOCCE, EREL, BATHEUS, and CORIN all faced some financial challenges in 2020, due to the competitive and decreasing market. STINA and DUSAKABINCIM are selling a significant number of products via the internet.

VILLEROY & BOCH did not declare any synthetic bath and ceramic shower tray sales for 2020.

Forecasts

In the first four months of 2021, bath as well as hydromassage bath sales have continued to decline. The ongoing challenge for these markets, i.e. shifting away from baths towards showering, has intensified and continued to negatively affect overall sales. On top of that, the number of new housing completions, which is the most important market driver for these sales, is expected to decline further in 2021.

However, on the other hand the share of walk-in showers started to decline again, as Turks had bad experiences such as water leaking due to using low-quality materials, which is supportive for the overall shower tray sales. On top of that, the ongoing COVID-19 pandemic, along with the growing replacements, continued in the first quarter of 2021 and affected the total bath and shower tray sales positively. As a result, the decline rate of the overall bath sales slowed down between January and April 2021. On the other hand, the declining rate of walk-in showers supported the shower tray market, which means that the declining trend of the overall shower tray market stopped, and the market started to grow again. However, no major changes are expected in the coming quarters and the overall bath market should remain negative, whereas the shower tray market should grow slightly on the whole in 2021.

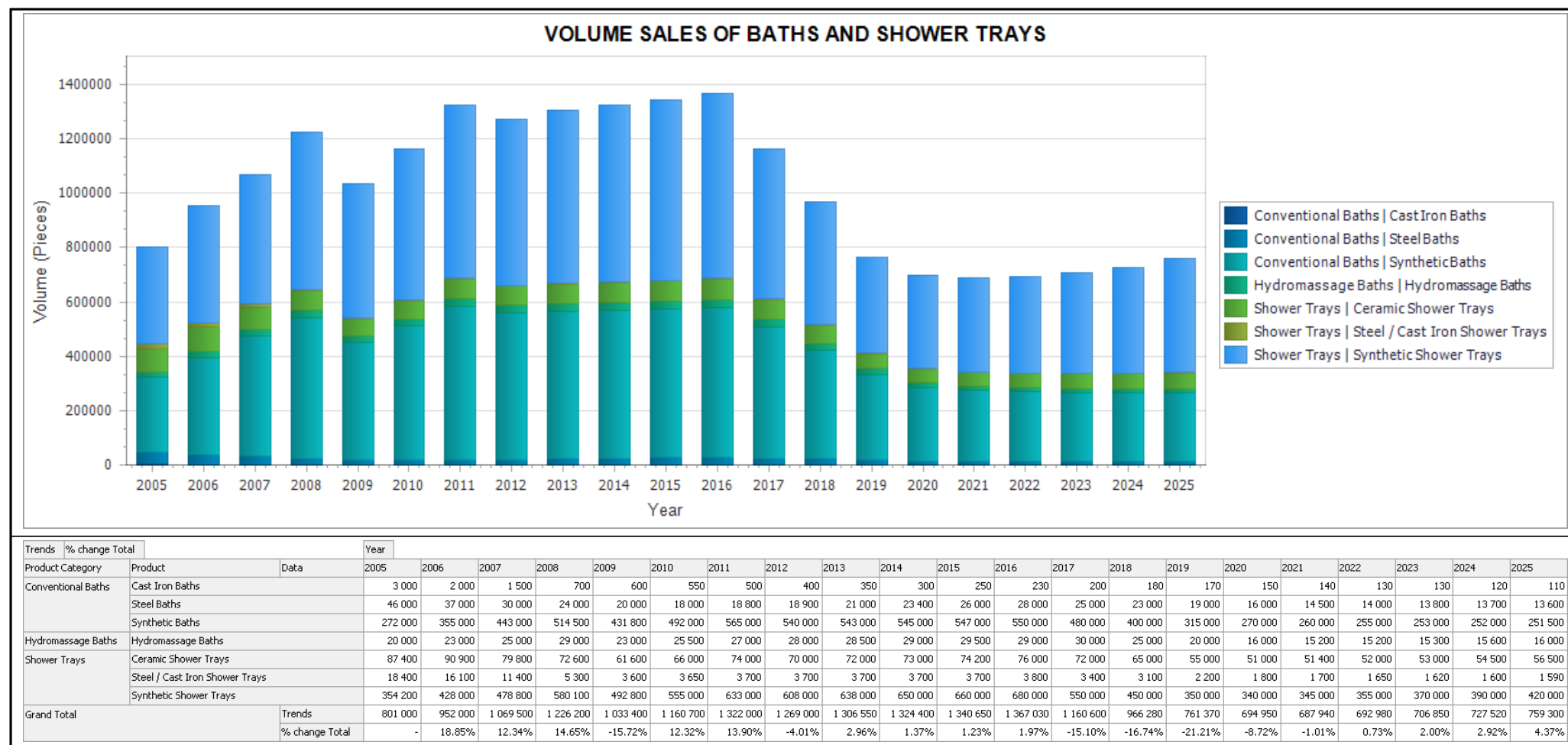
The mid-term outlook for the Turkish bath and shower tray market is very similar to the first quarter of 2021, with growing shower tray and declining bath sales. The substitution trend from baths to showers should still be prevalent and is unlikely to slow down, given that the factors for the trend are not set to reverse anytime soon, such as smaller living spaces, busier lives and an ageing population. Hence, all bath types are expected to decline during the forecast period. However, the negative rate is predicted to decrease by 2025, thanks to replacement activity in

the residential segment.

The outlook for shower trays is even brighter for the forecast period. As mentioned before, the popularity of walk-in showers is expected to decline further in the coming years, which should affect the shower tray market positively. Hence, the overall shower tray market is predicted to grow stronger annually until 2025. This scenario should be supported by an increasingly brighter outlook for the new construction market. However, this counts on a positive end of the coronavirus pandemic. The overall market should mostly be supported by synthetic models, which operate in the mass market. On the other hand, steel / cast iron shower tray customers primarily prefer design and quality, with cost playing a secondary role. A better economic environment could encourage consumers to choose a more luxury-priced item.

1.2 Historical Trends and Forecasts

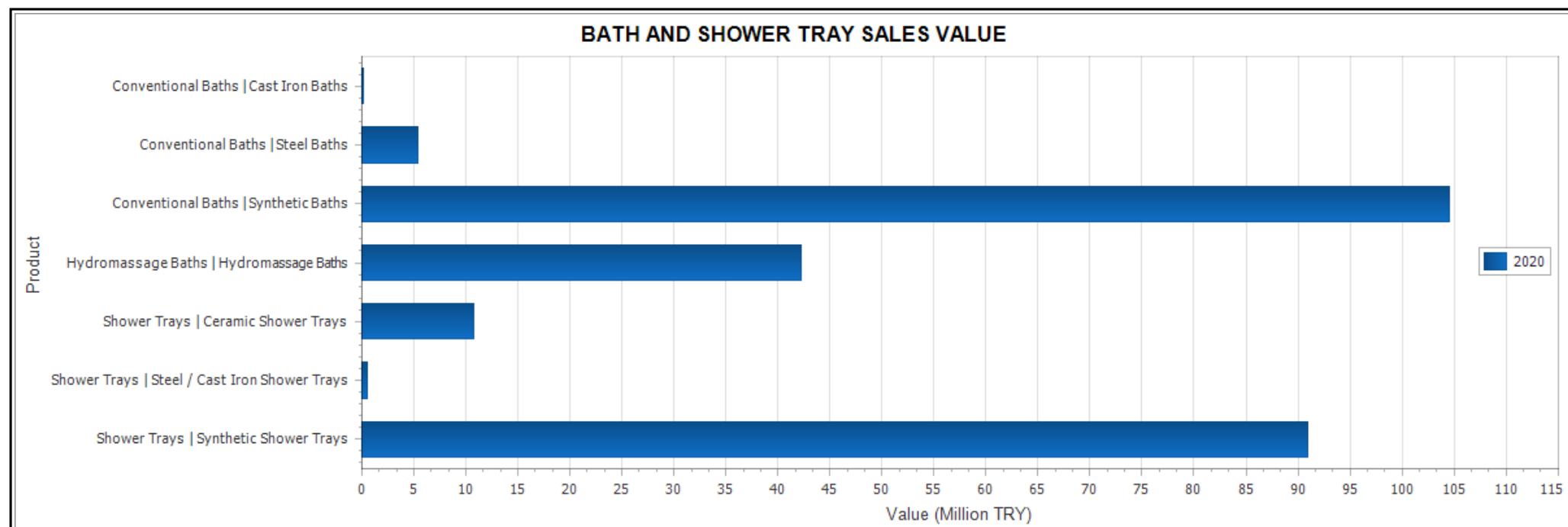
TR - SUMMARY OF TRENDS IN THE BATH AND SHOWER TRAY MARKET : HISTORICAL AND FORECASTS



Source: BRG based on industry interviews.

1.3.1 Prices and Market Values : Baths and Shower Trays

TR - ESTIMATED VALUES OF THE BATH AND SHOWER TRAY MARKET

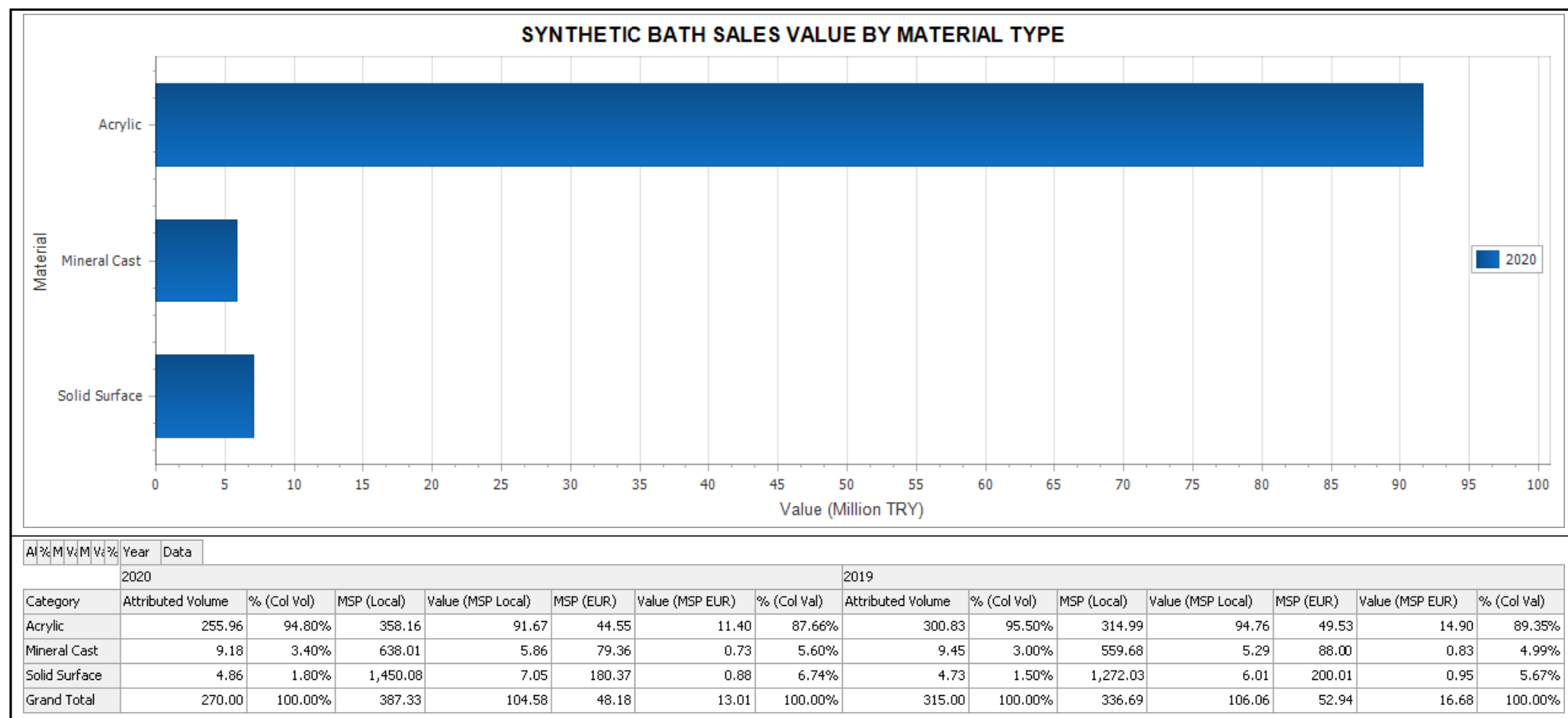


Volumes	MSP (Local)	Value (MSP)	MSP (EUR)	Value (MSP)	Year	Data						
2020							2019					
Product Category	Product	Volumes	MSP (Local)	Value (MSP Local)	MSP (EUR)	Value (MSP EUR)	Volumes	MSP (Local)	Value (MSP Local)	MSP (EUR)	Value (MSP EUR)	
Conventional Baths	Cast Iron Baths	150	1,063.95	0.16	132.34	0.02	170	934.16	0.16	146.88	0.02	
	Steel Baths	16 000	338.46	5.42	42.10	0.67	19 000	297.52	5.65	46.78	0.89	
	Synthetic Baths	270 000	387.34	104.58	48.18	13.01	315 000	336.70	106.06	52.94	16.68	
Hydromassage Baths	Hydromassage Baths	16 000	2,639.61	42.23	328.33	5.25	20 000	2,318.73	46.37	364.58	7.29	
Shower Trays	Ceramic Shower Trays	51 000	209.51	10.69	26.06	1.33	55 000	187.30	10.30	29.45	1.62	
	Steel / Cast Iron Shower Trays	1 800	281.22	0.51	34.98	0.06	2 200	249.76	0.55	39.27	0.09	
	Synthetic Shower Trays	340 000	267.39	90.91	33.26	11.31	350 000	234.43	82.05	36.86	12.90	
Grand Total		694 950	366.21	254.49	45.55	31.66	761 370	329.86	251.15	51.87	39.49	

Source: BRG based on industry interviews. Local Prices Shown in TRY. Values Shown in Millions (at MSP = Manufacturers' Selling Price).

1.3.2 Prices and Market Values : Synthetic Baths

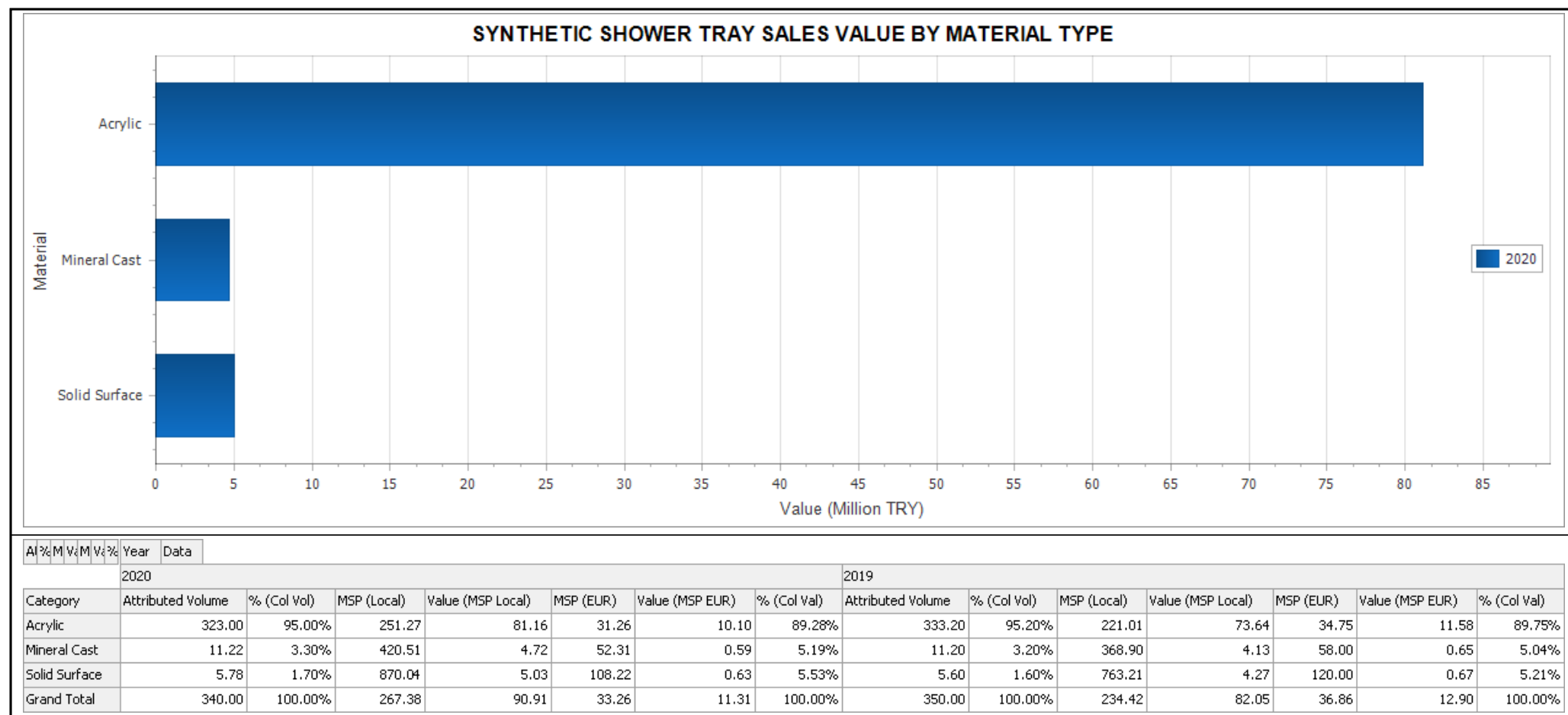
TR - ESTIMATED VALUES OF THE SYNTHETIC BATH MARKET BY TYPE OF MATERIAL



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Local Prices Shown in TRY. Values Shown in Millions (at MSP = Manufacturers' Selling Price)

1.3.3 Prices and Market Values : Synthetic Shower Trays

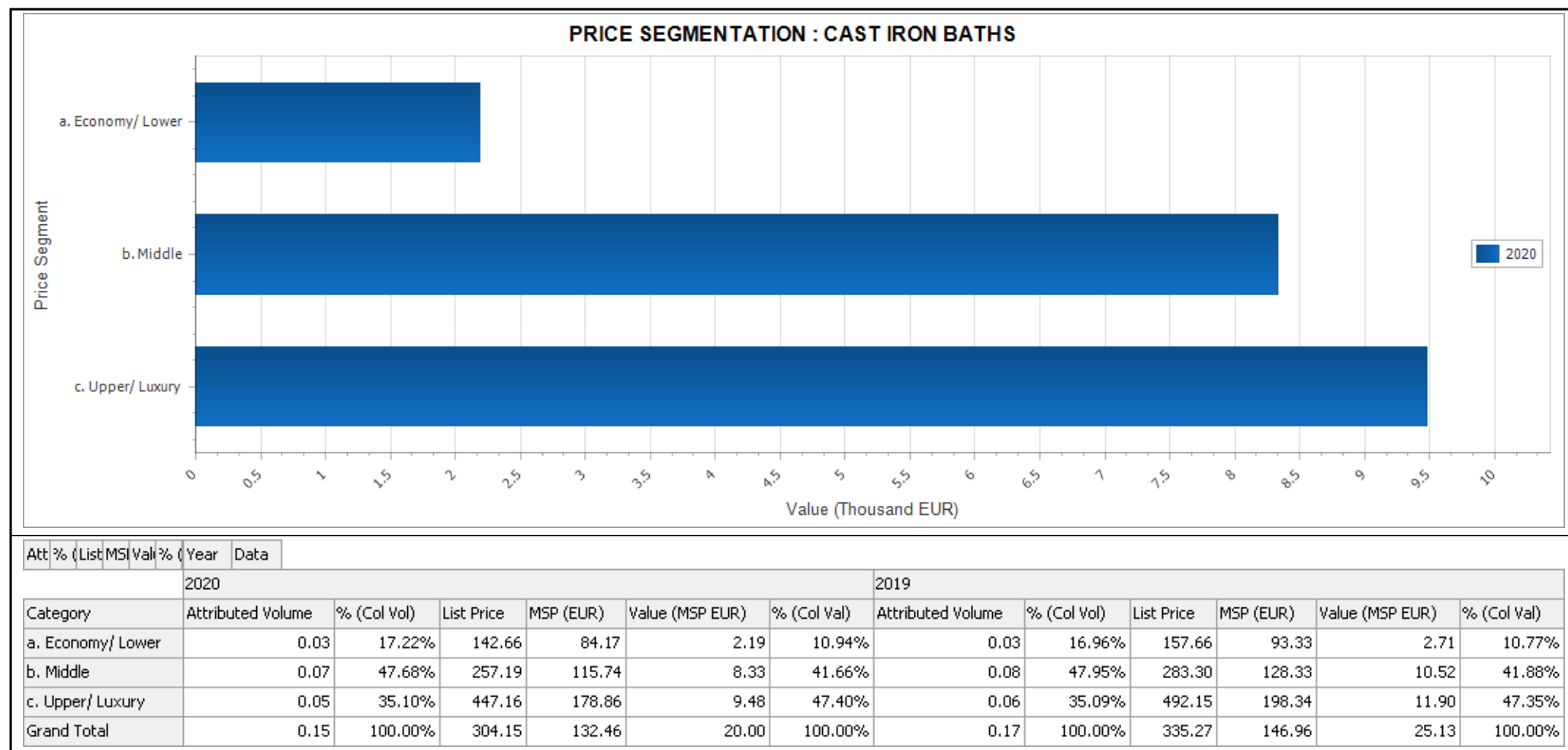
TR - ESTIMATED VALUES OF THE SYNTHETIC SHOWER TRAYS MARKET BY TYPE OF MATERIAL



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Local Prices Shown in TRY. Values Shown in Millions (at MSP = Manufacturers' Selling Price)

1.3.4 Price Segmentation : Cast Iron Baths

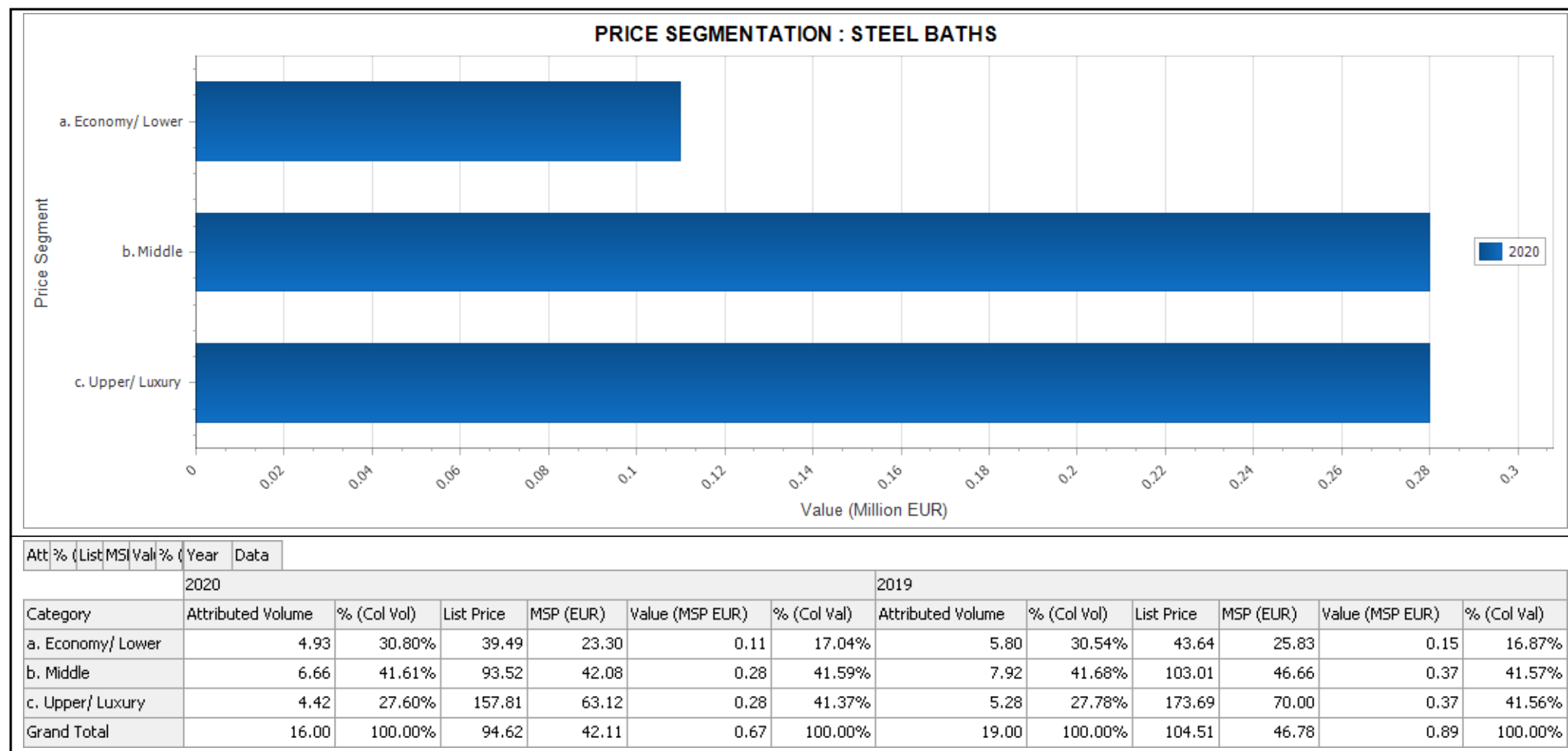
TR - ESTIMATED PRICE SEGMENTATION FOR CAST IRON BATHS



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Volumes and Values Shown in Thousands (at MSP = Manufacturers' Selling Price). Economy/ Lower <€100, Middle €100-€150, Upper/ Luxury >€150

1.3.5 Price Segmentation : Steel Baths

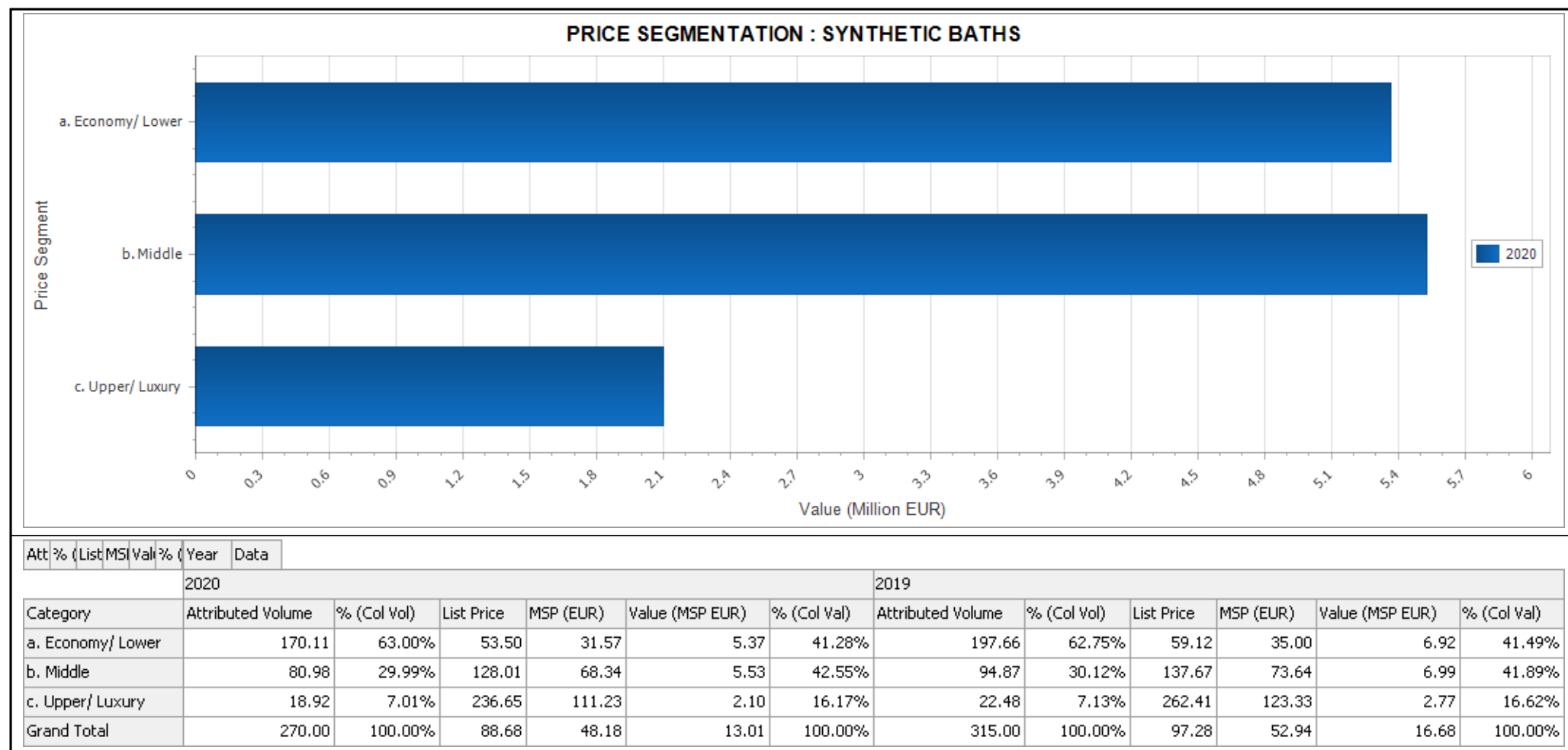
TR - ESTIMATED PRICE SEGMENTATION FOR STEEL BATHS



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€30, Middle €30-€50, Upper/ Luxury >€50

1.3.6 Price Segmentation : Synthetic Baths

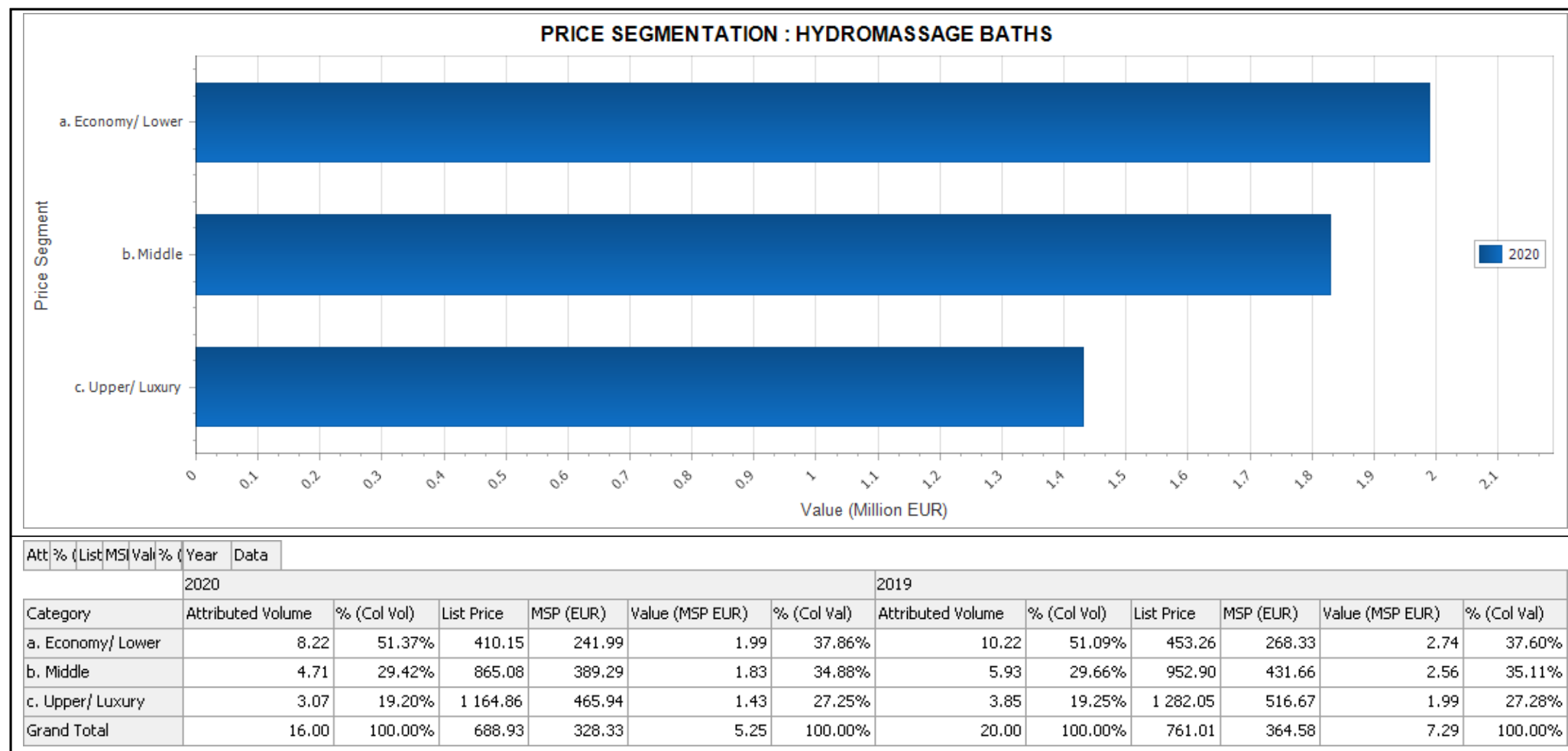
TR - ESTIMATED PRICE SEGMENTATION FOR SYNTHETIC BATHS



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€40, Middle €40-€80, Upper/ Luxury >€80

1.3.7 Price Segmentation : Hydromassage Baths

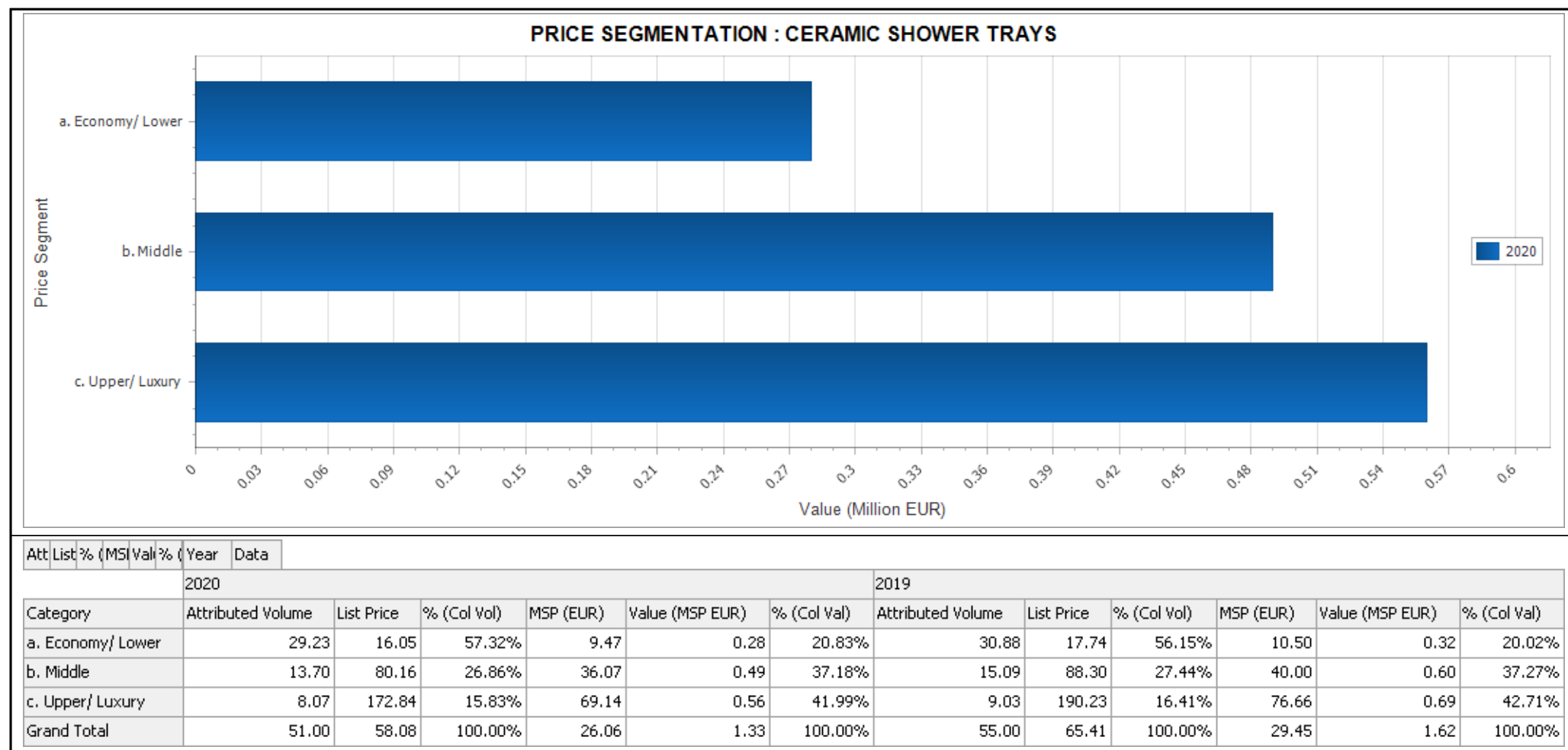
TR - ESTIMATED PRICE SEGMENTATION FOR HYDROMASSAGE BATHS



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€300, Middle €300-€450, Upper/ Luxury >€450

1.3.8 Price Segmentation : Ceramic Shower Trays

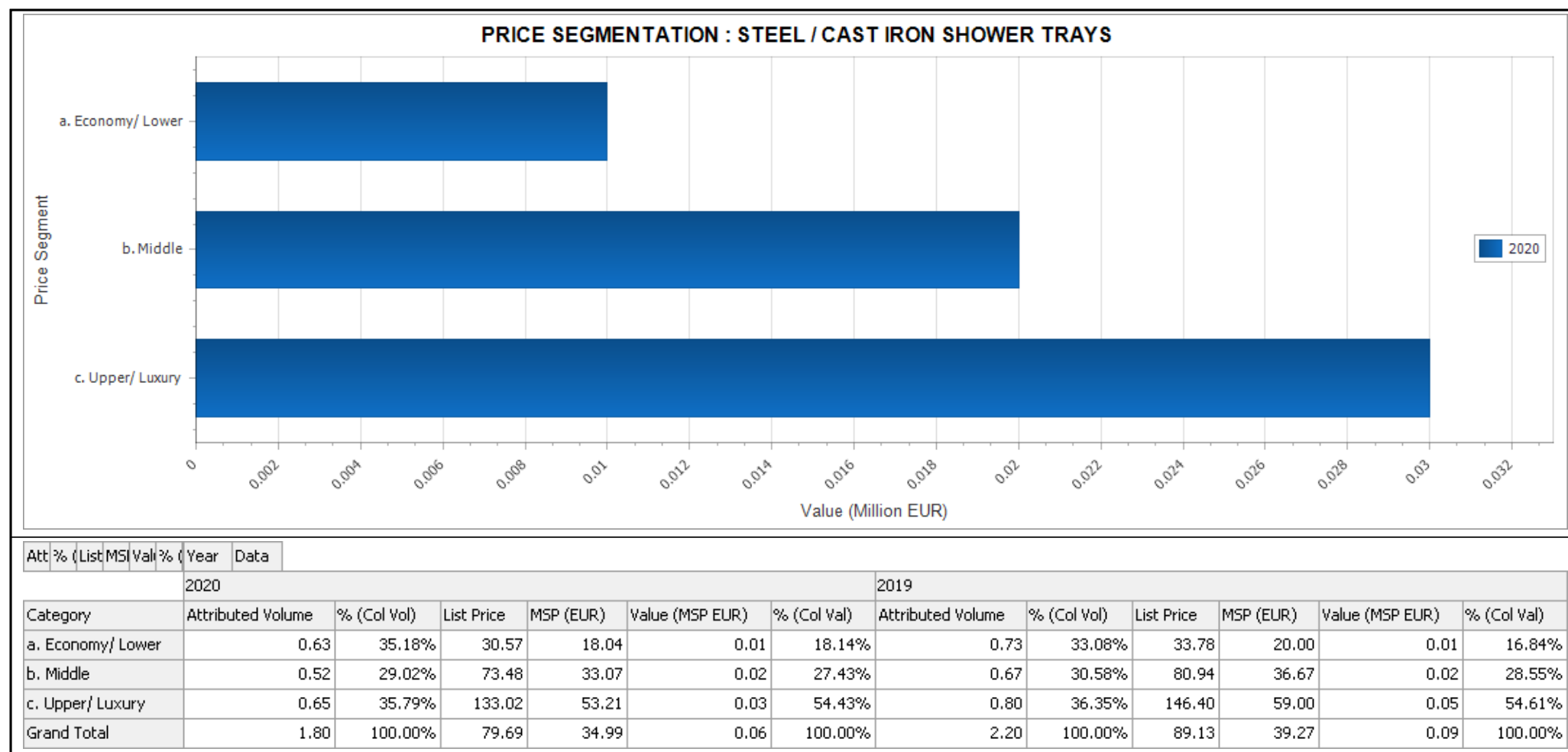
TR - ESTIMATED PRICE SEGMENTATION FOR CERAMIC SHOWER TRAYS



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€25, Middle €25-€50, Upper/ Luxury >€50

1.3.9 Price Segmentation : Steel/ Cast Iron Shower Trays

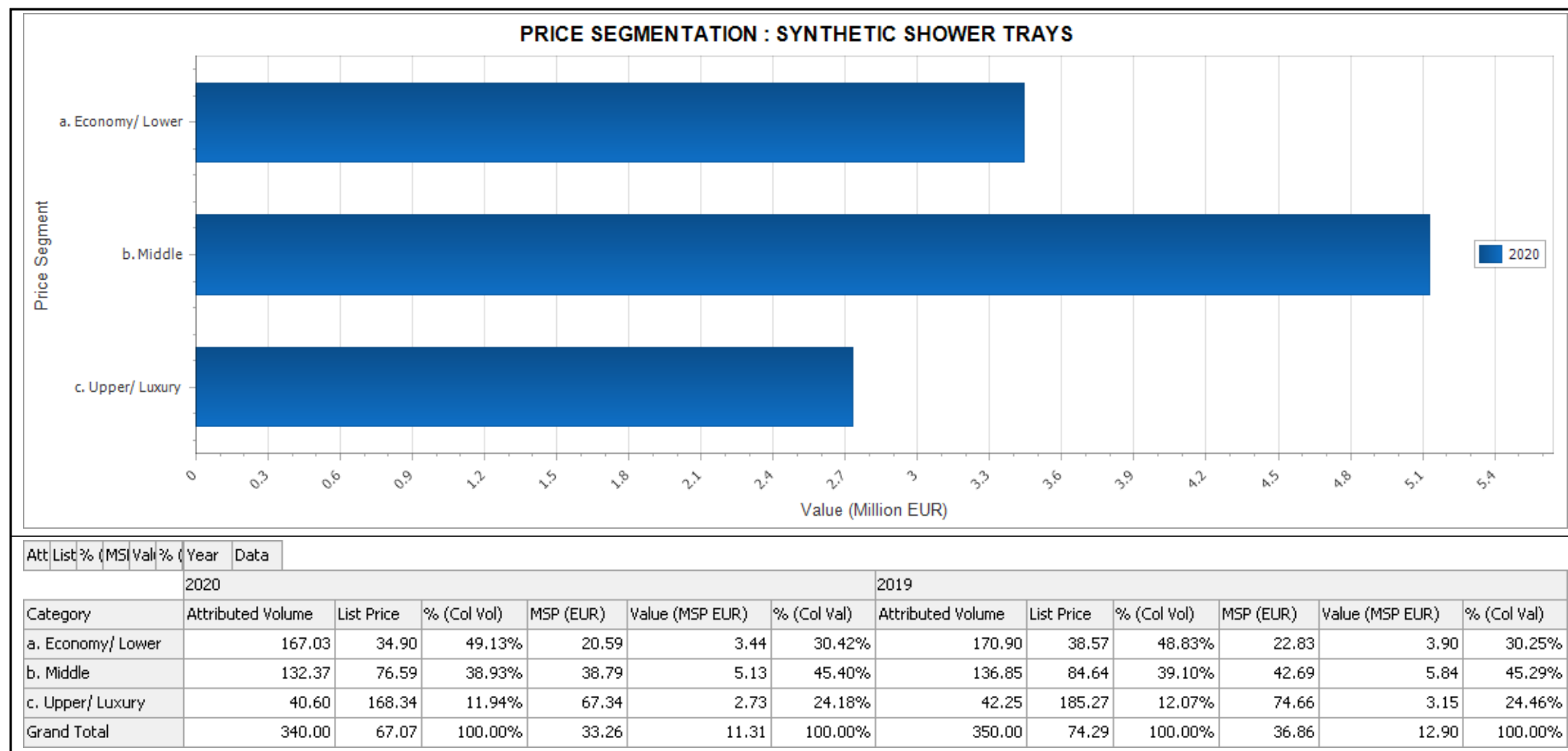
TR - ESTIMATED PRICE SEGMENTATION FOR STEEL / CAST IRON SHOWER TRAYS



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€25, Middle €25-€40, Upper/ Luxury >€40

1.3.10 Price Segmentation : Synthetic Shower Trays

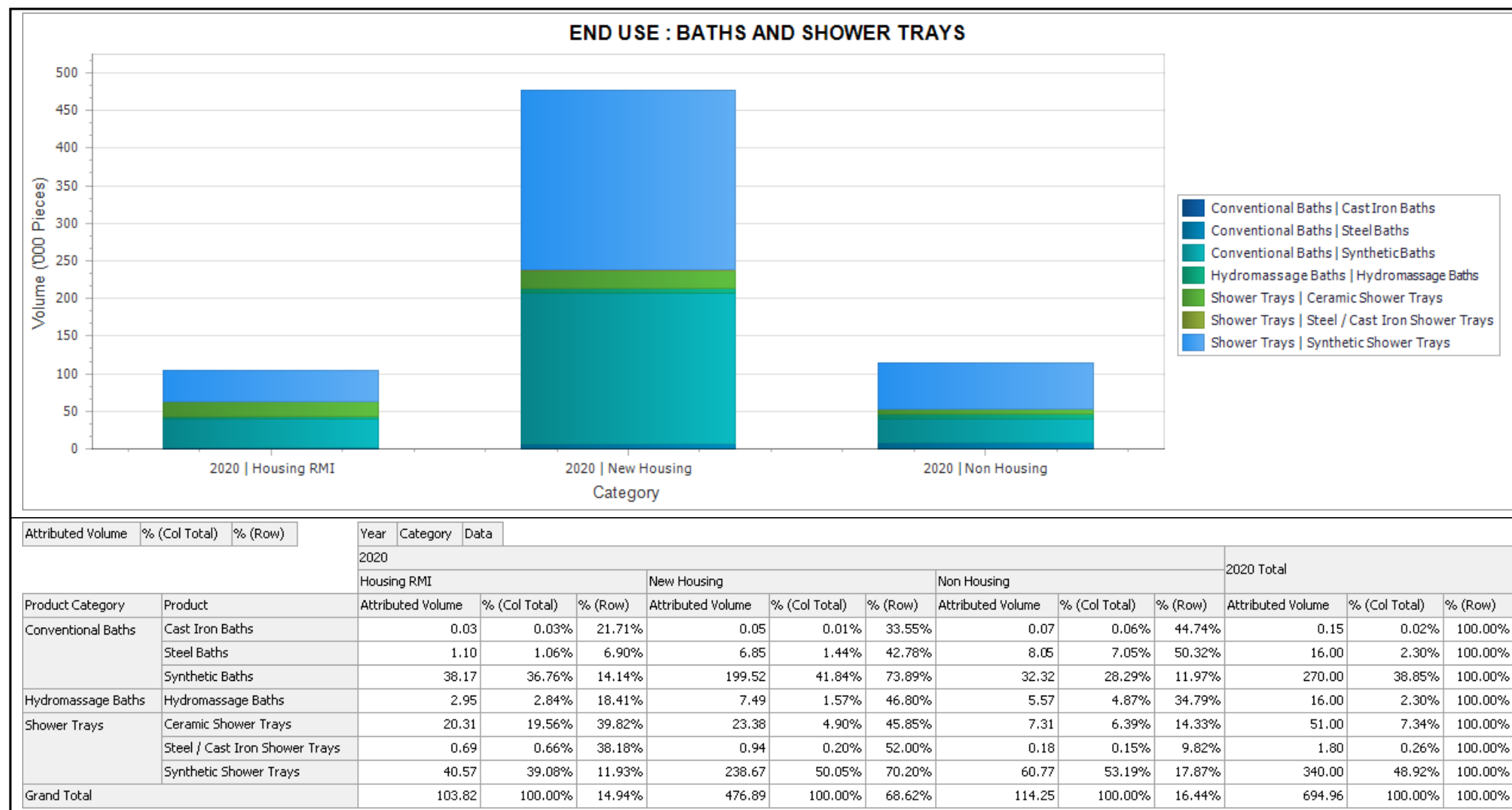
TR - ESTIMATED PRICE SEGMENTATION FOR SYNTHETIC SHOWER TRAYS



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€30, Middle €30-€50, Upper/ Luxury >€50

1.4.1 End Use : Base Year

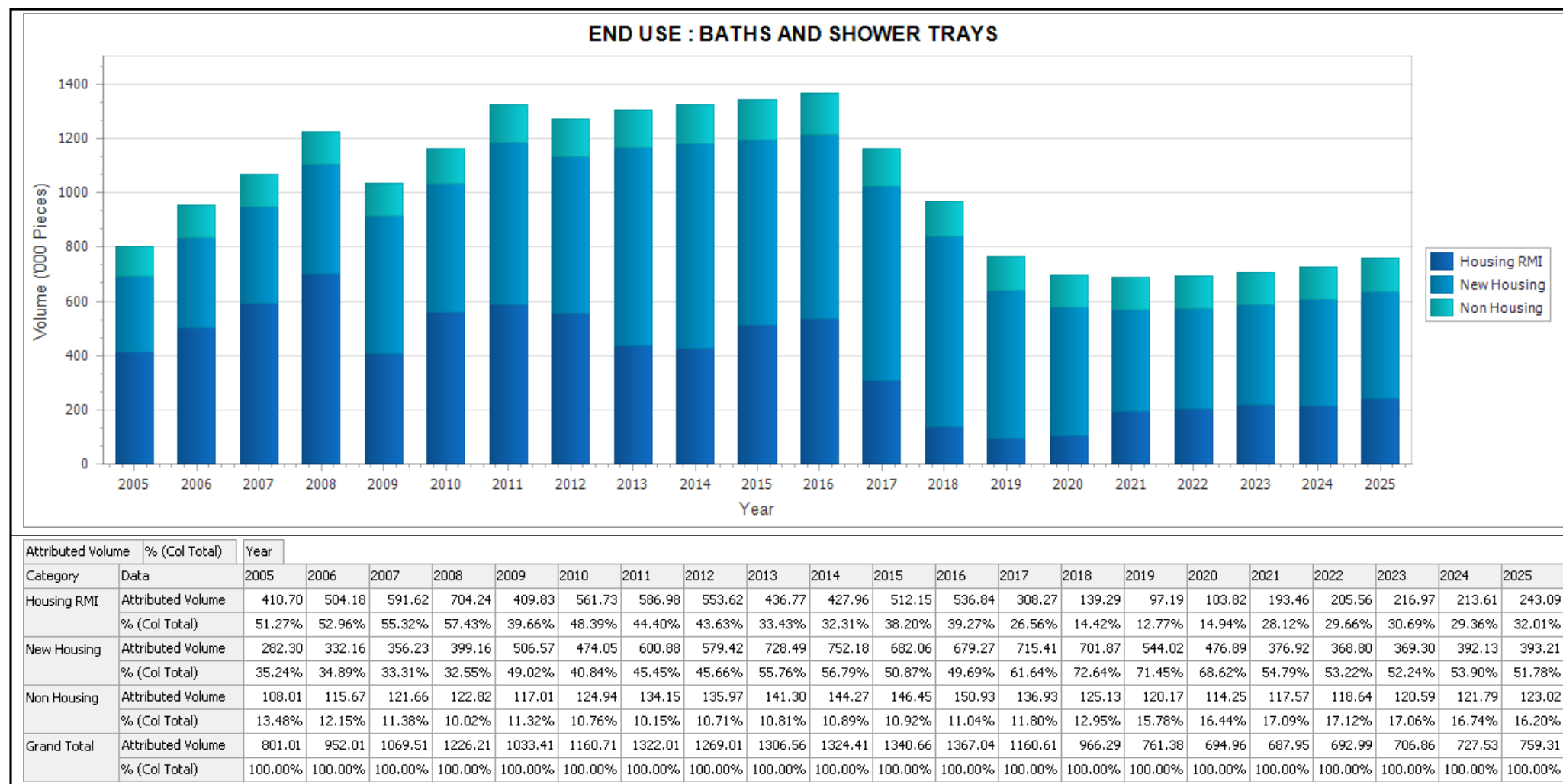
TR - SALES OF BATHS AND SHOWER TRAYS BY TYPE AND END USE SEGMENT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

1.4.2 End Use : Historical and Forecasts

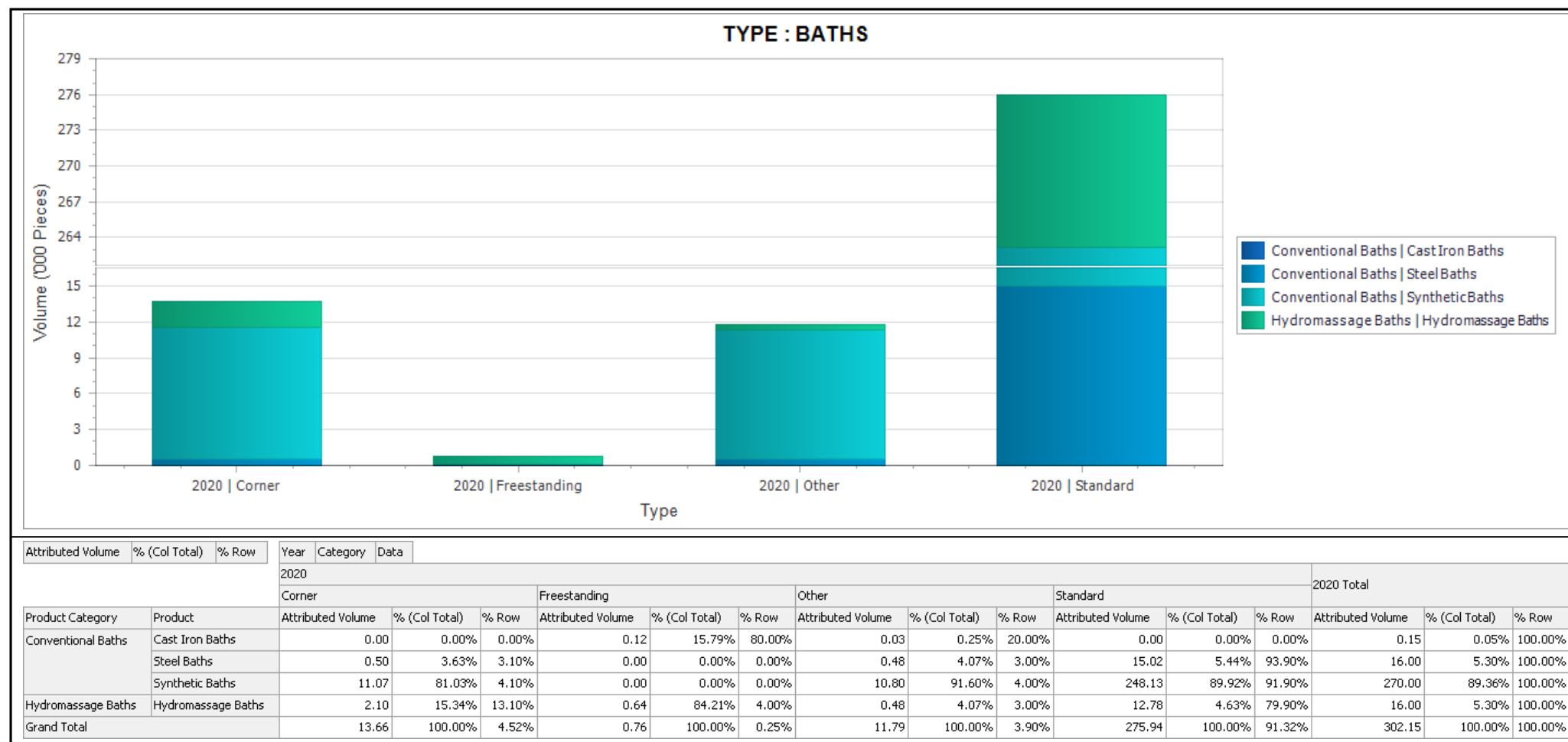
TR - SALES OF BATHS AND SHOWERS TRAYS BY END USE SEGMENT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

1.4.3 Sales by Type : Baths

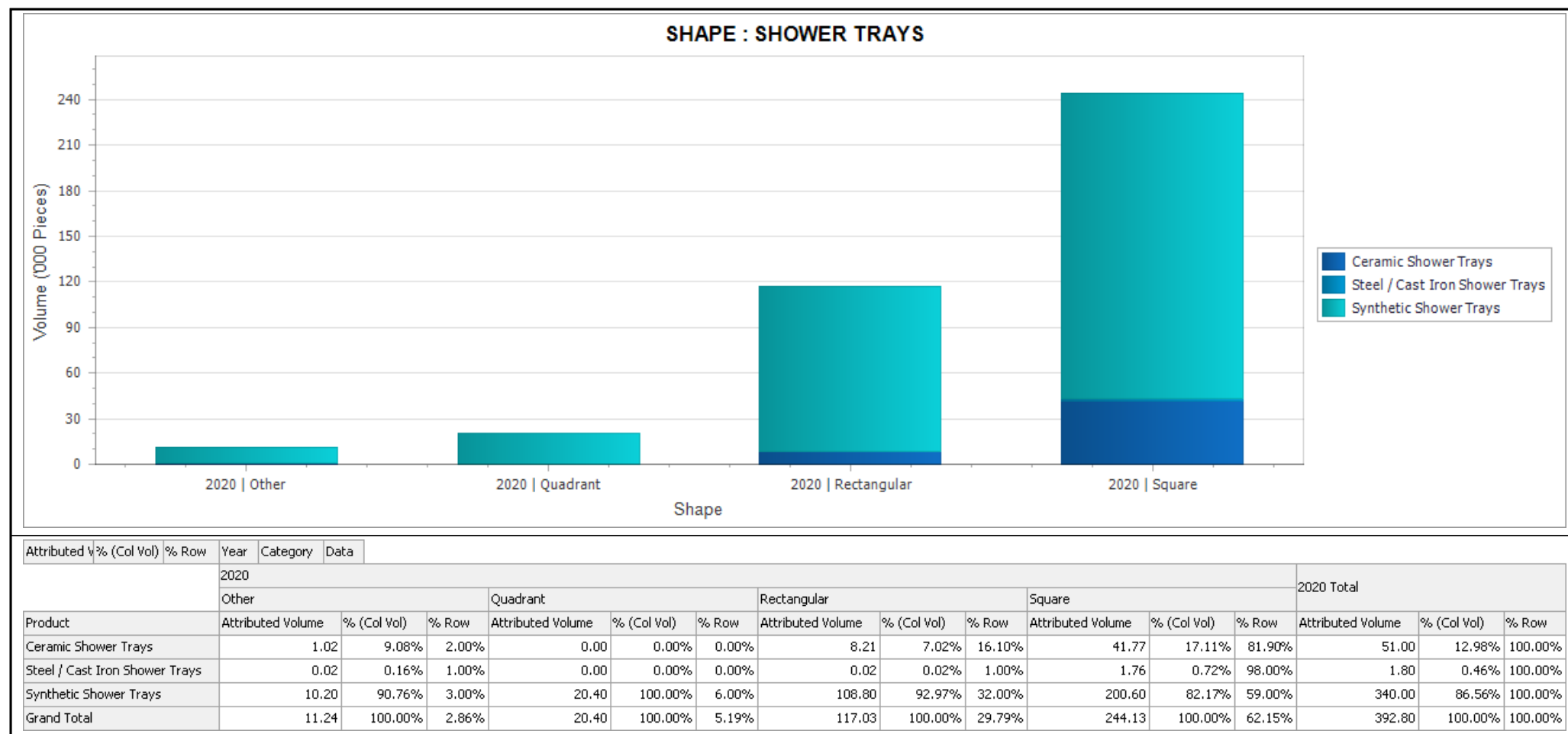
TR - BATH SALES BY TYPE ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

1.4.4 Sales by Shape : Shower Trays

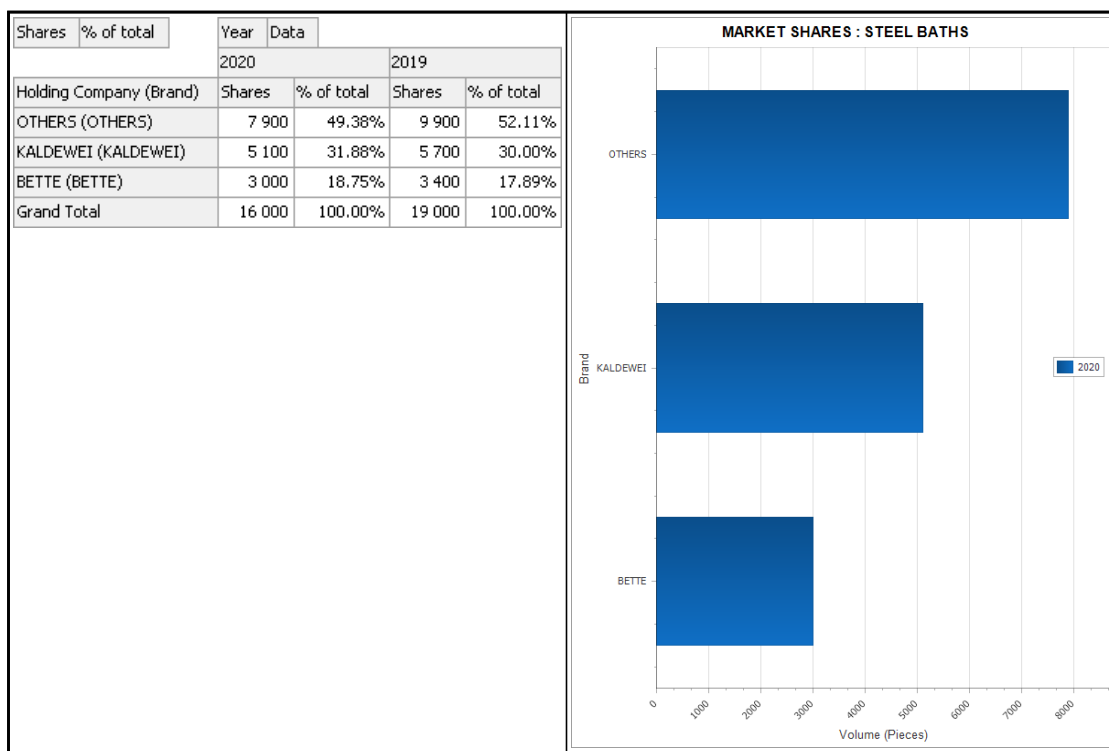
TR - SHOWER TRAY SALES BY SHAPE ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

1.5.1 Market Shares : Steel Baths

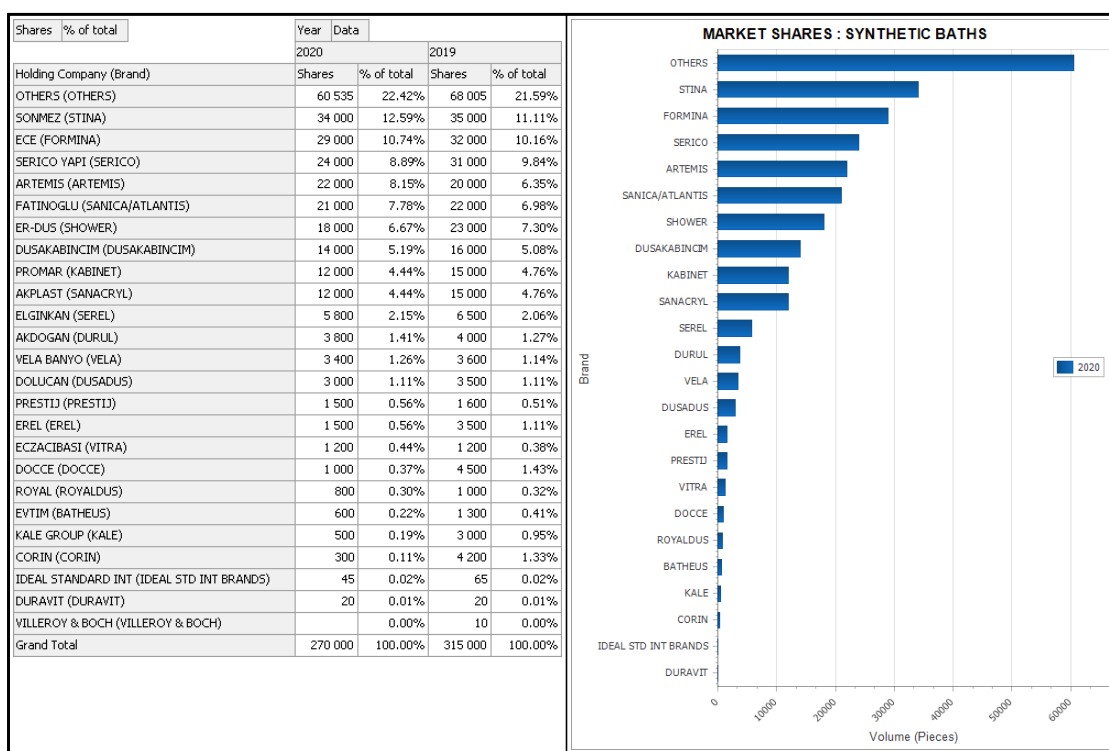
TR - ESTIMATED MARKET SHARES FOR STEEL BATHS



Source: BRG based on industry interviews.

1.5.2 Market Shares : Synthetic Baths

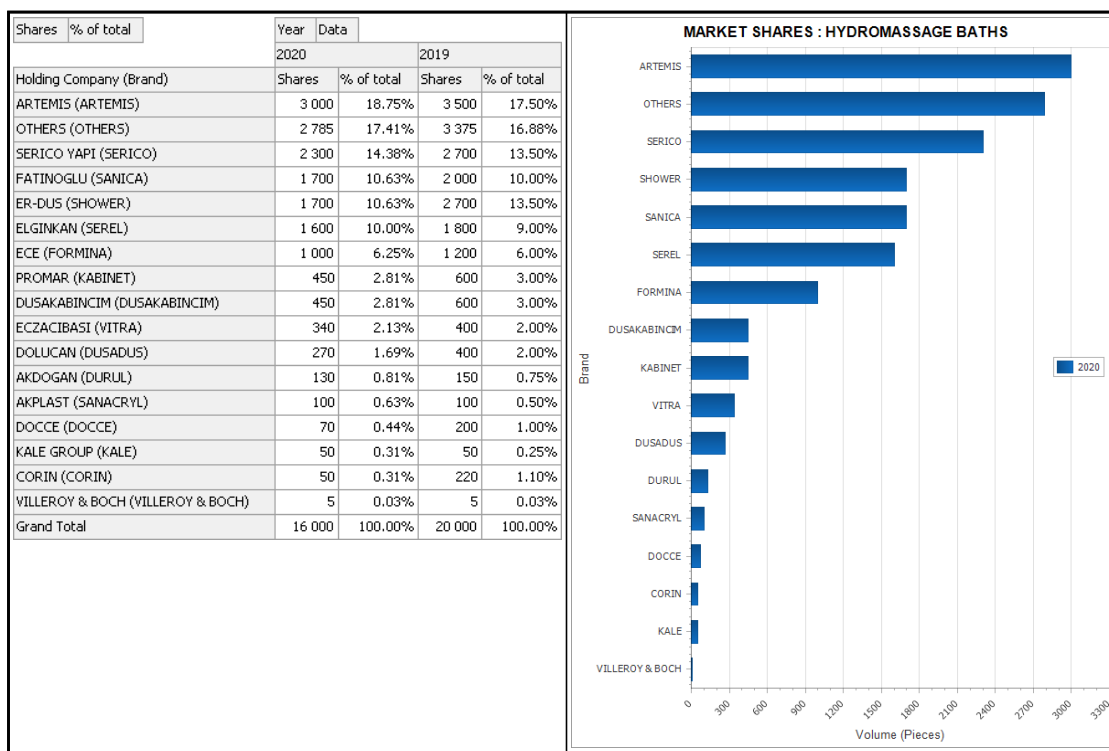
TR - ESTIMATED MARKET SHARES FOR SYNTHETIC BATHS



Source: BRG based on industry interviews.

1.5.3 Market Shares : Hydromassage Baths

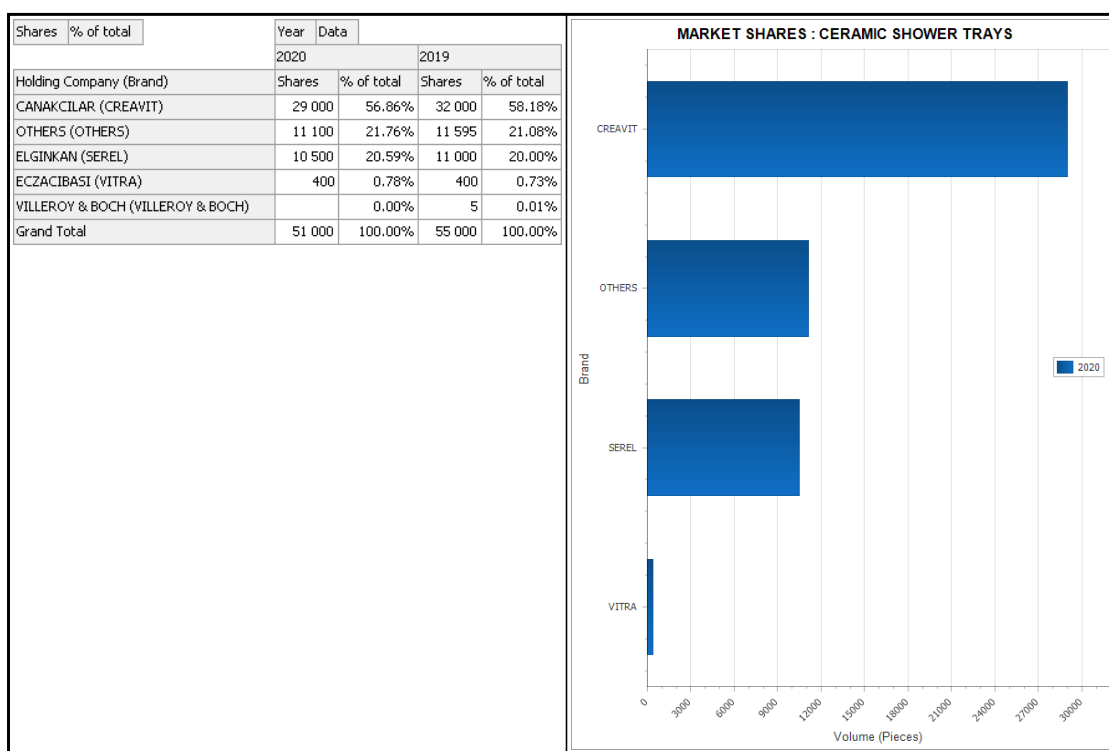
TR - ESTIMATED MARKET SHARES FOR HYDROMASSAGE BATHS



Source: BRG based on industry interviews.

1.5.4 Market Shares : Ceramic Shower Trays

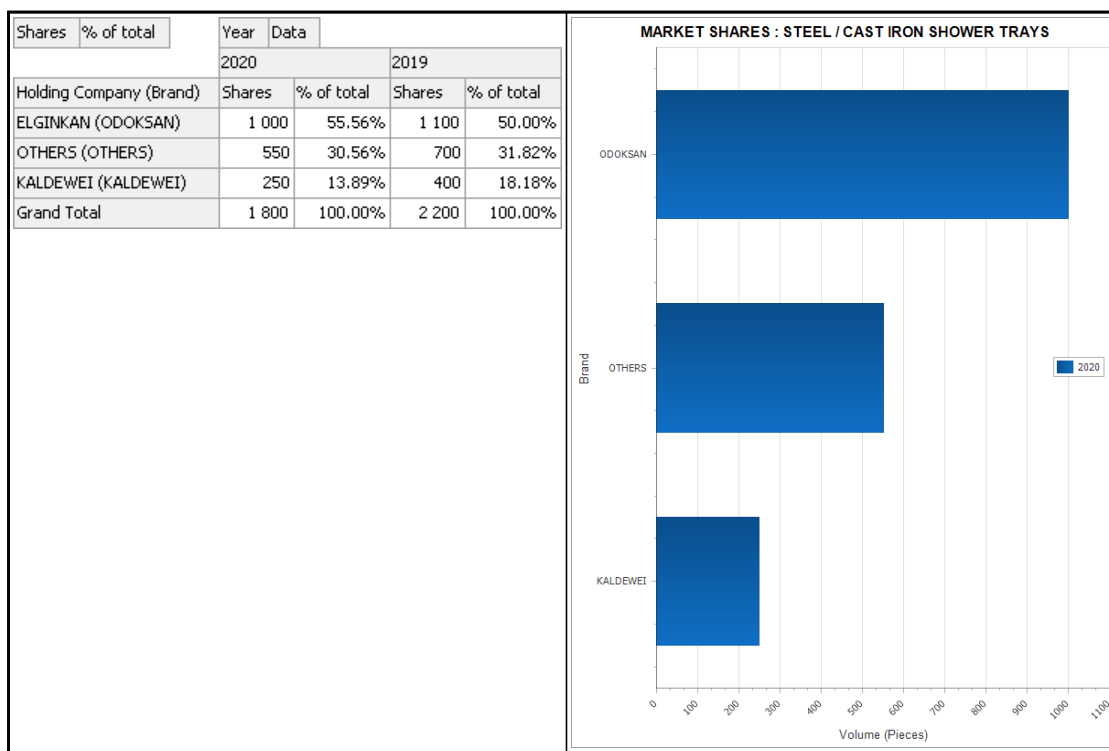
TR - ESTIMATED MARKET SHARES FOR CERAMIC SHOWER TRAYS



Source: BRG based on industry interviews

1.5.5 Market Shares : Steel / Cast Iron Shower Trays

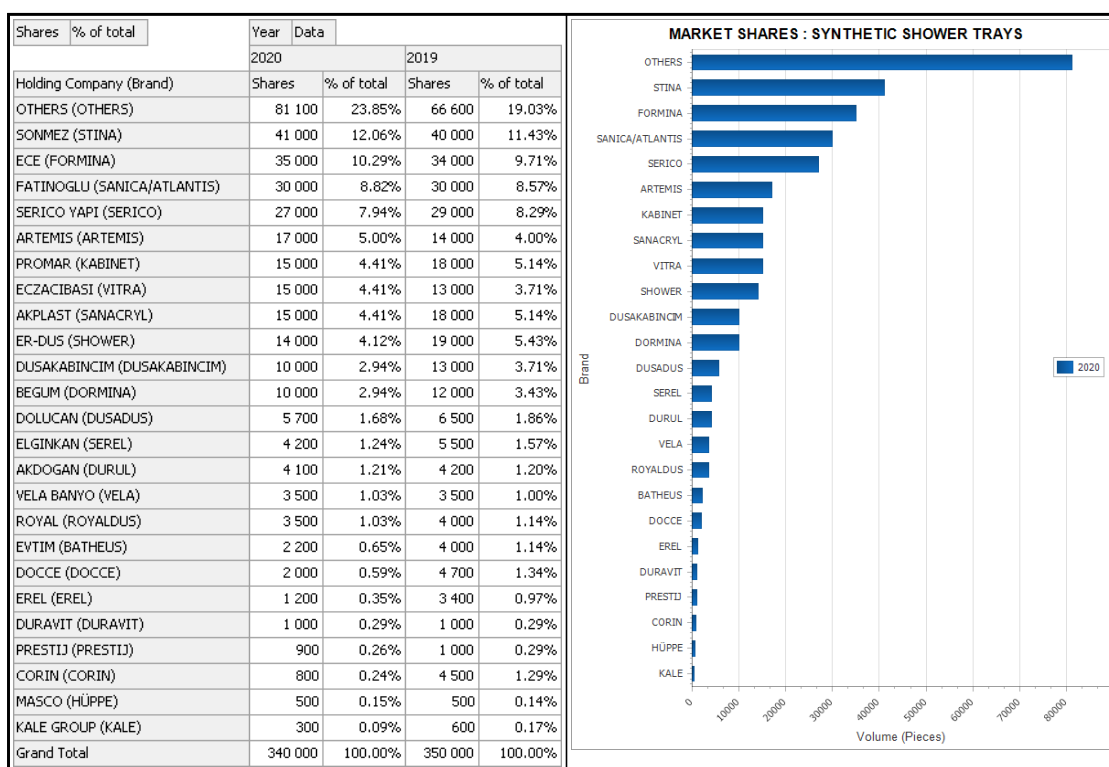
TR - ESTIMATED MARKET SHARES FOR STEEL / CAST IRON SHOWER TRAYS



Source: BRG based on industry interviews.

1.5.6 Market Shares : Synthetic Shower Trays

TR - ESTIMATED MARKET SHARES FOR SYNTHETIC SHOWER TRAYS



Source: BRG based on industry interviews.

1.6.1 TR - IMPORTS OF BATHS AND SHOWER TRAYS - CAST IRON 2020

Source Country	2020				2019			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
OTHERS	0	100.0%	10	100.0%	0	100.0%	8	100.0%
Total	0	100.0%	10	100.0%	0	100.0%	8	100.0%

Source : Trademap

1.6.2 TR - EXPORTS OF BATHS AND SHOWER TRAYS - CAST IRON 2020

Destination	2020				2019			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
Nigeria	3	37.5%	2	12.5%	0	0.0%	0	0.0%
Serbia	3	37.5%	3	18.8%	0	0.0%	0	0.0%
France	2	25.0%	2	12.5%	0	0.0%	0	0.0%
OTHERS	0	0.0%	9	56.3%	0	0.0%	1	0.0%
Total	8	100.0%	16	100.0%	0	0.0%	1	0.0%

Source : Trademap

1.6.3 TR - IMPORTS OF BATHS AND SHOWER TRAYS - STEEL 2020

Source Country	2020				2019			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
Germany	28	75.7%	283	81.6%	4	33.3%	37	56.9%
Portugal	8	21.6%	21	6.1%	3	25.0%	7	10.8%
Finland	1	2.7%	39	11.2%	0	0.0%	0	0.0%
OTHERS	0	0.0%	4	1.2%	5	41.7%	21	32.3%
Total	37	100.0%	347	100.0%	12	100.0%	65	100.0%

Source : Trademap

1.6.4 TR - EXPORTS OF BATHS AND SHOWER TRAYS - STEEL 2020

Destination	2020				2019			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
Germany	80	28.9%	509	31.6%	28	10.7%	158	9.5%
United Kingdom	16	5.8%	99	6.1%	1	0.4%	1	0.1%
Libya, State of	15	5.4%	73	4.5%	8	3.1%	40	2.4%
France	12	4.3%	28	1.7%	5	1.9%	23	1.4%
Azerbaijan	10	3.6%	32	2.0%	2	0.8%	15	0.9%
Italy	10	3.6%	57	3.5%	16	6.1%	99	6.0%
Romania	10	3.6%	75	4.6%	7	2.7%	42	2.5%
United Arab Emirates	10	3.6%	57	3.5%	44	16.8%	256	15.4%
Qatar	9	3.2%	62	3.8%	7	2.7%	54	3.2%
Ukraine	9	3.2%	43	2.7%	10	3.8%	50	3.0%
Greece	7	2.5%	43	2.7%	13	5.0%	74	4.4%
OTHERS	89	32.1%	535	33.2%	121	46.2%	851	51.2%
Total	277	100.0%	1 613	100.0%	262	100.0%	1 663	100.0%

Source : Trademap

1.6.5 TR - IMPORTS OF BATHS AND SHOWER TRAYS - SYNTHETIC 2020

Source Country	2020				2019			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
China	54	41.2%	302	35.9%	63	28.8%	439	28.9%
Bulgaria	14	10.7%	65	7.7%	16	7.3%	71	4.7%
India	9	6.9%	40	4.8%	9	4.1%	43	2.8%
Slovakia	9	6.9%	68	8.1%	0	0.0%	4	0.3%
Egypt	8	6.1%	62	7.4%	11	5.0%	39	2.6%
Italy	8	6.1%	82	9.8%	8	3.7%	74	4.9%
Germany	6	4.6%	49	5.8%	73	33.3%	551	36.3%
Poland	5	3.8%	27	3.2%	10	4.6%	47	3.1%
Korea, Republic of	4	3.1%	26	3.1%	0	0.0%	3	0.2%
France	3	2.3%	25	3.0%	5	2.3%	38	2.5%
Netherlands	2	1.5%	17	2.0%	19	8.7%	153	10.1%
OTHERS	9	6.9%	78	9.3%	5	2.3%	57	3.8%
Total	131	100.0%	841	100.0%	219	100.0%	1 519	100.0%

Source : Trademap

1.6.6 TR - EXPORTS OF BATHS AND SHOWER TRAYS - SYNTHETIC 2020

Destination	2020				2019			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
United Kingdom	1 517	15.8%	4 107	15.4%	948	9.1%	3 017	10.4%
Israel	1 464	15.3%	3 948	14.8%	1 328	12.8%	3 768	13.0%
Libya, State of	783	8.2%	1 710	6.4%	1 578	15.2%	3 492	12.1%
Germany	569	5.9%	1 994	7.5%	716	6.9%	2 181	7.5%
Romania	318	3.3%	543	2.0%	177	1.7%	234	0.8%
Georgia	310	3.2%	752	2.8%	609	5.8%	1 381	4.8%
Italy	287	3.0%	759	2.8%	374	3.6%	1 054	3.6%
Saudi Arabia	267	2.8%	541	2.0%	206	2.0%	487	1.7%
Egypt	240	2.5%	966	3.6%	80	0.8%	453	1.6%
United Arab Emirates	218	2.3%	668	2.5%	293	2.8%	891	3.1%
Portugal	207	2.2%	569	2.1%	339	3.3%	900	3.1%
OTHERS	3 400	35.5%	10 087	37.9%	3 765	36.2%	11 038	38.2%
Total	9 580	100.0%	26 644	100.0%	10 413	100.0%	28 896	100.0%

Source : Trademap

1.7.1

Route to Market: Baths
TR - ESTIMATED DISTRIBUTION FOR BATHS: 2020

Manufacturer / Importer

Manufacturers / Importers
100.0%

Wholesale Level

74.0%
Wholesalers / Merchants
74.0%

Retail Level

Manufacturer	Wholesaler
3.4%	0.2%
Internet Retailers	
3.6%	

Manufacturer	Wholesaler
9.0%	3.1%
DIY Retailers	
12.1%	

Manufacturer	Wholesaler
6.6%	5.0%
Specialist Retailers	
11.6%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
7.0%	30.0%	2.2%	6.5%	10.4%
Installers				
56.1%				

End Use Level

Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
0.0%	25.7%	1.4%	5.6%	1.0%	51.1%
Residential					
84.8%					
New Build			Residential RMI		
70.8%			14.0%		
Project	Non-Project		Project	Non-Project	
55.0%	15.8%		4.0%	10.0%	

Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
0.0%	10.0%	0.0%	0.0%	0.2%	5.0%
Non-Residential					
15.2%					
New Build			Non-Residential RMI		
8.0%			7.2%		
Project	Non-Project		Project	Non-Project	
5.3%	2.7%		3.9%	3.3%	

1.7.2

Route to Market: Baths
TR - ESTIMATED DISTRIBUTION FOR BATHS: 2025

Manufacturer / Importer

Manufacturers / Importers
100.0%

Wholesale Level

73.8%
Wholesalers / Merchants
73.8%

Retail Level

Manufacturer	Wholesaler
3.7%	0.3%
Internet Retailers	
4.0%	

Manufacturer	Wholesaler
9.1%	3.2%
DIY Retailers	
12.3%	

Manufacturer	Wholesaler
6.5%	5.0%
Specialist Retailers	
11.5%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
6.9%	30.0%	2.3%	6.6%	10.3%
Installers				
56.1%				

End Use Level

Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
0.0%	25.3%	1.7%	5.7%	1.0%	48.7%
Residential					
82.4%					
New Build			Residential RMI		
49.1%			33.3%		
Project	Non-Project		Project	Non-Project	
40.0%	9.1%		10.0%	23.3%	

Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
0.0%	10.0%	0.0%	0.0%	0.2%	7.4%
Non-Residential					
17.6%					
New Build			Non-Residential RMI		
8.3%			9.3%		
Project	Non-Project		Project	Non-Project	
5.5%	2.8%		5.0%	4.3%	

Source: BRG Based on Industry Interviews. - Totals may vary due to roundings

1.7.3 Route to Market - Consolidated View : Baths

TR - ESTIMATED DISTRIBUTION FOR BATHS : 2020

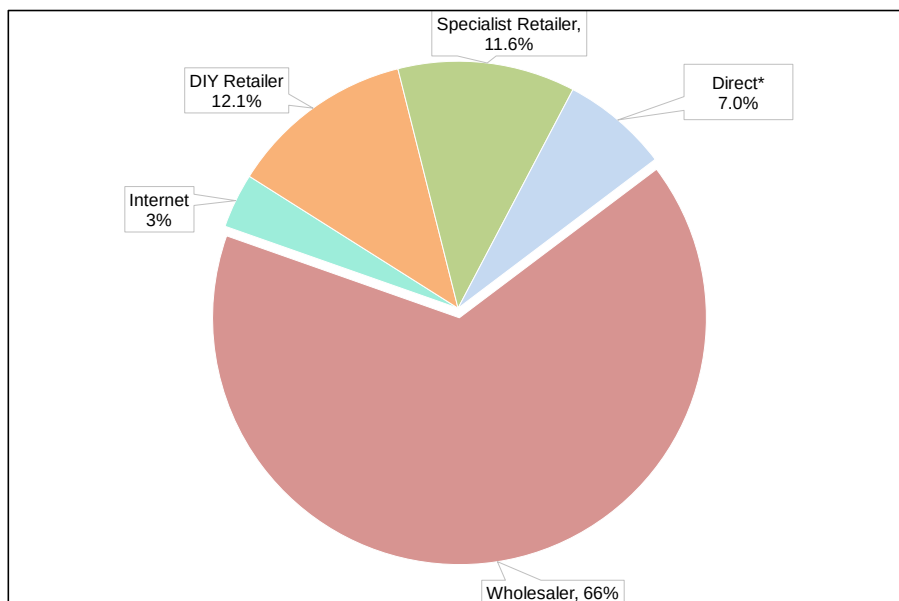
Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
65.7%	3.6%	12.1%	11.6%	7.0%	59.0%	9.2%	25.8%	6.0%
100%					68.2%		31.8%	

1.7.4 Route to Market - Consolidated View : Baths

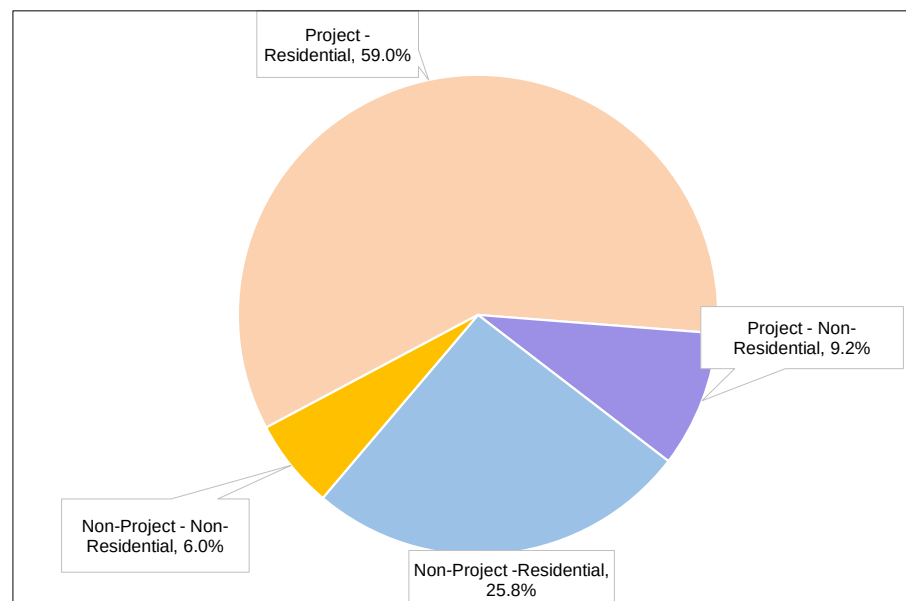
TR - ESTIMATED DISTRIBUTION FOR BATHS : 2025

Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
65.3%	4.0%	12.3%	11.5%	6.9%	50.0%	10.5%	32.4%	7.1%
100%					60.5%		39.5%	

TR - SALES OF BATHS BY DISTRIBUTION CHANNEL: 2020



TR - SALES OF BATHS BY POINT OF DECISION AND END-USE SECTOR: 2020



Source: BRG Based on Industry Interviews.

*Direct includes sales going directly from manufacturers to installers, contractors and private customers.

1.7.5 Route to Market: Shower Trays
TR - ESTIMATED DISTRIBUTION FOR SHOWER TRAYS: 2020

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	70.2%
	Wholesalers / Merchants
	70.2%

Retail Level	Manufacturer 4.1%	Wholesaler 1.0%	Manufacturer 8.7%	Wholesaler 4.1%	Manufacturer 7.7%	Wholesaler 4.1%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	5.1%		12.8%		11.8%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
9.3%	33.8%	1.5%	11.1%	10.6%
Installers				
66.3%				

End Use Level	Manufacturer 0.0%	Wholesaler 19.8%	Internet 3.6%	DIY 1.7%	Specialist 1.0%	Installer 56.5%	Manufacturer 0.0%	Wholesaler 7.4%	Internet 0.0%	DIY 0.0%	Specialist 0.2%	Installer 9.8%
	Residential						Non-Residential					
	82.6%						17.4%					
	New Build			Residential RMI			New Build			Non-Residential RMI		
	67.0%			15.7%			8.7%			8.7%		
	Project	Non-Project		Project	Non-Project		Project	Non-Project		Project	Non-Project	
	40.0%	27.0%		7.0%	8.7%		6.4%	2.3%		4.8%	3.9%	

1.7.6 Route to Market: Shower Trays
TR - ESTIMATED DISTRIBUTION FOR SHOWER TRAYS: 2025

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	70.0%
	Wholesalers / Merchants
	70.0%

Retail Level	Manufacturer 4.3%	Wholesaler 1.1%	Manufacturer 8.9%	Wholesaler 4.2%	Manufacturer 7.6%	Wholesaler 4.0%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	5.4%		13.1%		11.6%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
9.2%	33.7%	1.6%	11.2%	10.4%
Installers				
66.1%				

End Use Level	Manufacturer 0.0%	Wholesaler 20.0%	Internet 3.8%	DIY 1.9%	Specialist 1.0%	Installer 57.9%	Manufacturer 0.0%	Wholesaler 7.0%	Internet 0.0%	DIY 0.0%	Specialist 0.2%	Installer 8.2%
	Residential						Non-Residential					
	84.6%						15.4%					
	New Build			Residential RMI			New Build			Non-Residential RMI		
	53.3%			31.3%			7.7%			7.7%		
	Project	Non-Project		Project	Non-Project		Project	Non-Project		Project	Non-Project	
	37.0%	16.3%		14.0%	17.3%		5.2%	2.5%		4.3%	3.4%	

Source: BRG Based on Industry Interviews. - Totals may vary due to roundings

1.7.7 Route to Market - Consolidated View : Shower Trays

TR - ESTIMATED DISTRIBUTION FOR SHOWER TRAYS : 2020

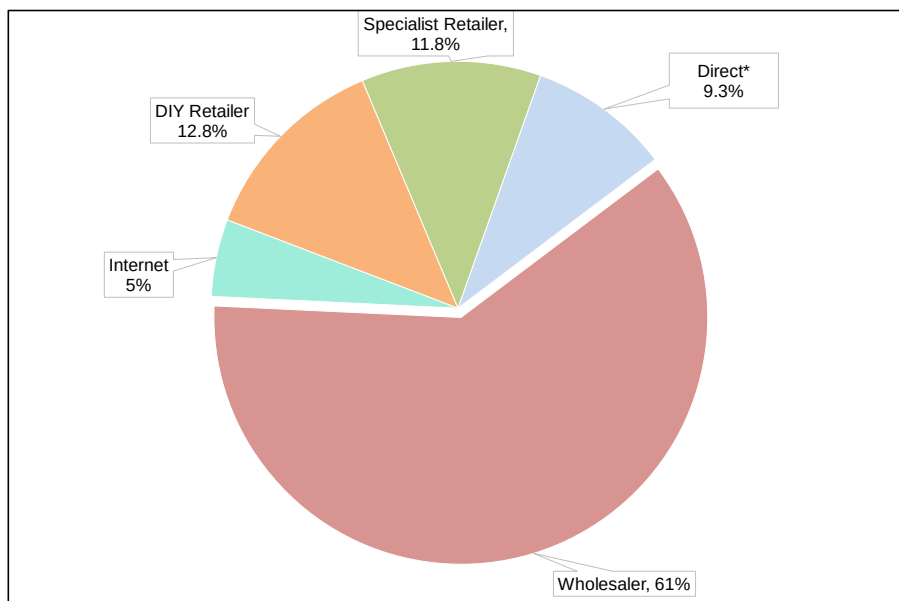
Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
61.0%	5.1%	12.8%	11.8%	9.3%	47.0%	11.2%	35.6%	6.2%
100%					58.2%		41.8%	

1.7.8 Route to Market - Consolidated View : Shower Trays

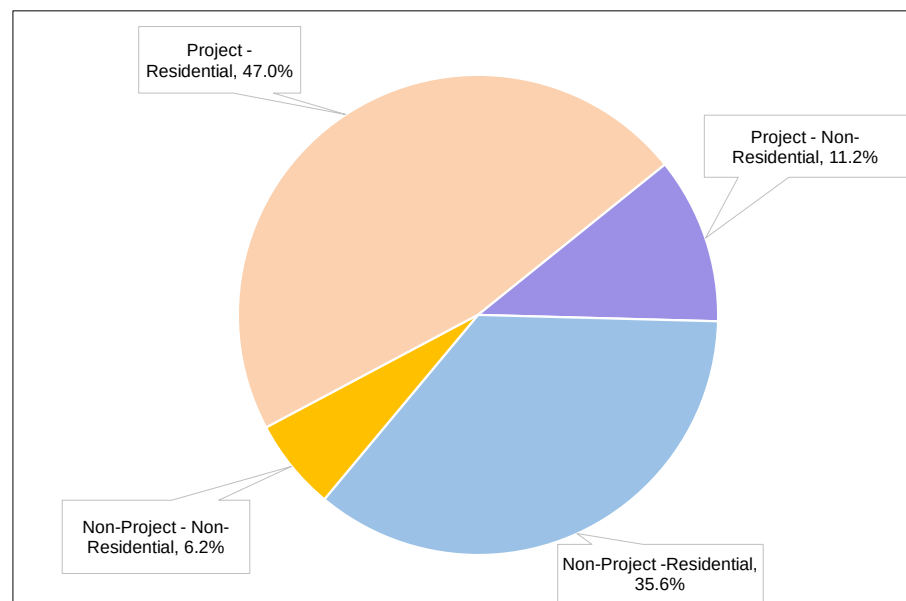
TR - ESTIMATED DISTRIBUTION FOR SHOWER TRAYS : 2025

Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
60.7%	5.4%	13.1%	11.6%	9.2%	51.0%	9.5%	33.6%	5.9%
100%					60.5%		39.5%	

TR - SALES OF SHOWER TRAYS BY DISTRIBUTION CHANNEL: 2020



TR - SALES OF SHOWER TRAYS BY POINT OF DECISION AND END-USE SECTOR: 2020



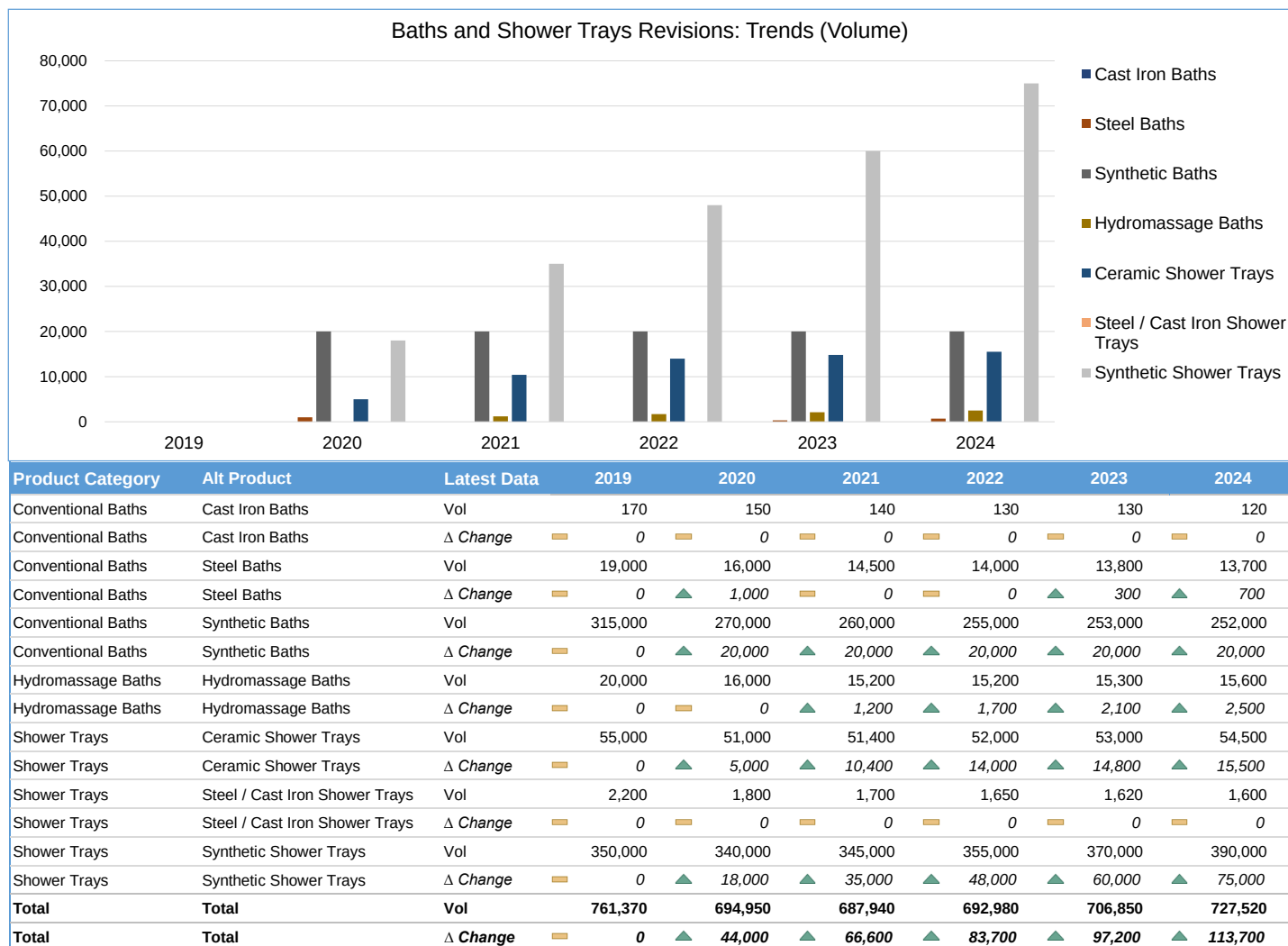
Source: BRG Based on Industry Interviews.

*Direct includes sales going directly from manufacturers to installers, contractors and private customers.

1.8 Revisions in the Turkish Baths and Shower Trays Market (April 2021)

TR - REVISIONS IN THE TURKISH BATHS AND SHOWER TRAYS MARKET

Data Comparison: October 2020



No revisions have been made to historical volumes and values for this market.

TURKEY

SANITARY WARE

THE EUROPEAN BATHROOM & KITCHEN PRODUCT MARKETS



Market Characteristics and Product Mix

Turkey has a substantial ceramic sanitary ware (CSW) industry, which began production in the 1960s with ECZACIBASI. Annual capacities have grown rapidly ever since, particularly in the early 1990s, but it is important to note that it is difficult to judge the total market capacity as there are a number of newly-established producers for which no reliable background data is available.

Considering the size and the production capacity of CSW in Turkey, its consumption, on a per capita basis, is lower than in any of the EU countries, even Switzerland and Germany (which have a low penetration of ceramics against other materials and a high proportion of rented accommodation). The reason for this is still the high number of so-called wet rooms, which are not trendy but basic and cheap, when compared to Europe. Consumption has grown significantly over the years and there is still considerable scope for further growth. There is particularly strong potential in the home improvement market, as income is increasing for many Turks and they are gradually moving away from traditional equipment, such as the *toilettes à la Turque* (Turkish toilets), towards more modern bathrooms.

Even though the popularity has fallen massively, floor-standing WCs still account for a large share of ceramic sanitary ware sales. However, in recent years the share of floor-standing WCs, which are installed with low-level wall-hung plastic cisterns, have continued to fall strongly, with wall-hung WCs gaining share. Looking at WC sales by type, the Turkish WC market is clearly dominated by sales of standard open-rim models. There are no known sales of closed-rim products on the Turkish market.

Among the strong domestic players, ECZACIBASI, CREAVIDIT, and SEREL sell a wide range of rimless WCs. More and more domestic players are entering the market, but none have recorded significant sales at this time. Awareness for rimless WCs is rather low in Turkey, with the products only being well known in the upper and high-end segments. However, awareness is increasing strongly thanks to growing product knowledge resulting from the efforts of manufacturers to promote rimless products. A key driver of the rimless market is the increasing penetration of wall-hung WCs, thanks to the growing number of middle-class families with higher purchasing power who are influenced by factors concerning hygiene, ease of cleaning and maintenance. The market of *toilettes à la Turque* / squat toilets is decreasing further. These models use either plastic cisterns or *passo rapido* type simple valves.

Washbasins hold a large share of total sales in Turkey, with a small proportion of hand washbasins. Especially regarding bowl washbasins, there is a clear trend towards coloured products, or ottoman-designed patterns. Ceramic sinks represent a niche market whereby sales have registered more or less stable volumes over the past five years. A very big number of sinks sold in Turkey are made of non-ceramic materials.

There is no tradition of using bidets in Turkey and the market, therefore, is negligible. Ceramic urinals are only sold to the non-housing segment. Overall, the market size for other CSW, including pedestals and shelves, decreased significantly in previous years. Pedestals, for example, are less needed as basins are more and more installed on countertops.

The share of concealed cisterns used to be small, as Turkish consumers found them expensive and the installation complex. However, with growing wall-hung toilet sales, Turks are increasingly more focused on design aspects. Hence, in 2019, the share of concealed plastic cisterns increased again even though the total cistern market decreased, and it currently represents the majority.

Recent Market Trends

Overall, the developments on the Turkish sanitary ware market in 2020 were largely a continuation of the sales trend seen in 2019. However, the trends of the sanitary ware products experienced largely unexpected developments in 2020, due to the economic challenges starting in 2019, which intensified in 2020 due to the worldwide COVID-19 pandemic. Moreover, the Turkish Lira lost its value again very strongly in 2020 against foreign currencies, along with an increasing inflation rate within a few weeks. Hence the purchasing power of Turks decreased significantly, which slowed down the demand for new housing in the country. As a result, many contractors could not finish their large residential projects and, therefore, all sales of sanitary ware products were challenged by the economic difficulties. As a result, the overall sanitary ware market was very negative for 2020 as a whole, with very strong decline rates.

The first lockdowns led to some restrictions regarding conventional distribution channels, such as specialists and installers finishing their jobs. However, after the COVID-19 lockdown many Turks escaped from big cities and settled down in rural areas at the seaside and the mountains, often in holiday homes designed to be used for only several weeks per year, with limited maintenance work over the years. However, in order to spend more time here, many Turks started, after the ease of restrictions, to renovate their holiday homes. As a result, the demand for sanitary ware overall was bigger, and the overall sanitary ware decline was not as big as previously forecast.

Regarding the product mix, the category 'other ceramic sanitary ware' has been decreasing for years – and this continued in 2020. The category consists of products such as pedestals and half pedestals. Pedestal sales by major manufacturers have almost entirely vanished and been replaced by bathroom furniture. Half pedestals can still be found on the market, e.g. in secondary bathrooms or low-cost hotels (both being places that usually do not feature bathroom furniture). Urinals are rarely used in residential houses, and their demand primarily depends on the development of non-housing construction. As was expected, the share of floor-standing WCs, which mostly use a low-level wall-hung plastic cistern, continued to fall while wall-hung WCs gained share.

The market volumes of washbasins and WCs declined further in 2020, after new housing completions decreased very strongly due to the currency crisis that started in June 2019 and continued in 2020. These markets are fully driven by construction and replacement activities, as they cannot be subsidised. Exposed plastic cisterns continued their declining trend further in 2020, as expected, due to the growing share of wall-hung toilets. Similarly to the washbasin and overall WC markets, concealed plastic cistern sales also declined again in 2020, due to the economic and construction recessions in the year.

Prices and Market Values

As the market is very competitive with a high number of domestic manufacturers and low labour costs, the Turkish ceramic sanitary ware average MSP is one of the lowest in Europe. Sanitary ware pricing can often depend on the locality of the manufacturer and how wide their distribution channels are structured. Local manufacturers are more sensitive to local price shifts and sensitivity, while larger manufacturers are able to maintain fixed nationwide price strategies. Sanitary ware usually provides dealers with a margin of approximately 40% of the list price; this figure can increase to up to 60% if dealers sign one-year contracts with manufacturers in advance.

Turkey is still an emerging country, and hence the share of the economy price point segment is very high. However, as the Turkish economy grew and purchasing power with it, the shift from the economy segments towards the middle and upper segments continued up to 2017. Then, due to the weakening Turkish currency, inflation increased and purchasing power decreased. As a result, after 2019 and again in 2020, the shift towards the luxury and upper segments stopped and the economy segment increased in share.

Regarding the MSP, all products have recorded increases in Turkish lira, especially after increasing inflation in the country. Imports of sanitary ware play only a limited role in Turkey, but energy and raw materials, which are very often imported, suddenly became very expensive for manufacturers as the Turkish Lira lost significant value against the US Dollar. Hence producers had to increase their MSPs, but at a lower rate than the inflation rate as the market was under extreme pressure, due to declining sales and the competition.

New Housing

There are often typical product installations for new housing, depending on contractors and projects. Big housing projects, such as Urban Transformation or TOKI, managed by the government, install almost exclusively low-end products with cheap ceramic sanitary ware, including floor-standing toilets and exposed plastic cisterns. Other small and lesser-known private contractors often do the same. Big and well-known private contractors, such as AGAOGLU, typically install upper-class CSW, including wall-hung toilets and concealed plastic cisterns.

New housing accounts for a high share of total CSW sales in Turkey, in contrast to other European markets. However, the share of new housing, which is around half of the overall market, is expected to decline slightly in the short term, as more and more bathrooms are expected to be renovated in the coming years.

The new build market suffered in 2020. Sanitary ware sales into this sector decreased significantly, driven by the weakening of the Turkish currency, high inflation, and the decreasing purchasing power of Turks. To further compound sanitary ware market difficulties, apartments have gradually smaller bathrooms on average and, therefore, are equipped with a smaller number of sanitary ware products. Other headwinds continued in 2020, including the development of the bathroom furniture market and the trend away from pedestal towards countertop washbasins (and to a much lesser extent washbasin only). Old-fashioned products such as floor-standing WCs and exposed cisterns still see a notable portion of the new housing segment, as they are comparatively cheap and installation can be achieved quickly, enabling contractors to save money on labour costs.

As far as sales of plastic cisterns, in 2020 demand for concealed instead of exposed models continued to be the key characteristic of a market increasingly focused on quality and design. Such a trend was particularly visible in new housing, as consumers prefer the minimalist designs that concealed plastic cisterns offer. Exposed plastic cistern sales decreased as expected in 2020, but concealed plastic cistern market volumes declined in line with the negative trend of wall-hung WCs. As mentioned above, the economic challenges for Turkey in 2019, and then 2020, left many new dwellings unsold and contractors could not finish their new projects. New housing should see a clear move away from exposed towards plastic cisterns in the foreseeable future, with the volume sales of exposed plastic cisterns clearly expected to follow a downward trend from 2021 to 2025.

Housing RMI

The refurbishment and replacement segment of the CSW market accounts for a very large share of sales volumes in the country. As the quality of the products is not as good as in Western Europe, CSW has shorter lifespans in Turkey. Overall sales in this segment declined as a whole in 2020, but it was less dramatic than previously forecast.

With ongoing lockdown periods over 2020, Turks started to spend more time at home and renovate their homes, which did not happen as much before, due to a lack of time. At the same time, many Turks escaped from big cities towards their homes in rural areas, where they invested in their homes – from essential and/or old facilities to new and modern bathrooms. As a result, housing RMI became a very important driver for the overall bathroom products market.

Regarding CSW product mix, floor-standing toilets are increasingly replaced by wall-hung models. Rimless WCs are used only in new housing, and the replacement market for these WC models is effectively non-existent yet, as rimless

toilets entered the Turkish market later than in Europe. The sales number of pedestals and half pedestals is strongly decreasing, due to the fact that replaced washbasins are gradually installed either without pedestals or in countertops.

Plastic cistern sales show slight differences. Concealed plastic cisterns have a slightly lower share compared to exposed, as the greater part of installed concealed plastic cisterns is not currently a mature market like the exposed models, and the products do not need replacement yet.

Non-Housing

International investments declined strongly, due to the economic recession in 2020. As a result of decreasing investments in the commercial and public sectors, total sales of CSW and plastic cisterns to the non-housing sector were expected to decline strongly further in 2020.

However, many restaurants, cafes and hotels took the advantage of closing their businesses during the lockdowns and renovated them, and purchased new bathroom products. As a result, the overall sanitary ware decline was less than previously forecast.

Regarding product mix, no major changes have been registered for 2020. Wall-hung WCs, washbasins and urinals are the most common sanitary ware products to be installed in non-residential buildings. From 2021 onwards, sales into the non-housing segment are predicted to resume growth, following significant GDP growth.

Supply Structure

In Turkey, it is thought that over 40 producers operate in the sector. Of these, nine produce on an industrial scale and/or are the sister companies of large ceramic tile manufacturers, namely VITRA, SEREL, and KALE.

A very big number of Turkish manufacturers, however, have increased their production capacities over the last decade. The growth in production capacities is estimated to exceed the growth in consumption and this might cause an increase in competition, mainly affecting companies operating in the low and middle price segments. Foreign brands still play a very limited role within the sanitary ware market in Turkey.

Most Turkish market players have gradually supplied the market with complete solutions, which means more and more manufacturers include in their portfolio CSW, cisterns, flushing valves, tiles, and taps and mixers. Hence, a significant number of companies have grown horizontally.

KALE originally developed from Turkey's leading tiles supplier holding company, the KALE GROUP. The company has followed the trend of growing horizontally, supplying different complementary products. KALE was established very well in the

CSW market, due to aggressive marketing and offering low-cost products. However, in 2019 and then 2020, the company experienced some additional challenges. Generally, the import rate of sanitary ware is very limited in Turkey. With the exception of KALE, which imports its products from China, all manufacturers produce in Turkey. Due to the financial crisis in the country, the government greatly increased the import tax for building products, especially CSW. On top of that, the weakening of the Turkish currency meant that imports became very expensive. These two factors pushed the MSPs of KALE products higher. As a result, the company's competitiveness decreased very strongly as the overall market became even more price-driven after the decline of the purchasing power of Turks. Hence, KALE experienced a very significant sales decline in 2020.

CREAVIT, TURKUAZ, and SANOVIT have all experienced very strong growth for the last few years. CREAVIT's main focus used to be CSW, but the company also increasingly supplies cisterns, flush valves, and taps and mixers. However, even though growing very strongly, its share in these markets is still not as large as in the CSW market. CREAVIT increased its production capacity after opening a new production line in its factory in 2018.

TURKUAZ, which is traditionally a washbasin manufacturer, also recently launched CSW WCs and complementary products. TURKUAZ also increased its export rate, benefitting from the increasing demand for cheap Turkish CSW products resulting from a weakening of the Turkish currency.

SANOVIT, another traditional CSW washbasin manufacturer, was expected to start selling WC and other CSW products in 2020, but could not realise that due to the very challenging market environment.

In comparison to other competitors, BOCCHI is not a very big CSW manufacturer, but operates very successfully and efficiently in niche markets. The company has a very high market share in the coloured CSW products market and the disabled bathroom market. It is also strong in online sales, and is constantly increasing its production capacity.

JAPAR, VISAM, TEMA, and SIAMP appear to be the leading suppliers of exposed plastic cisterns. It is difficult to quantify the exact size of this market, as there are several local suppliers scattered all around the country.

Forecasts

The ongoing COVID-19 pandemic, along with the growing replacements, continued in the first quarter of 2021 and affected the total sanitary ware sales positively. As a result, the decline rate of the overall sanitary ware sales slowed down between January 2021 and April 2021. Moreover, washbasin, WC and concealed plastic cistern demand started to grow again, mainly driven by the ongoing replacement activity. The renovation segment is highly benefitting from the ongoing lockdowns, along with people spending more time at home as well as the big movement away from cities towards rural areas. Exposed plastic cistern and other CSW sales,

which have experienced declining popularity in the last few years, have continued to shrink and not benefit from the very strong replacement activity.

The mid-term outlook for the Turkish sanitary ware market is projected to benefit from the advanced situation, and grow further by 2025. However, different CSW products are predicted to follow their old trends and develop again in different directions. Floor-standing toilets should lose share to increasing sales of wall-hung toilets. BRG expects that standard open-rim models will still have accounted for the majority of WC sales in 2020, even though the share of rimless WCs is expected to have increased very strongly. Without-ledge models should further dominate all rimless WCs by 2020, as the market is dominated by domestic players that are focused mainly on this type of rimless WC.

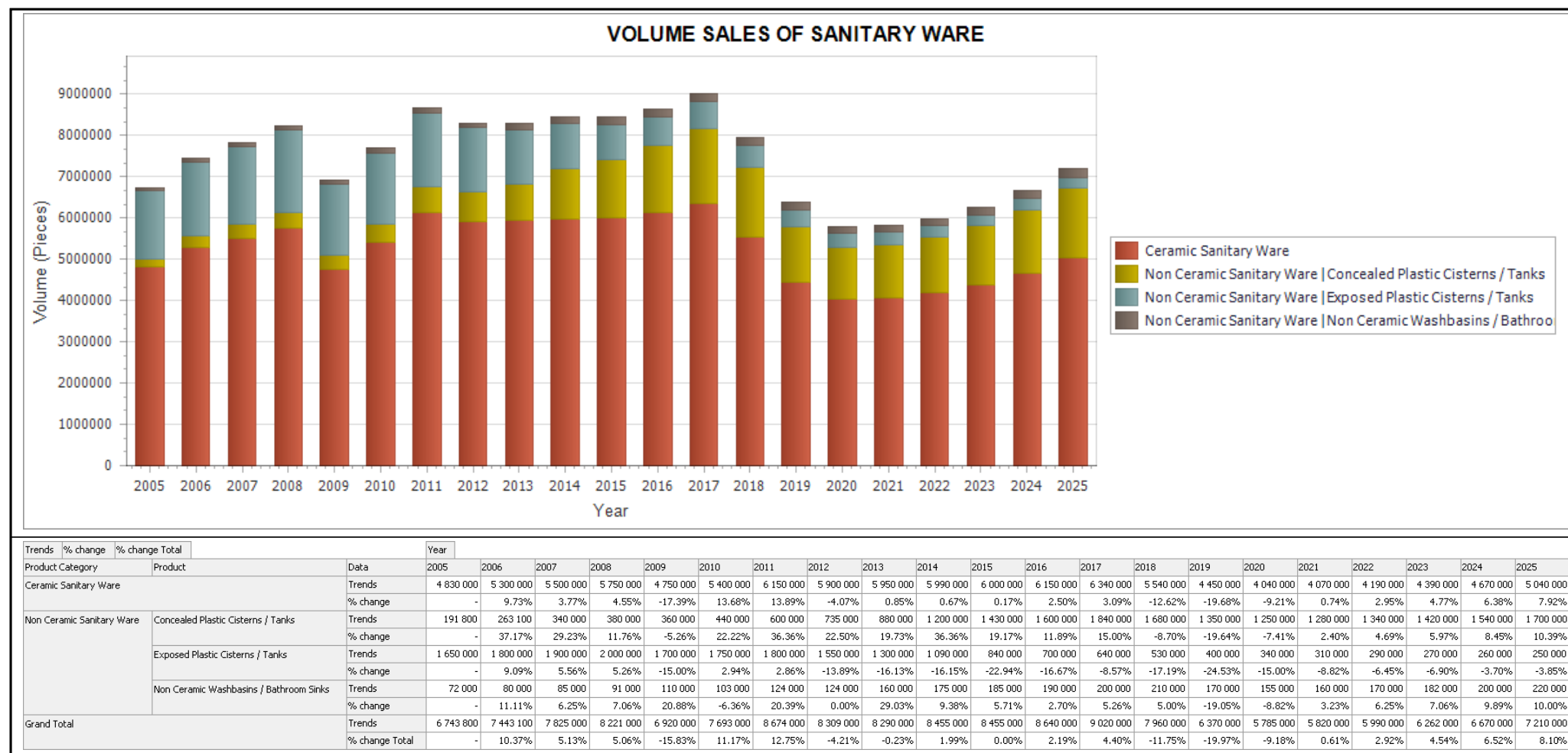
'Other CSW' product sales, which include amongst others pedestals and half pedestals, are projected to decline further very strongly. This is due to the fact that washbasins are increasingly installed without any pedestals, directly to the wall or on countertops.

In the cistern segment, the shift towards concealed models is expected to grow further while sales of exposed cisterns are predicted to decline. Net sales growth should remain positive, as the economy recovers along with new housing activity.

Political instability could negatively affect the economy and construction output, which is likely to lead to a decrease in the sanitary ware market. The large new housing share leaves the sector very sensitive to construction activity. A stable political environment leads to consumer confidence and higher spending (which translates into replacement, renovation and construction, leading to more installations). However, during periods of political/economic instability, people tend to spend less, which causes a decrease in sanitary ware sales. The purchasing power of the general population remains lower than in most Western European countries. As a result, regarding replacement activity, not everyone can afford an upgrade of sanitary ware installations, particularly those in rural areas. Therefore, strong economic growth in the coming years and the increasing purchasing power of households should positively affect sales.

2.2.1 Historical Trends and Forecasts : Sanitary Ware

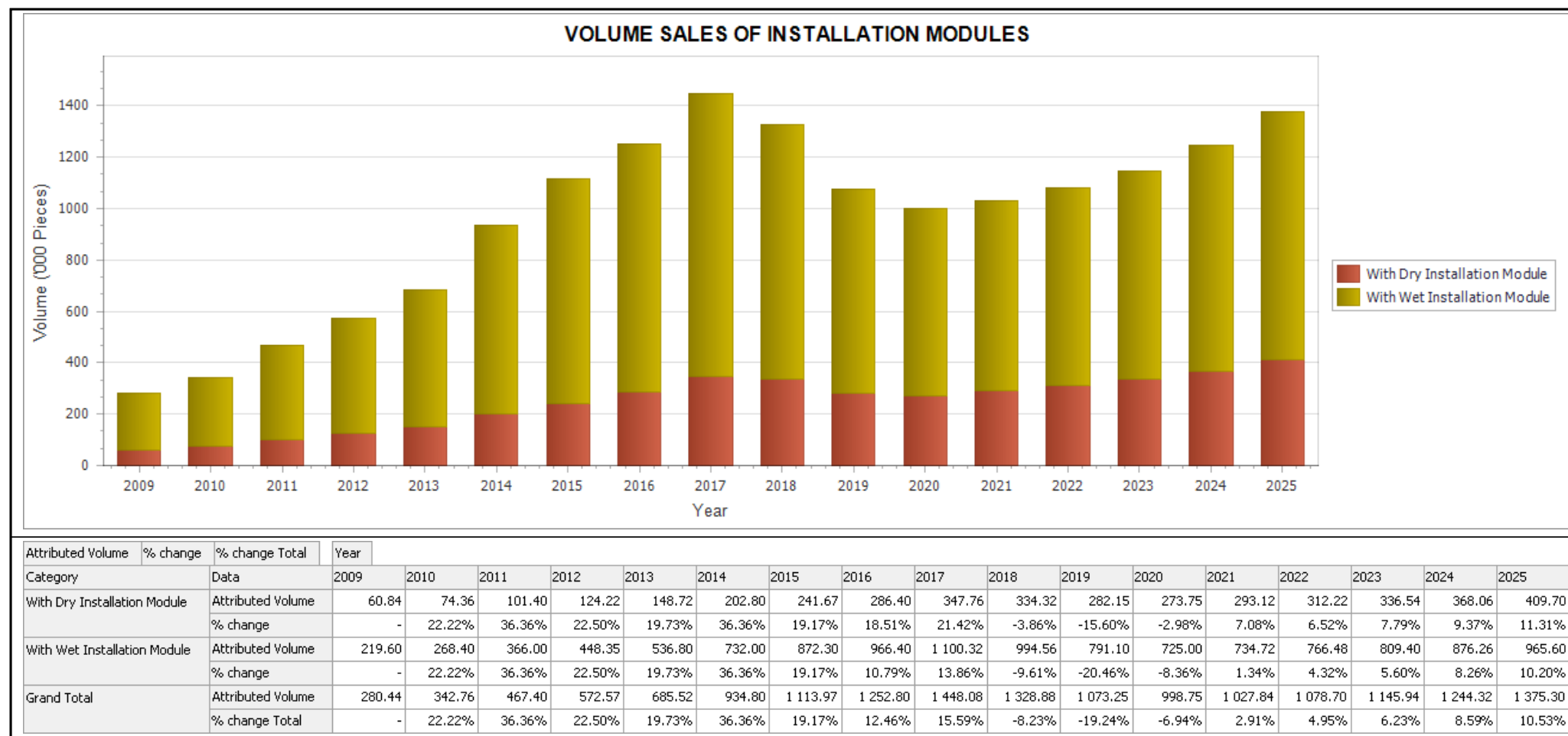
TR - SUMMARY OF TRENDS IN THE SANITARY WARE MARKET : HISTORICAL AND FORECASTS



Source: BRG based on industry interviews. NB: Ceramic Sanitary Ware Includes Ceramic Shower Trays.

2.2.2 Historical Trend and Forecasts: Installation Mod.

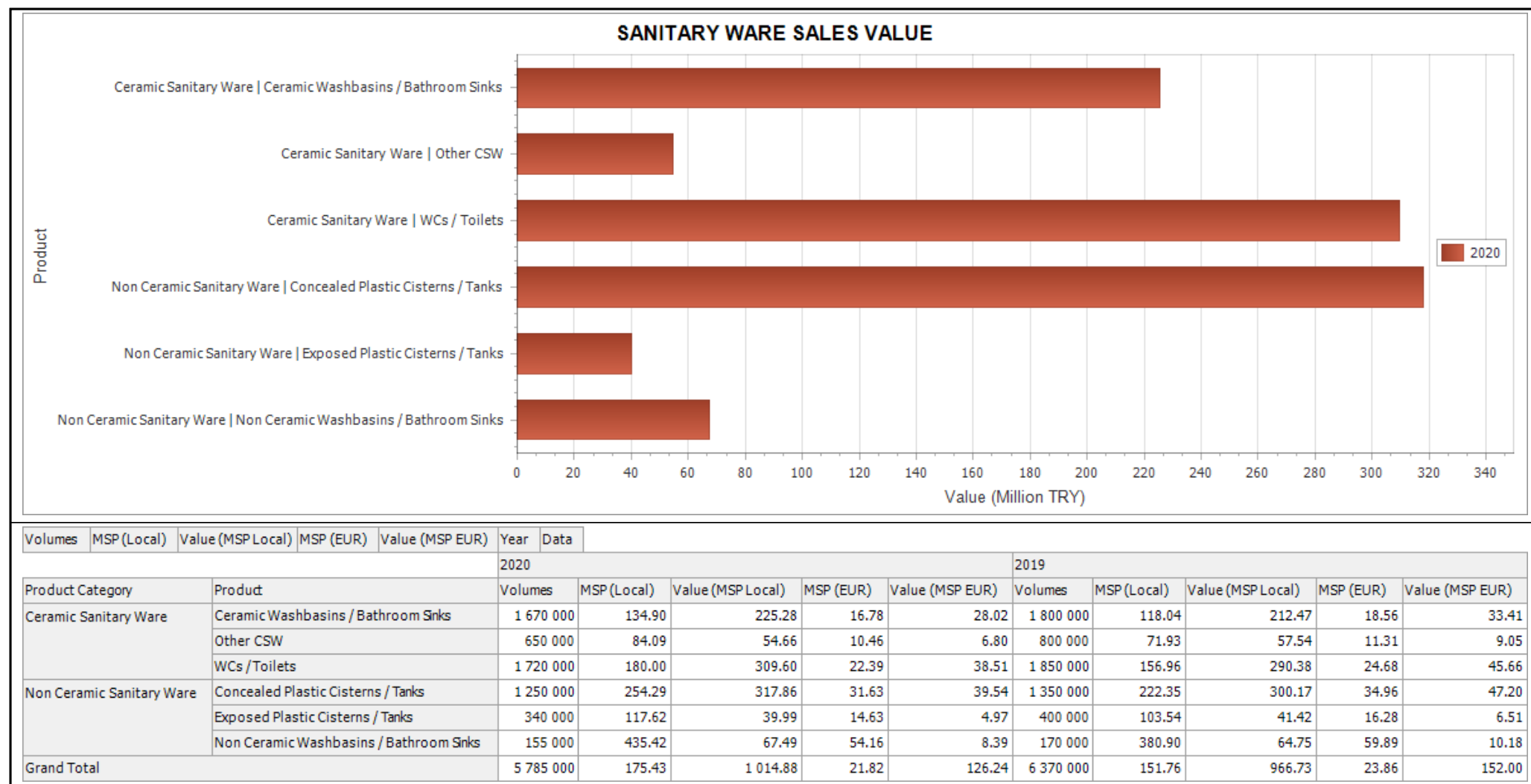
TR - SUMMARY OF TRENDS IN THE INSTALLATION MODULE MARKET : HISTORICAL AND FORECASTS ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

2.3.1 Prices and Market Values : Sanitary Ware

TR - ESTIMATED VALUES OF THE SANITARY WARE MARKET



Source: BRG based on industry interviews. Local Prices Shown in TRY. Values Shown in Millions (at MSP = Manufacturers' Selling Price)

2.3.2 Prices and Market Values by Type : CSW

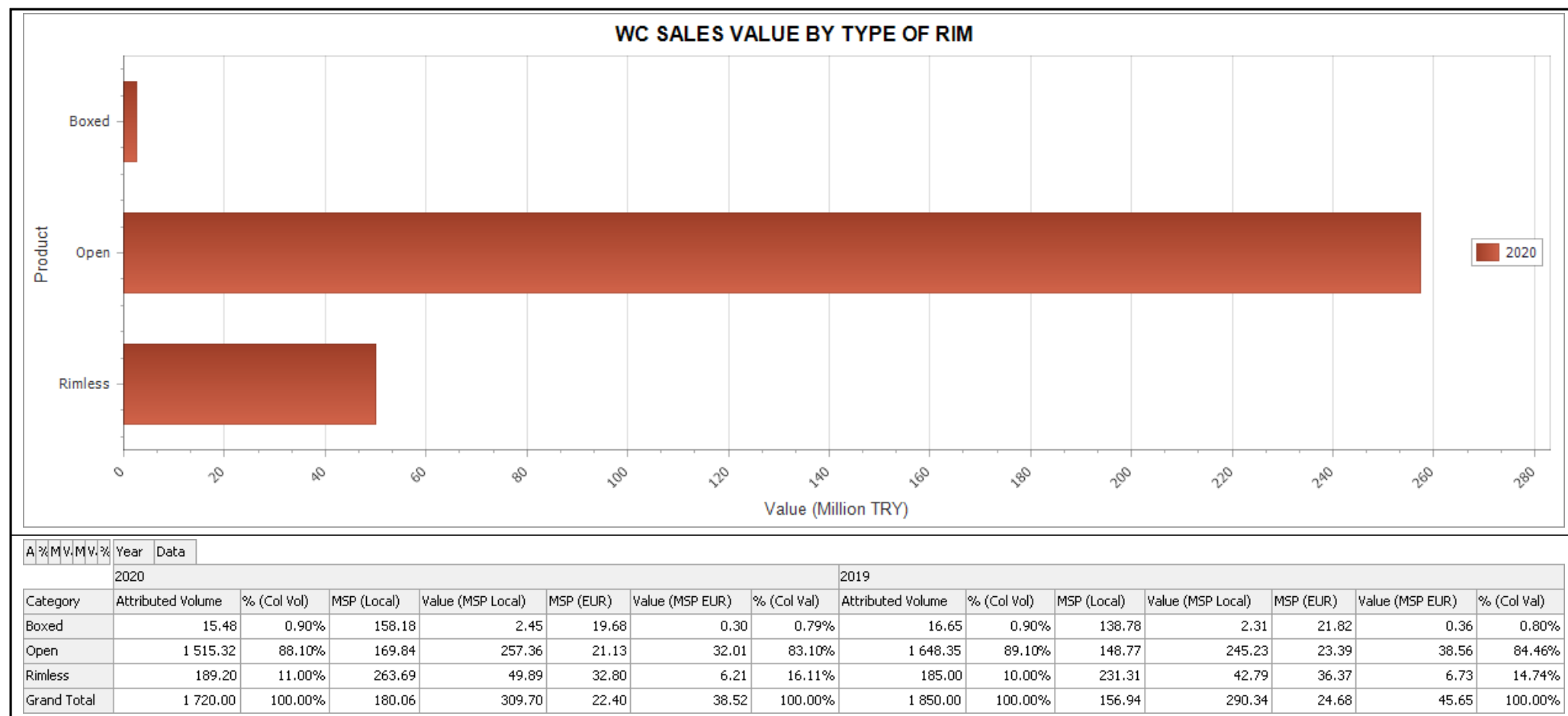
TR - ESTIMATED VALUES OF THE CERAMIC SANITARY WARE MARKET BY TYPE OF PRODUCT

Attribute	% (Col Vol)	MSP (Local)	Value (MSP Local)	MSP (EUR)	Value (MSP EUR)	% (Col Val)	Year	Data																
							2020									2019								
Alt Product		Category	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	MSP (EUR)	Value (MSP EUR)	% (Col Val)	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	MSP (EUR)	Value (MSP EUR)	% (Col Val)								
Ceramic Washbasins / Bathroom Sinks		Countertop Washbasins	567.80	34.00%	143.76	81.63	17.88	10.15	36.24%	576.00	32.00%	126.12	72.65	19.83	11.42	34.20%								
		Hand Washbasins	133.60	8.00%	109.98	14.69	13.68	1.83	6.52%	153.00	8.50%	96.47	14.76	15.17	2.32	6.95%								
		Inset Washbasins	16.70	1.00%	141.39	2.36	17.59	0.29	1.05%	18.00	1.00%	124.04	2.23	19.50	0.35	1.05%								
		Pedestal Washbasins	400.80	24.00%	132.94	53.28	16.54	6.63	23.66%	450.00	25.00%	116.61	52.47	18.33	8.25	24.70%								
		Undermount Washbasins	1.67	0.10%	136.57	0.23	16.99	0.03	0.10%	1.80	0.10%	119.79	0.22	18.83	0.03	0.10%								
		Wall Hung Washbasins	549.43	32.90%	132.94	73.04	16.54	9.09	32.43%	601.20	33.40%	116.61	70.10	18.33	11.02	33.00%								
Other CSW		Bidets	45.50	7.00%	194.56	8.85	24.20	1.10	16.20%	49.60	6.20%	170.67	8.47	26.83	1.33	14.71%								
		Close-Coupled Cisterns	257.10	39.55%	67.63	17.39	8.41	2.16	31.82%	332.20	41.52%	59.34	19.71	9.33	3.10	34.25%								
		Others	24.05	3.70%	47.13	1.13	5.86	0.14	2.07%	30.40	3.80%	41.34	1.26	6.50	0.20	2.18%								
		Pedestals & Half Pedestals	195.00	30.00%	61.63	12.02	7.67	1.49	21.99%	248.00	31.00%	54.07	13.41	8.50	2.11	23.30%								
		Shower Trays	51.00	7.85%	209.47	10.68	26.05	1.33	19.55%	55.00	6.88%	187.33	10.30	29.45	1.62	17.90%								
		Urinals	77.35	11.90%	59.18	4.58	7.36	0.57	8.38%	84.80	10.60%	51.93	4.40	8.16	0.69	7.65%								
WCs /Toilets		Close-Coupled WC Bowls	180.60	10.50%	173.99	31.42	21.64	3.91	10.15%	203.50	11.00%	152.63	31.06	24.00	4.88	10.70%								
		Floor Standing WC Bowls	249.40	14.50%	161.93	40.39	20.14	5.02	13.04%	305.25	16.50%	142.04	43.36	22.33	6.82	14.93%								
		Squat Pans	172.00	10.00%	118.42	20.37	14.73	2.53	6.58%	194.25	10.50%	103.88	20.18	16.33	3.17	6.95%								
		Wall Hung WC Bowls	1 118.00	65.00%	194.52	217.48	24.20	27.05	70.23%	1 147.00	62.00%	170.65	195.74	26.83	30.78	67.42%								
Grand Total			4 040.00	100.00%	145.92	589.54	18.15	73.33	100.00%	4 450.00	100.00%	125.91	560.32	19.80	88.10	100.00%								

Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Local Prices Shown in TRY. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price).

2.3.3 Prices and Market Values by Type of Rim : WCs

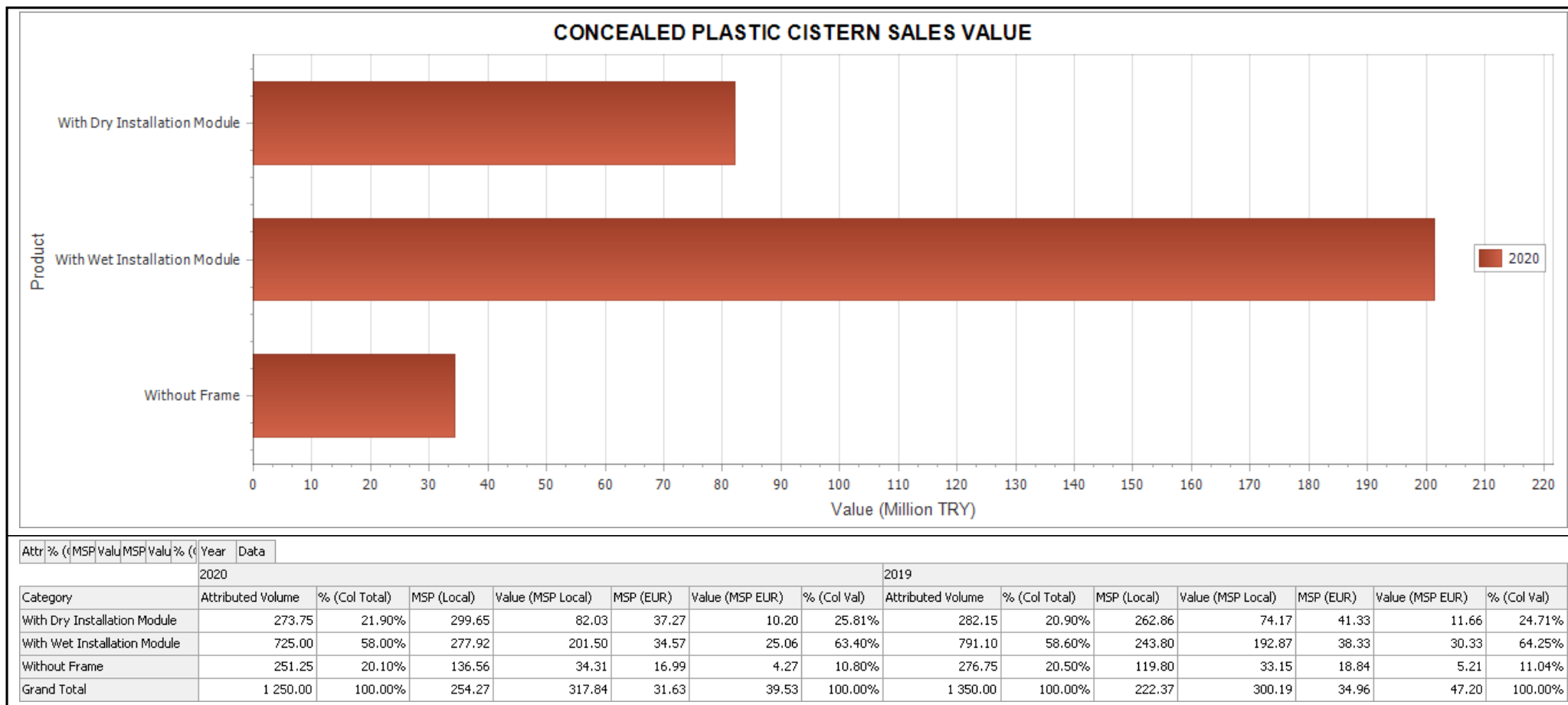
TR - ESTIMATED VALUES OF THE WC MARKET BY TYPE OF RIM



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Local Prices Shown in TRY. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price).

2.3.4 Prices and Market Values by Type : Conc. Cisterns

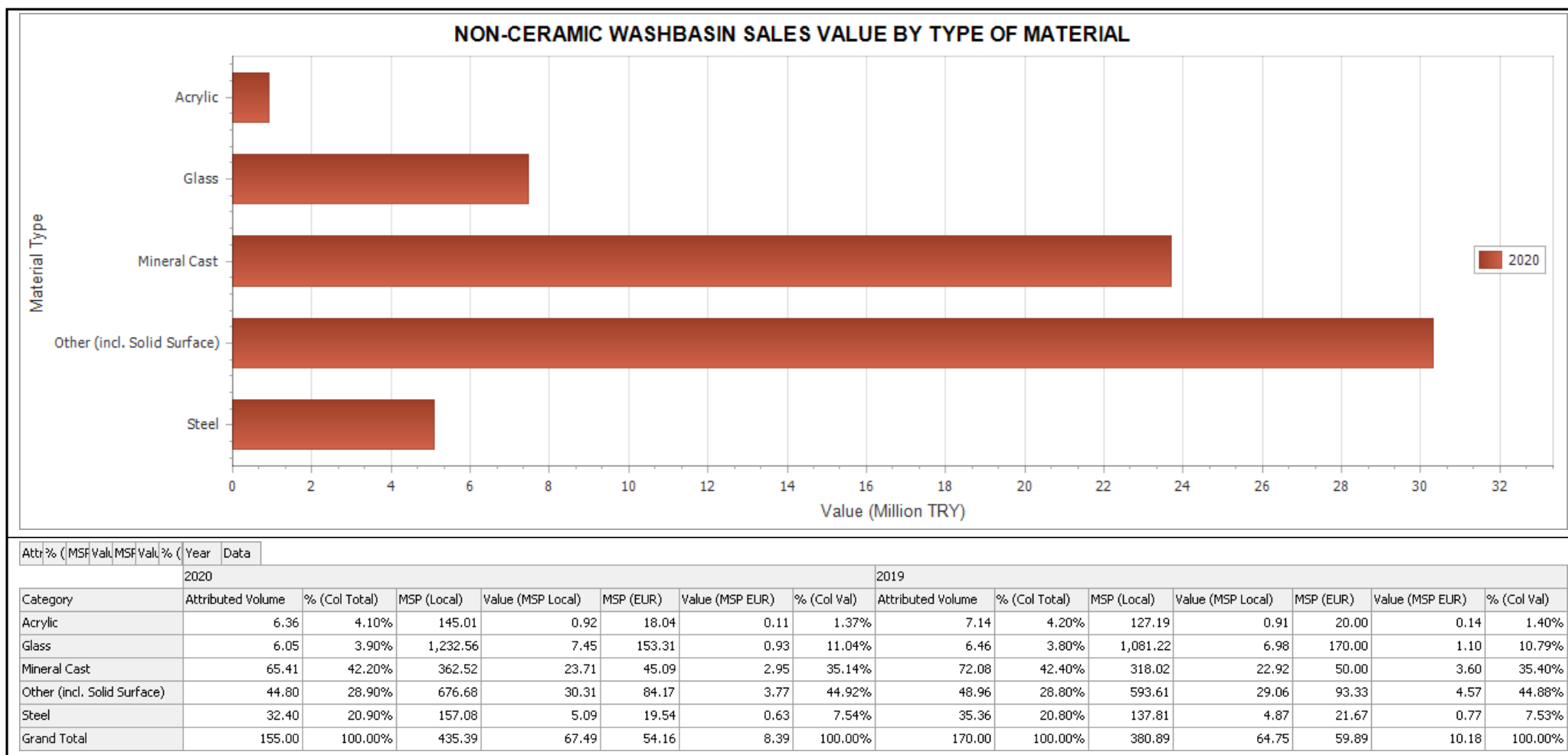
TR - ESTIMATED VALUE OF THE CONCEALED PLASTIC CISTERNS MARKET BY TYPE OF INSTALLATION



Source: BRG based on industry interviews. Local Prices Shown in TRY. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price).

2.3.5 Prices and Market Values : Non-Ceramic Washbasins

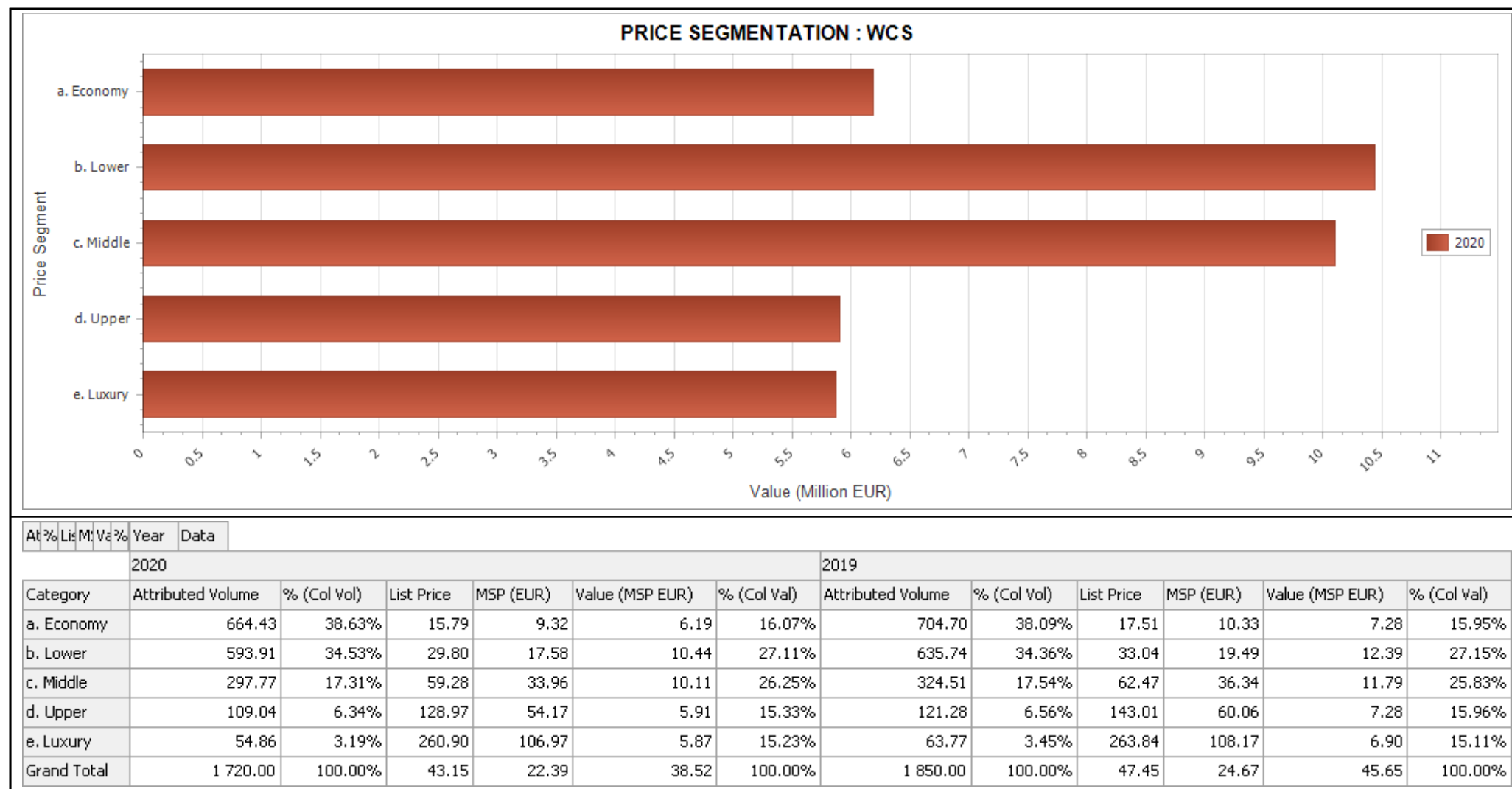
TR - ESTIMATED VALUE OF THE NON-CERAMIC WASHBASIN MARKET BY MATERIAL TYPE



Source: BRG based on industry interviews. Local Prices Shown in TRY. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price).

2.3.6 Price Segmentation : WCs

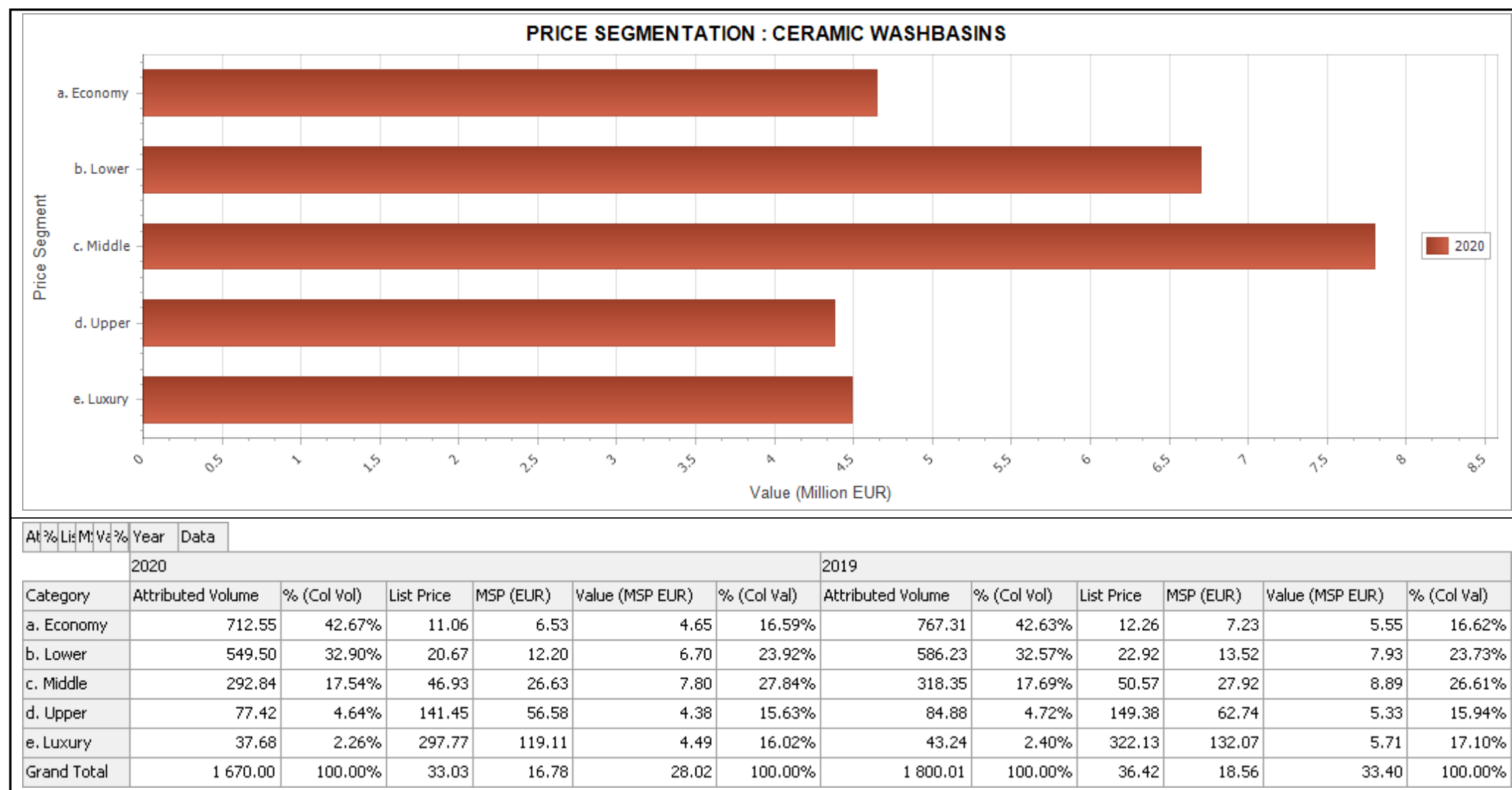
TR - ESTIMATED PRICE SEGMENTATION FOR WC



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Includes all WC Types. Volumes Shown in Thousands and Values Shown in Millions. Economy <€15, Lower €15-€25, Middle €25-€50, Upper €50-€100, Luxury >€100

2.3.7 Price Segmentation : Ceramic Washbasins

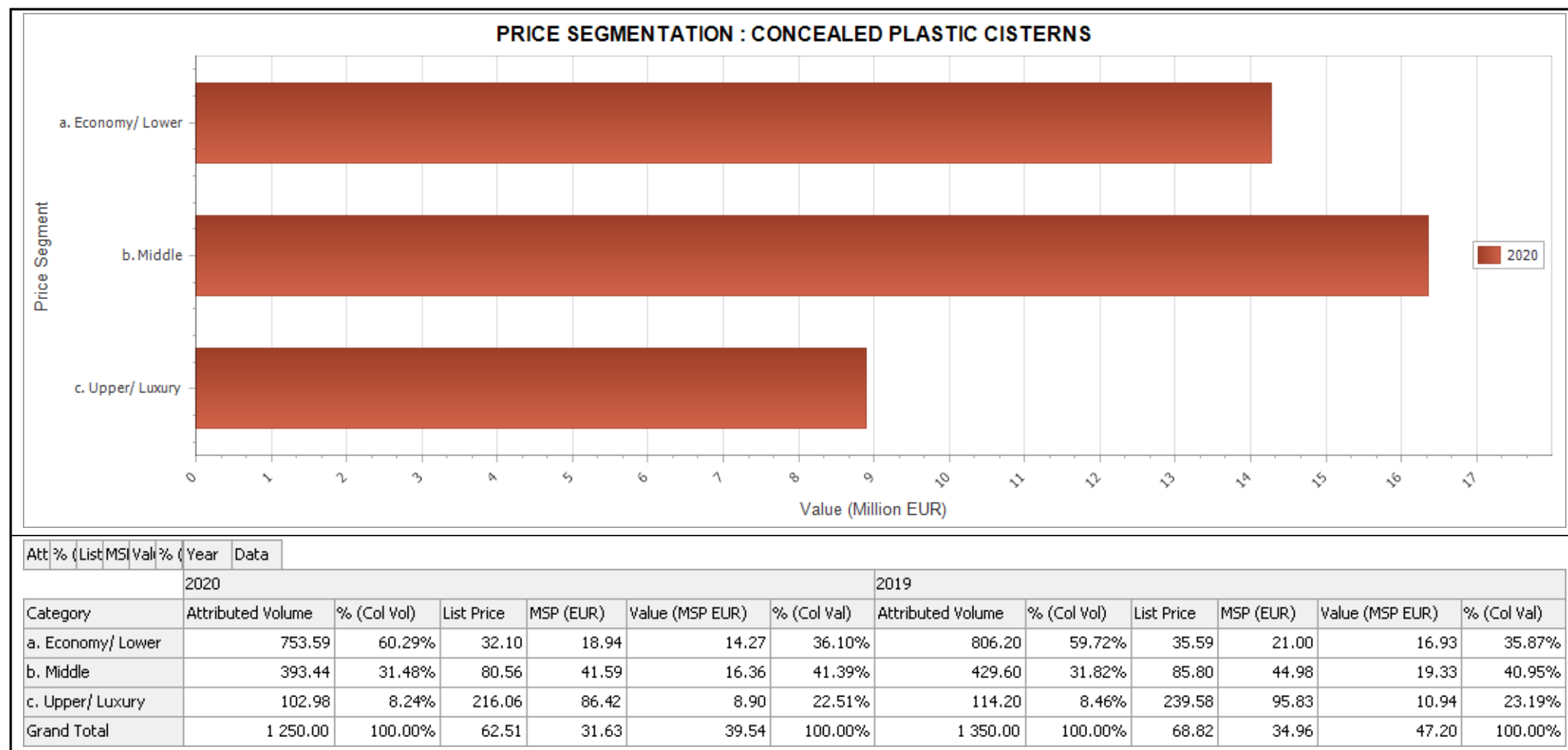
TR - ESTIMATED PRICE SEGMENTATION FOR CERAMIC WASHBASINS



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Includes all Washbasin Types. Volumes Shown in Thousands and Values Shown in Millions. Economy <€10, Lower €10-€20, Middle €20-€40, Upper €40-€80, Luxury >€80

2.3.8 Price Segmentation : Concealed Plastic Cisterns

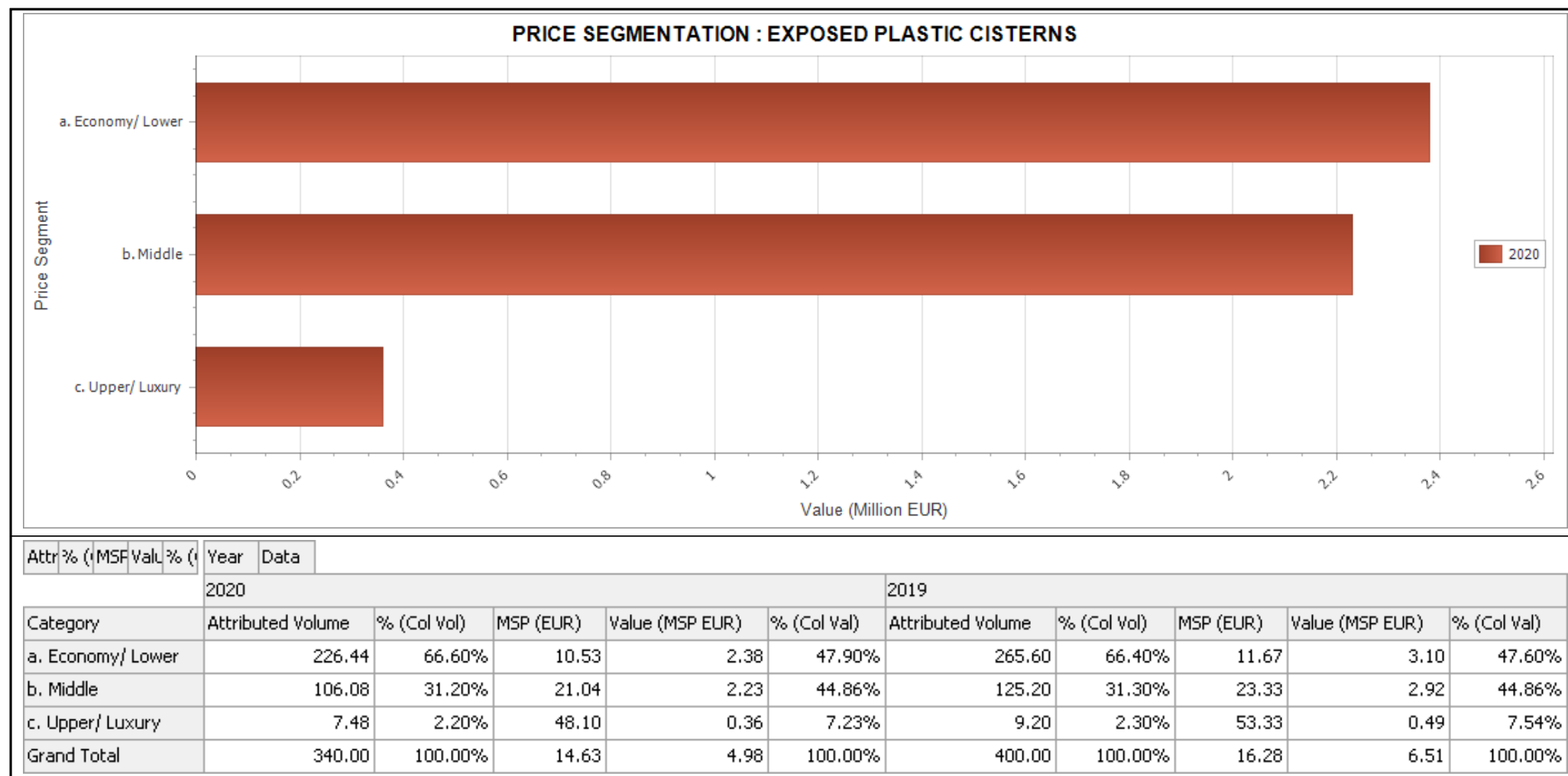
TR - ESTIMATED PRICE SEGMENTATION FOR CONCEALED PLASTIC CISTERNS



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Includes all WC Types. Volumes Shown in Thousands and Values Shown in Millions. Economy/ Lower <€30, Middle €30-€60, Upper/ Luxury >€60

2.3.9 Price Segmentation : Exposed Plastic Cisterns

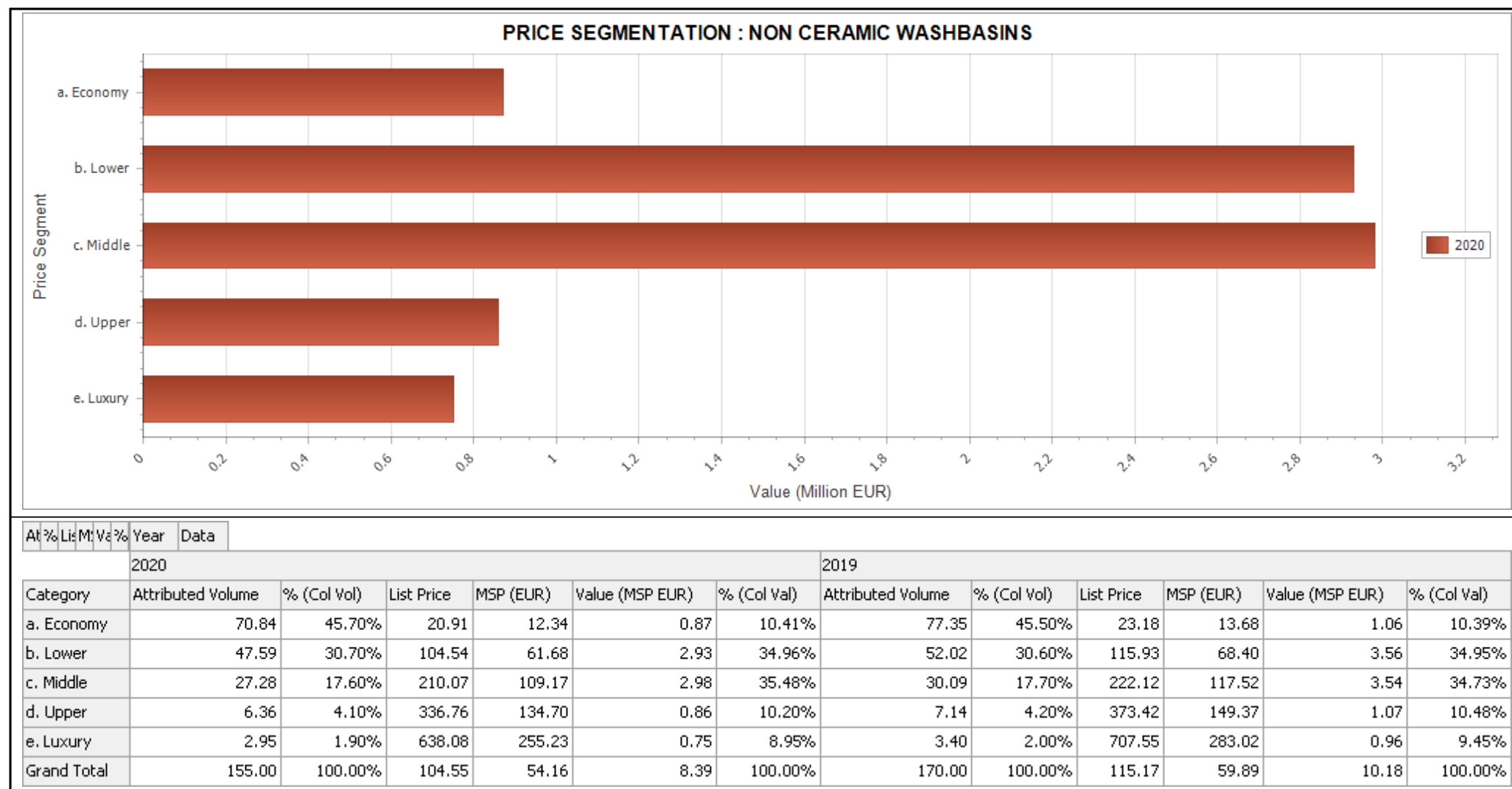
TR - ESTIMATED PRICE SEGMENTATION FOR EXPOSED PLASTIC CISTERNS



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Includes all WC Types. Volumes Shown in Thousands and Values Shown in Millions. Economy/ Lower <€15, Middle €15-€30, Upper/ Luxury >€30

2.3.10 Price Segmentation : Non Ceramic Washbasins

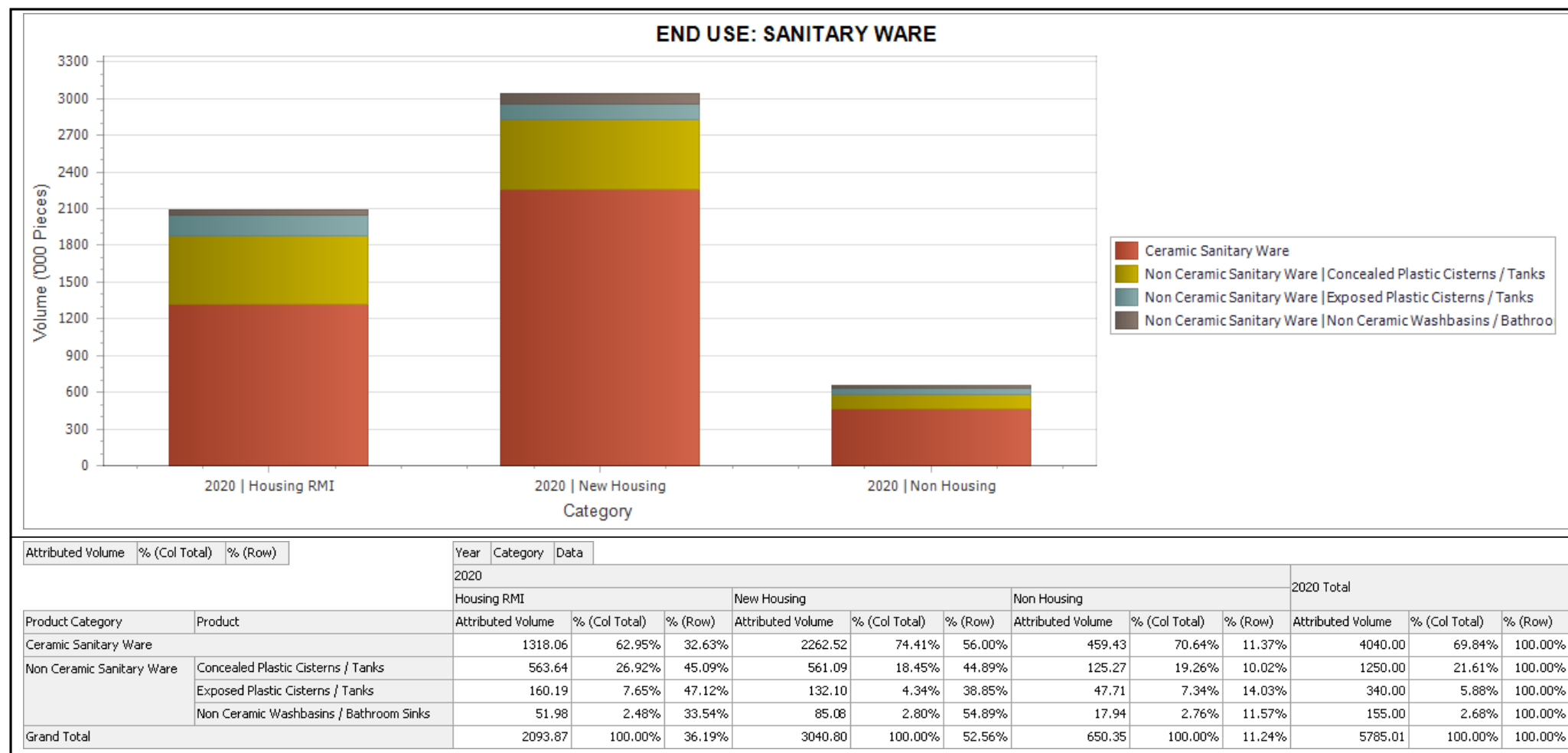
TR - ESTIMATED PRICE SEGMENTATION FOR NON CERAMIC WASHBASINS



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Includes all Washbasin Types. Volumes Shown in Thousands and Values Shown in Millions. Economy <€40, Lower €40-€80, Middle €80-€120, Upper €120-€180, Luxury >€180

2.4.1 End Use : Base Year

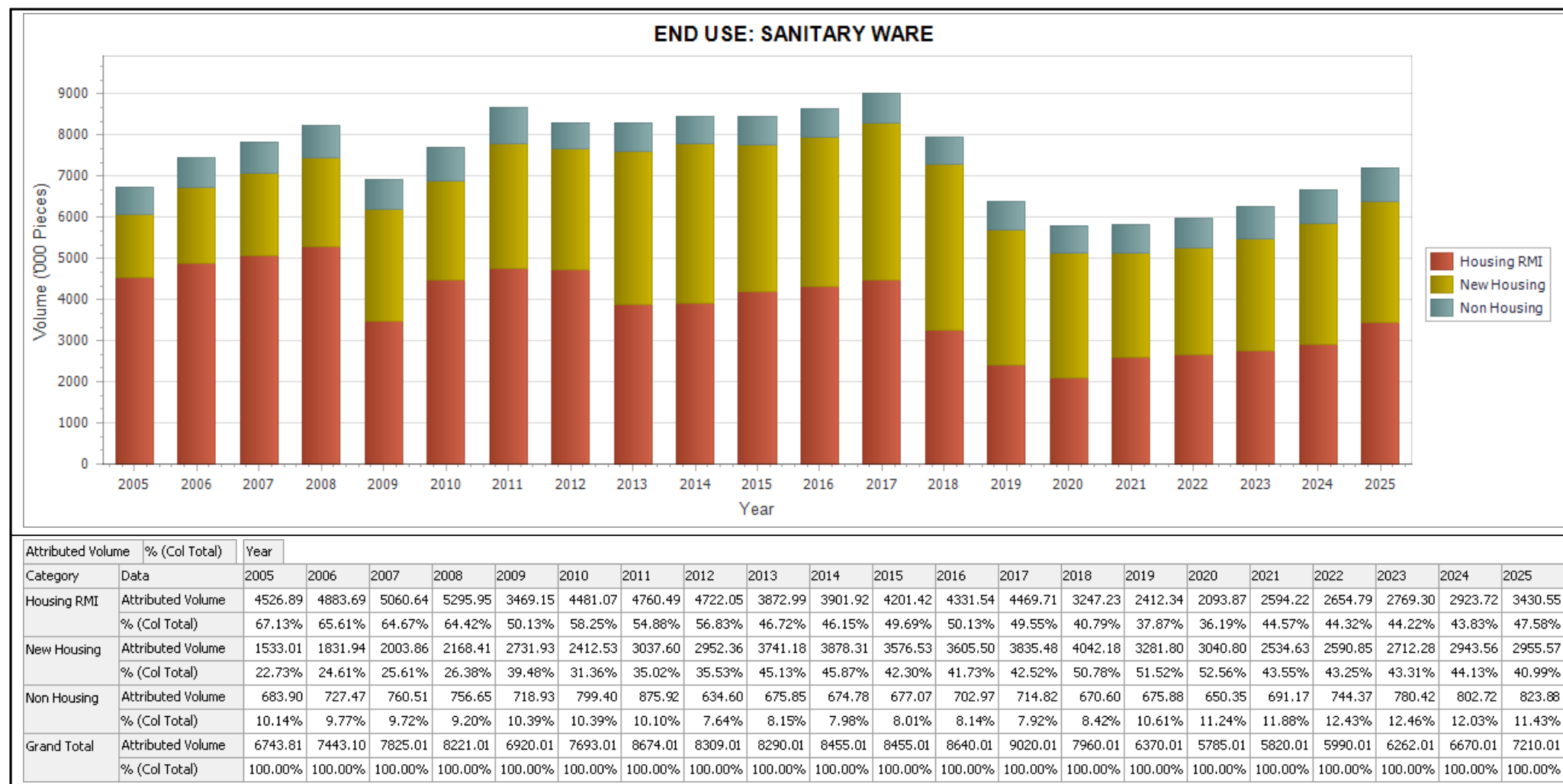
TR - SALES OF SANITARY WARE BY TYPE AND END USE SEGMENT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

2.4.2 End Use : Historical and Forecasts

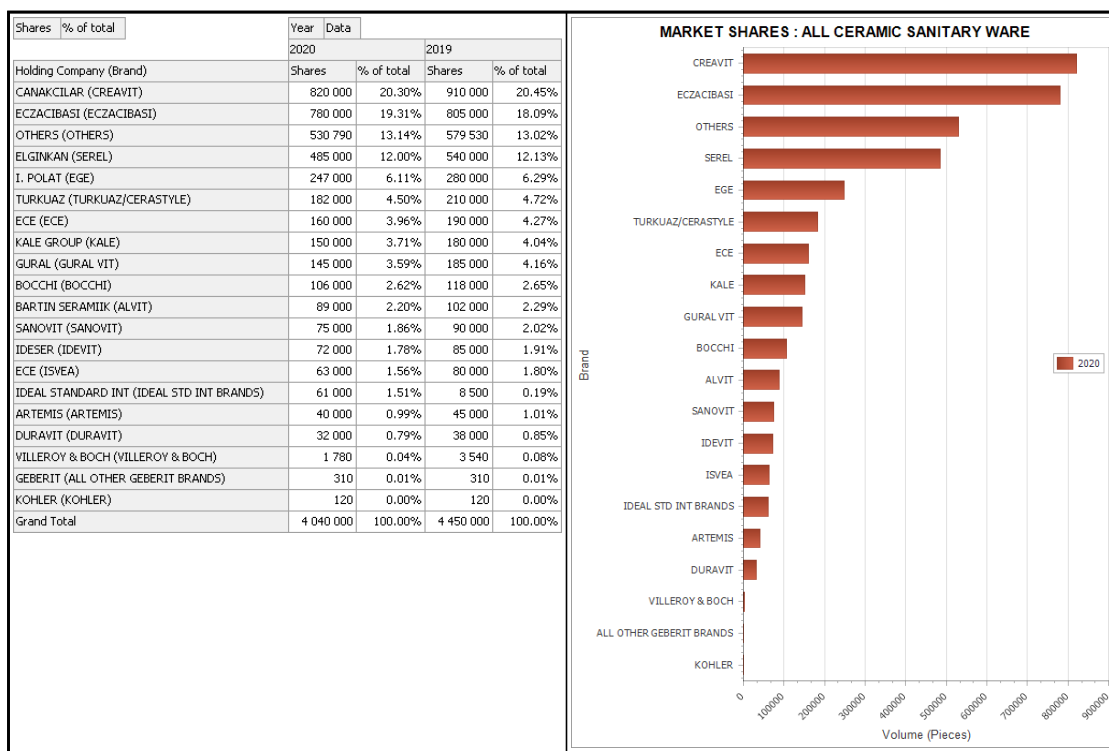
TR - SALES OF SANITARY WARE BY END USE SEGMENT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

2.5.1 Market Shares : All Ceramic Sanitary Ware

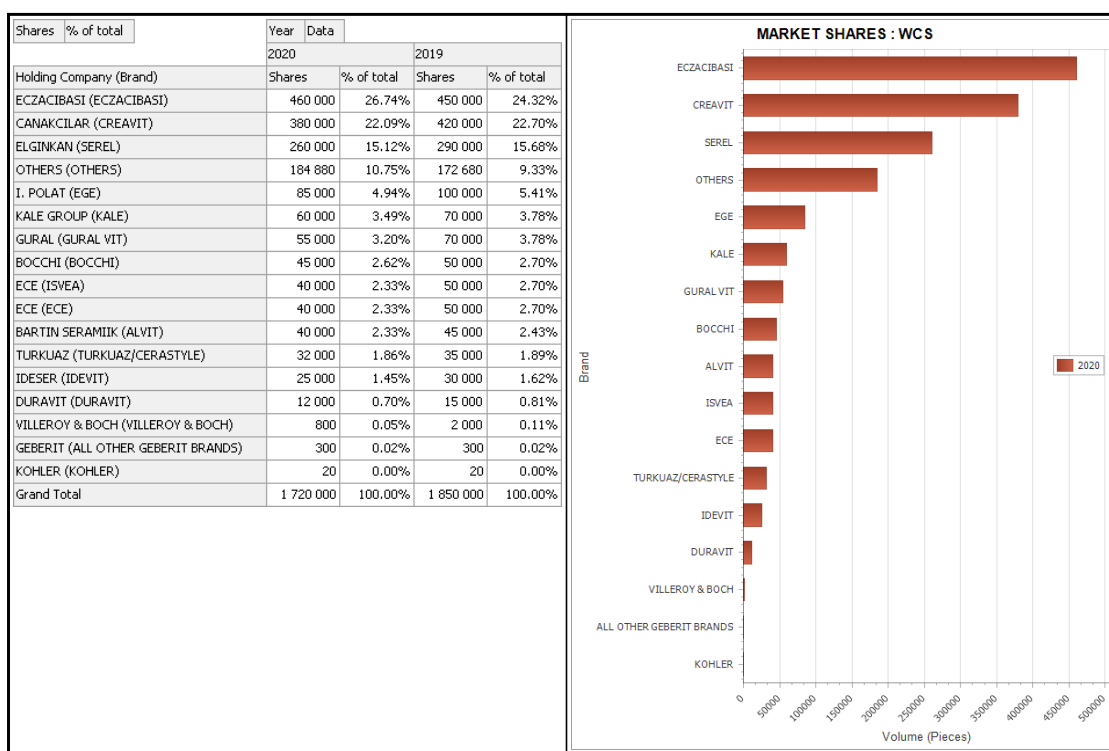
TR - ESTIMATED MARKET SHARES FOR ALL CERAMIC SANITARY WARE



Source: BRG based on industry interviews

2.5.2 Market Shares : WCs

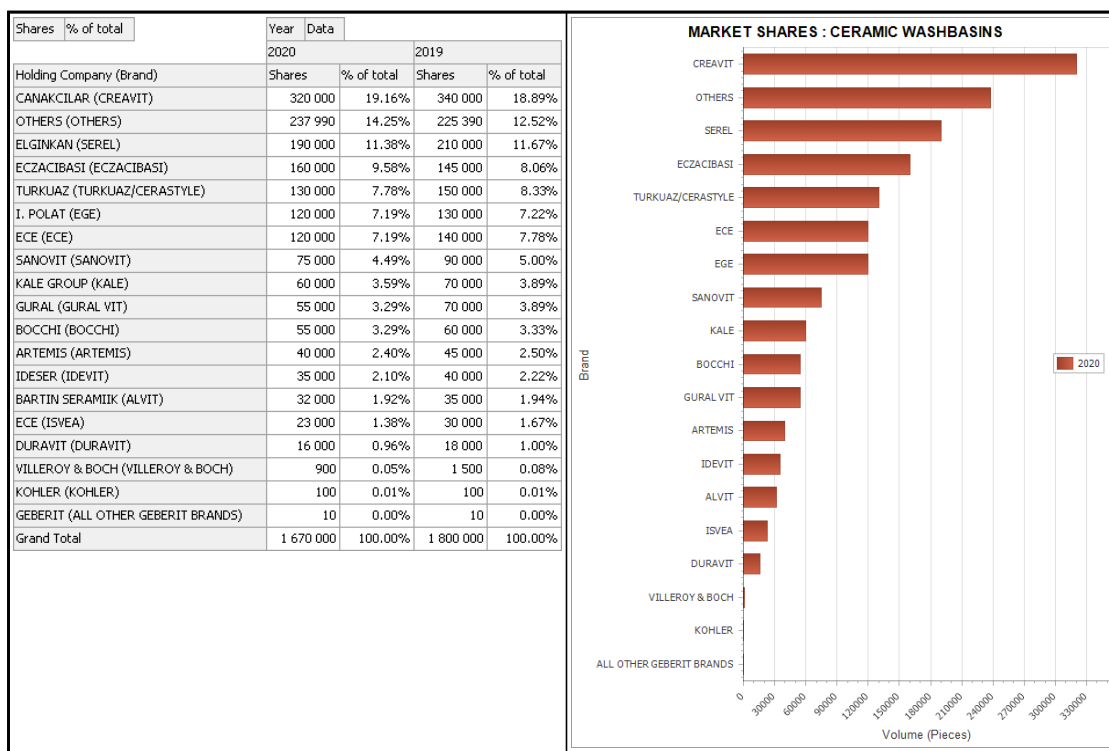
TR - ESTIMATED MARKET SHARES FOR WCS



Source: BRG based on industry interviews.

2.5.3 Market Shares : Ceramic Washbasins

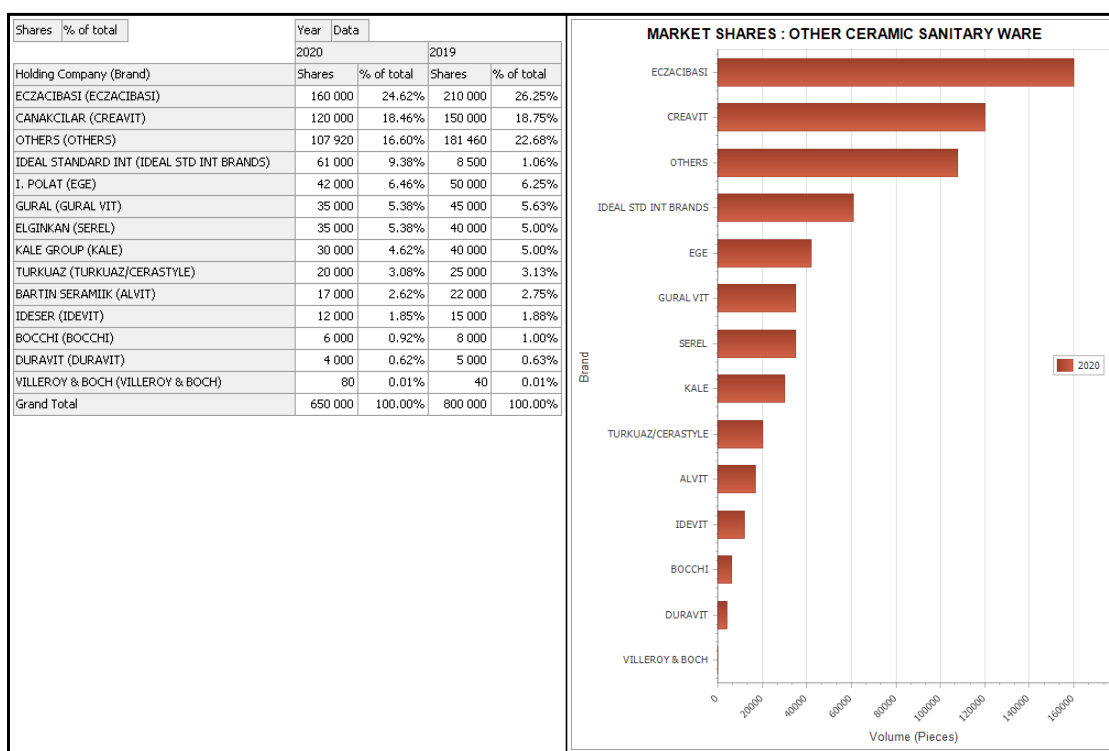
TR - ESTIMATED MARKET SHARES FOR CERAMIC WASHBASINS



Source: BRG based on industry interviews

2.5.4 Market Shares : Other Ceramic Sanitary Ware

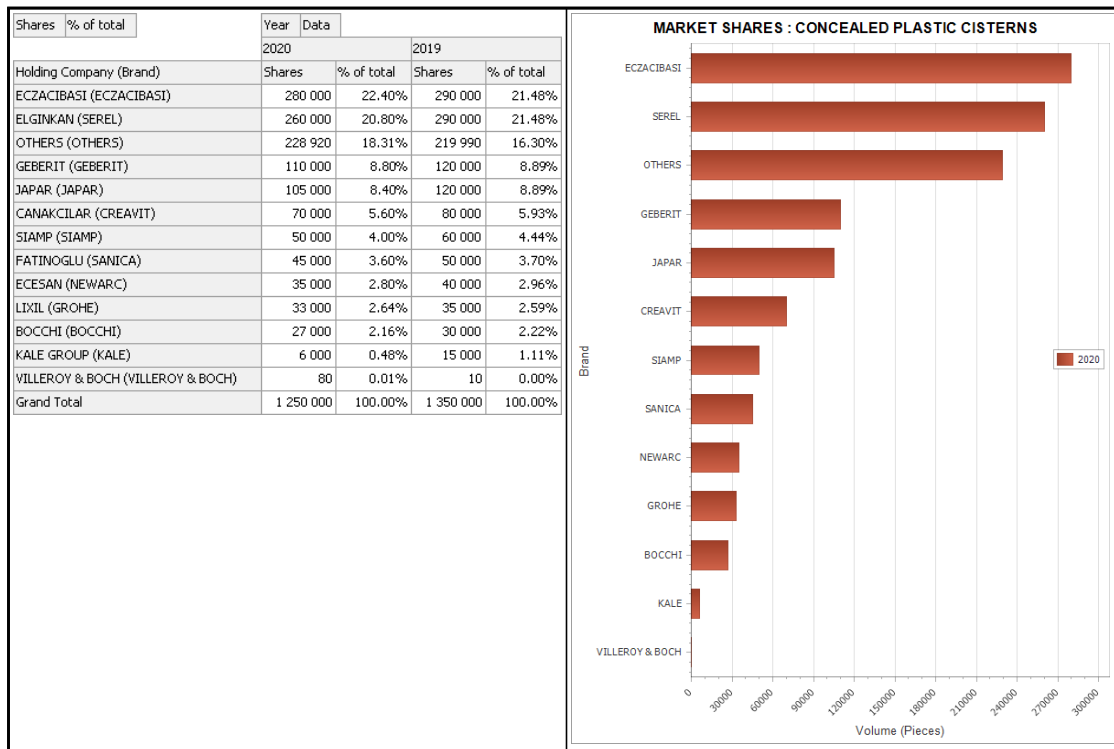
TR - ESTIMATED MARKET SHARES FOR OTHER CERAMIC SANITARY WARE



Source: BRG based on industry interviews

2.5.5 Market Shares : Concealed Plastic Cisterns

TR - ESTIMATED MARKET SHARES FOR CONCEALED PLASTIC CISTERNS



Source: BRG based on industry interviews

2.6.1 TR - IMPORTS OF ALL CERAMIC SANITARY WARE 2020

Source Country	2020				2019			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
China	553	26.2%	1 139	20.2%	421	29.4%	844	20.8%
Russian Federation	452	21.4%	528	9.4%	347	24.2%	440	10.9%
Germany	291	13.8%	1 030	18.3%	108	7.5%	667	16.5%
Egypt	147	7.0%	267	4.7%	38	2.7%	79	1.9%
India	84	4.0%	304	5.4%	80	5.6%	285	7.0%
Poland	82	3.9%	221	3.9%	6	0.4%	14	0.3%
France	71	3.4%	194	3.4%	24	1.7%	123	3.0%
Iran, Islamic Republic of	64	3.0%	68	1.2%	2	0.1%	1	0.0%
United Arab Emirates	57	2.7%	98	1.7%	0	0.0%	2	0.0%
Italy	55	2.6%	564	10.0%	105	7.3%	581	14.3%
Bulgaria	37	1.8%	87	1.5%	22	1.5%	44	1.1%
OTHERS	217	10.3%	1 140	20.2%	280	19.5%	973	24.0%
Total	2 110	100.0%	5 640	100.0%	1 433	100.0%	4 053	100.0%

Source : Trademap

2.6.2 TR - EXPORTS OF ALL CERAMIC SANITARY WARE 2020

Destination	2020				2019			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
France	27 147	16.3%	31 995	13.5%	26 911	16.3%	31 860	13.1%
Germany	22 908	13.8%	40 580	17.1%	20 187	12.2%	38 018	15.7%
Italy	11 777	7.1%	14 050	5.9%	11 704	7.1%	14 243	5.9%
United Kingdom	11 680	7.0%	18 034	7.6%	13 429	8.1%	20 226	8.3%
Israel	7 262	4.4%	9 795	4.1%	7 873	4.8%	8 881	3.7%
United States of America	6 800	4.1%	13 763	5.8%	6 263	3.8%	13 842	5.7%
Serbia	4 594	2.8%	4 407	1.9%	3 848	2.3%	3 894	1.6%
Spain	4 267	2.6%	6 490	2.7%	3 887	2.4%	6 026	2.5%
Poland	4 042	2.4%	5 929	2.5%	3 605	2.2%	5 651	2.3%
Morocco	3 661	2.2%	4 447	1.9%	2 919	1.8%	3 687	1.5%
United Arab Emirates	3 626	2.2%	4 332	1.8%	3 535	2.1%	5 172	2.1%
OTHERS	58 552	35.2%	83 750	35.3%	60 759	36.8%	90 879	37.5%
Total	166 316	100.0%	237 572	100.0%	164 920	100.0%	242 379	100.0%

Source : Trademap

2.6.3 TR - IMPORTS OF PLASTIC CISTERNS 2020

Source Country	2020				2019			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
Germany	1 562	68.8%	6 143	50.7%	1 244	62.0%	5 126	41.7%
Finland	252	11.1%	3 053	25.2%	325	16.2%	3 912	31.8%
China	157	6.9%	908	7.5%	197	9.8%	1 234	10.0%
France	102	4.5%	807	6.7%	59	2.9%	632	5.1%
Italy	48	2.1%	456	3.8%	52	2.6%	493	4.0%
Lithuania	39	1.7%	103	0.8%	0	0.0%	0	0.0%
OTHERS	111	4.9%	652	5.4%	129	6.4%	894	7.3%
Total	2 271	100.0%	12 122	100.0%	2 006	100.0%	12 291	100.0%

Source : Trademap

2.6.4 TR - EXPORTS OF PLASTIC CISTERNS 2020

Destination	2020				2019			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
Iraq	1 349	8.5%	3 980	7.3%	925	6.3%	3 304	6.5%
France	1 274	8.0%	3 484	6.4%	1 070	7.3%	3 003	5.9%
Germany	1 185	7.4%	6 408	11.7%	744	5.1%	4 005	7.9%
Georgia	908	5.7%	1 906	3.5%	824	5.6%	1 910	3.7%
Egypt	821	5.1%	2 302	4.2%	705	4.8%	2 103	4.1%
Bulgaria	747	4.7%	2 557	4.7%	954	6.5%	3 282	6.4%
Spain	713	4.5%	3 222	5.9%	333	2.3%	1 487	2.9%
Russian Federation	707	4.4%	1 907	3.5%	642	4.4%	1 791	3.5%
United Kingdom	702	4.4%	2 995	5.5%	522	3.6%	2 213	4.3%
Israel	585	3.7%	2 209	4.0%	452	3.1%	1 929	3.8%
Poland	572	3.6%	2 055	3.8%	347	2.4%	1 317	2.6%
OTHERS	6 380	40.0%	21 561	39.5%	7 137	48.7%	24 672	48.4%
Total	15 943	100.0%	54 586	100.0%	14 655	100.0%	51 016	100.0%

Source : Trademap

2.7.1 Route to Market: Ceramic Sanitary Ware
TR - ESTIMATED DISTRIBUTION FOR CERAMIC SANITARY WARE: 2020

Manufacturer / Importer

Manufacturers / Importers	
100.0%	

Wholesale Level

72.6%	
Wholesalers / Merchants	
72.6%	

Retail Level

Manufacturer	Wholesaler
3.7%	1.2%
Internet Retailers	
4.9%	

Manufacturer	Wholesaler
7.5%	3.1%
DIY Retailers	
10.6%	

Manufacturer	Wholesaler
7.8%	3.3%
Specialist Retailers	
11.1%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
8.4%	22.0%	1.3%	7.3%	10.5%
Installers				
49.5%				

End Use Level

Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
0.0%	38.5%	3.2%	2.9%	0.5%	43.5%
Residential					
88.6%					
New Build			Residential RMI		
56.0%			32.6%		
Project	Non-Project	Project	Non-Project		
33.4%	22.6%	11.7%	20.9%		

Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
0.0%	4.5%	0.4%	0.4%	0.1%	6.0%
Non-Residential					
11.4%					
New Build			Non-Residential RMI		
6.6%			4.8%		
Project	Non-Project	Project	Non-Project		
3.8%	2.8%	1.5%	3.3%		

2.7.2 Route to Market: Ceramic Sanitary Ware
TR - ESTIMATED DISTRIBUTION FOR CERAMIC SANITARY WARE: 2025

Manufacturer / Importer	Manufacturers / Importers				
	100.0%				
Wholesale Level	72.4%				
	Wholesalers / Merchants				
	72.4%				
Retail Level	Manufacturer		Wholesaler		
	3.9%		1.3%		
	Internet Retailers				
	5.2%				
	Manufacturer		Wholesaler		
	7.7%		3.2%		
	DIY Retailers				
	10.9%				
	Manufacturer		Wholesaler		
	7.6%		3.2%		
Specialist Retailers					
10.8%					

Source: BRG Based on Industry Interviews. - Totals may vary due to roundings

2.7.3 Route to Market - Consolidated View : Ceramic Sanitary Ware

TR - ESTIMATED DISTRIBUTION FOR CERAMIC SANITARY WARE : 2020

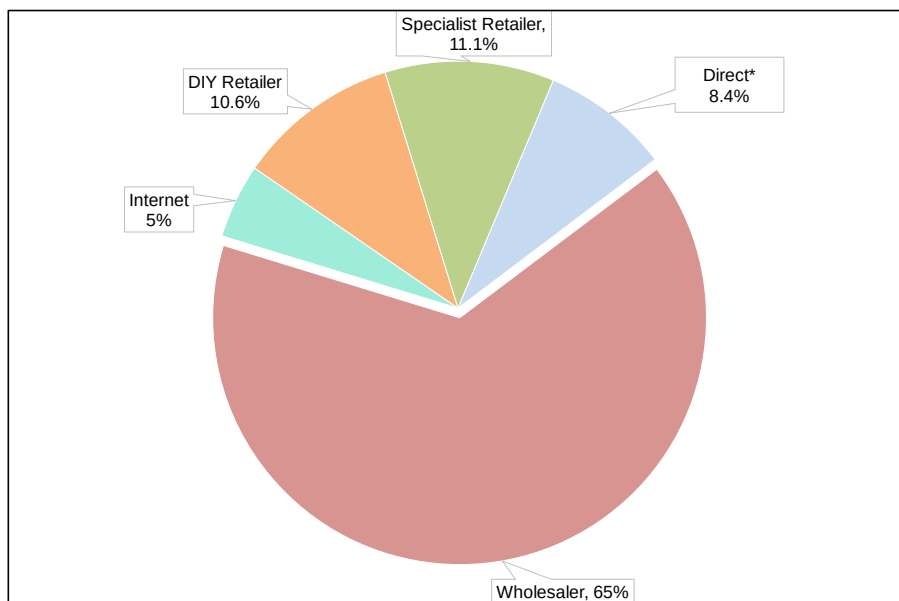
Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
65.0%	4.9%	10.6%	11.1%	8.4%	45.1%	5.3%	43.5%	6.1%
100%					50.4%		49.6%	

2.7.4 Route to Market - Consolidated View : Ceramic Sanitary Ware

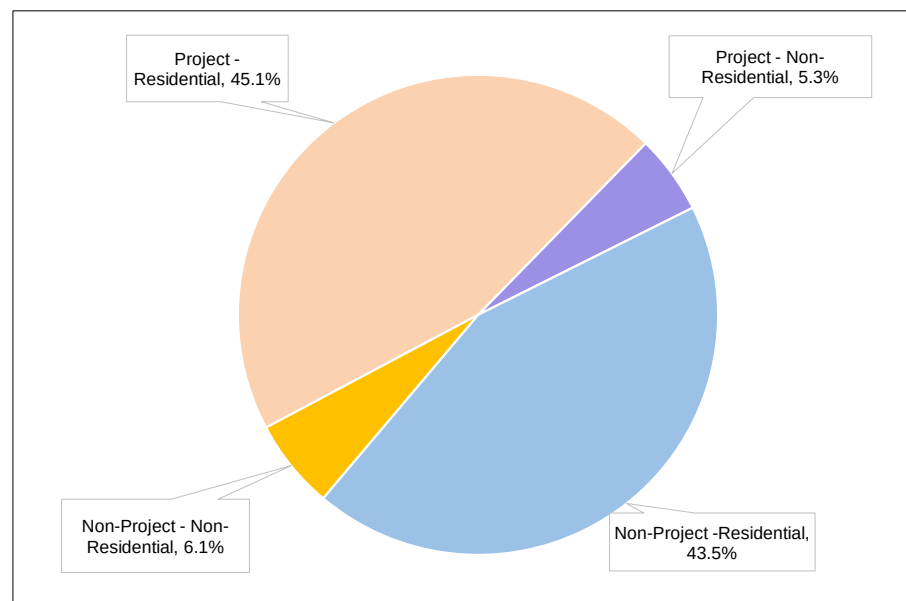
TR - ESTIMATED DISTRIBUTION FOR CERAMIC SANITARY WARE : 2025

Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
64.7%	5.2%	10.9%	10.8%	8.4%	40.0%	5.1%	48.8%	6.1%
100%					45.1%		54.9%	

TR - SALES OF CERAMIC SANITARY WARE BY DISTRIBUTION CHANNEL: 2020



TR - SALES OF CERAMIC SANITARY WARE BY POINT OF DECISION AND END-USE SECTOR: 2020



Source: BRG Based on Industry Interviews.

*Direct includes sales going directly from manufacturers to installers, contractors and private customers.

2.7.5 Route to Market: Plastic Cisterns
TR - ESTIMATED DISTRIBUTION FOR PLASTIC CISTERNS: 2020

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	61.7%
	Wholesalers / Merchants
	61.7%

Retail Level	Manufacturer	Wholesaler	Manufacturer	Wholesaler	Manufacturer	Wholesaler
	4.4%	3.6%	13.7%	4.1%	12.0%	4.1%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	8.0%		17.8%		16.1%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
8.1%	40.0%	1.5%	8.3%	15.7%
Installers				
73.6%				

End Use Level	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
	0.0%	6.6%	5.9%	8.4%	0.3%	67.8%	0.0%	3.3%	0.6%	1.1%	0.1%	5.8%
	Residential						Non-Residential					
	89.1%						10.9%					
	New Build			Residential RMI			New Build			Non-Residential RMI		
	43.6%			45.5%			5.0%			5.9%		
	Project		Non-Project	Project		Non-Project	Project		Non-Project	Project		Non-Project
	23.0%		20.6%	9.0%		36.5%	2.8%		2.2%	1.0%		4.9%

2.7.6 Route to Market: Plastic Cisterns
TR - ESTIMATED DISTRIBUTION FOR PLASTIC CISTERNS: 2025

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	59.0%
	Wholesalers / Merchants
	59.0%

Retail Level	Manufacturer	Wholesaler	Manufacturer	Wholesaler	Manufacturer	Wholesaler
	5.5%	3.7%	14.5%	4.2%	12.5%	4.0%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	9.2%		18.7%		16.5%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
8.5%	40.0%	1.6%	8.4%	16.2%
Installers				
74.7%				

End Use Level	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
	0.0%	3.8%	6.9%	9.1%	0.2%	67.8%	0.0%	3.3%	0.7%	1.2%	0.1%	6.9%
	Residential						Non-Residential					
	87.8%						12.2%					
	New Build			Residential RMI			New Build			Non-Residential RMI		
	37.1%			50.7%			5.5%			6.7%		
	Project		Non-Project	Project		Non-Project	Project		Non-Project	Project		Non-Project
	20.0%		17.1%	11.0%		39.7%	3.0%		2.5%	1.3%		5.4%

Source: BRG Based on Industry Interviews. N.B. Include both Exposed and Concealed Plastic Cisterns - Totals may vary due to roundings

2.7.7 Route to Market - Consolidated View : Plastic Cisterns

TR - ESTIMATED DISTRIBUTION FOR PLASTIC CISTERNS : 2020

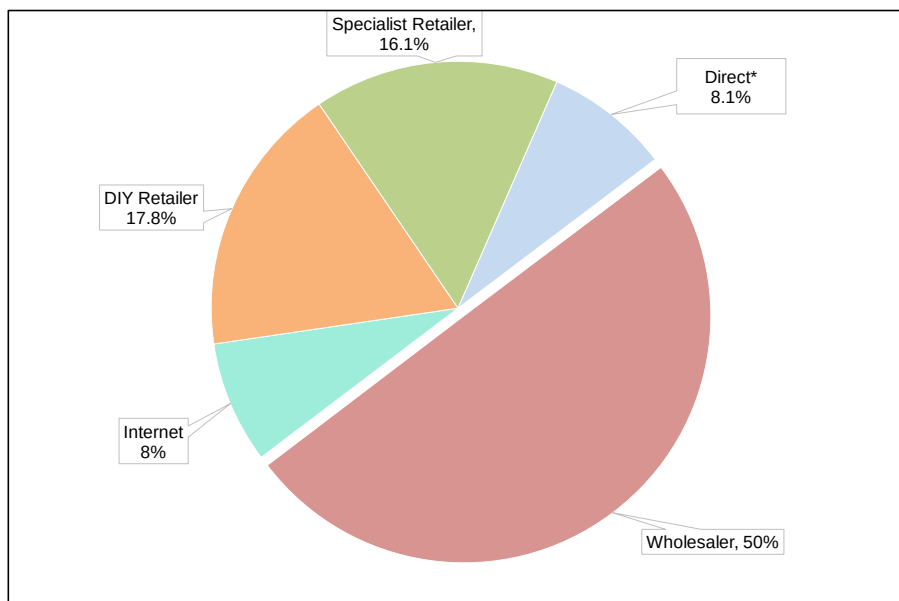
Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
49.9%	8.0%	17.8%	16.1%	8.1%	32.0%	3.8%	57.1%	7.1%
100%					35.8%		64.2%	

2.7.8 Route to Market - Consolidated View : Plastic Cisterns

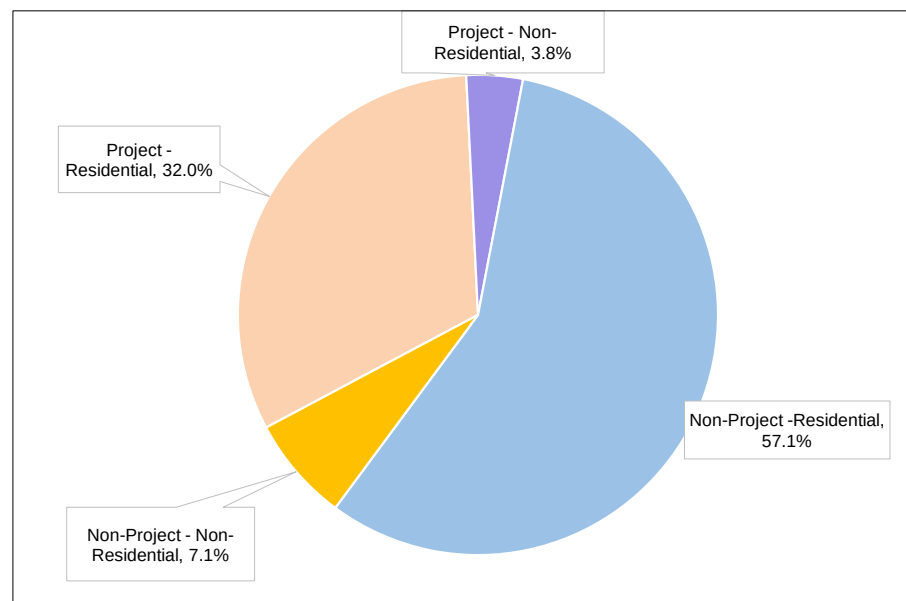
TR - ESTIMATED DISTRIBUTION FOR PLASTIC CISTERNS : 2025

Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
47.1%	9.2%	18.7%	16.5%	8.5%	31.0%	4.3%	56.8%	7.9%
100%					35.3%		64.7%	

TR - SALES OF PLASTIC CISTERNS BY DISTRIBUTION CHANNEL: 2020



TR - SALES OF PLASTIC CISTERNS BY POINT OF DECISION AND END-USE SECTOR: 2020



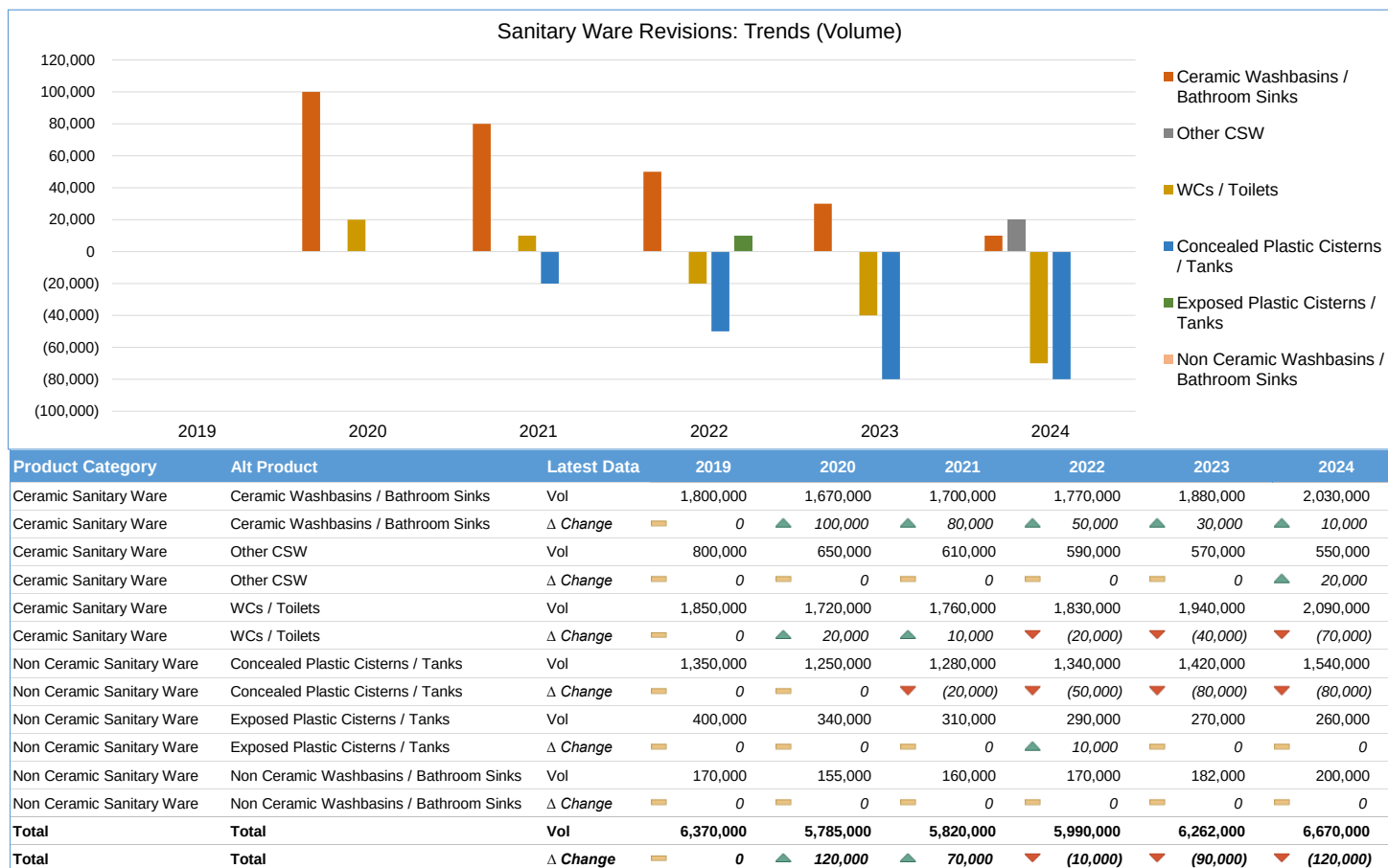
Source: BRG Based on Industry Interviews.

*Direct includes sales going directly from manufacturers to installers, contractors and private customers.

2.8 Revisions in the Turkish Sanitary Ware Market (April 2021)

TR - REVISIONS IN THE TURKISH SANITARY WARE MARKET

Data Comparison: October 2020



No revisions have been made to historical volumes and values for this market.

TURKEY

TAPS AND MIXERS

THE EUROPEAN BATHROOM & KITCHEN PRODUCT MARKETS



Market Characteristics and Product Mix

In terms of pieces, the taps and mixers market is the biggest within all bathroom products in Turkey. Despite declines in some years it grew relatively strongly after 2013, but dropped in 2018 and then again in 2019 and 2020, due to the economic recession. The positive trend between 2013 and 2017 could be attributed to the good economic development in Turkey until 2018. The increasing income of the Turkish population has boosted renovation activity, and new construction activity has increased strongly at the same time.

In Turkey, the average city dwelling often has a main bathroom and a small toilet room. In the main bathroom, it is usual to find a washbasin with a mixer and a mixer with a shower handset (over a bath, shower tray or a tiled floor enclosed with a curtain). In the small toilet, it is usual to find a hand washbasin and a mixer tap. If the dwelling has a Turkish/Arabic toilet, which is a very strongly declining market, there is also very often a pillar tap next to the squat toilet. The kitchen would also have a mixer over a sink. This gives, on average, three mixers and four pillar taps fitted into the standard Turkish home. However, replacements of pillar taps designed for hand washbasins are very often one-head mixers.

Today, one-head mixers account for the vast majority of sales. There has been a clear trend towards one-head mixers over the last few years, and this meant continuous growth for this product until 2018. Two-head mixer sales shares have continued to decline as a result of the preference towards one-head mixers; however, there is always a market for dual-head mixers, mainly dual-head/one-hole mixers. Dual-head mixers are increasingly a luxury product, consisting mainly of 'retro style' (Victorian style) mixers. The importance, in contrast to other European countries, of thermostatic mixers is rather low; however, the share has continued to increase slightly per annum in recent years. As the overall market grows, thermostatic mixer sales are also expected to grow, taking sales from one-pillar taps.

Since 2000, there has been a self-closing and electronic taps and mixers market in Turkey, but this market grew very slowly at the beginning. However, with an increase in epidemic diseases and a growing awareness of hygiene issues in recent years, sales have been increasing at a slightly faster rate per year. Yet it still remains a small market when compared to the other types of taps and mixers sold in Turkey. Sales of electronic taps and mixers have been positively influenced by growing investment into commercial buildings such as shopping malls, restaurants, hotels and hospitals. Self-closing taps and mixers gradually became old-fashioned, mainly cannibalised by electric types. Hence, the share of the self-closing type is not expected to grow significantly in the coming years.

Concealed models are still niche products in Turkey, supplying mostly high-end villas and hotels. The market of concealed taps and mixers is still very small, recording gradual but strong growth rates. Due to the high number of exposed

models, concealed taps and mixers are predicted to stay niche products in the coming years.

The overall taps and mixers market, compared to other bathroom products, reflects a very high share of replacements in Turkey. This is due to the low quality and the short lifespans of these products. Hence, replacement has become the most important driver for the Turkish taps and mixers market.

The WC and urinal flushing valve market is strongly linked to the development of the non-residential sector, as these products are often used in commercial and public buildings such as airports, schools and hotels.

Showering is becoming increasingly more popular than bathing in Turkey, as time saving has become more important to the Turks. This means that shower products, such as shower panels and hydrotherapy cubicles, are becoming more popular.

As purchasing power is increasing and showering becoming more trendy, Turks want to invest in their bathrooms. Shower panels are the cheapest way to make a difference, and this segment is growing gradually within the hydrotherapy market. Unlike in Europe, luxury shower pipes from foreign brands are not really a substitution product for shower pipes in Turkey, as shower panels are much cheaper than this type of luxury shower pipe.

Recent Market Trends

The declining trend of the overall taps and mixers market in 2019 continued in 2020. Due to weak new housing activity and the declining purchase power of Turks, along with growing inflation in the country, the overall demand for taps and mixers was negative in 2020. As a result, the overall tap and mixer sales are expected to have been very negative in 2020, with very strong decline rates. This projection was supported at the beginning of 2020, when the COVID-19 pandemic started in the first few months.

The first lockdowns led to some restrictions regarding conventional distribution channels such as specialists and installers finishing their jobs. However, after the COVID-19 lockdown many Turks escaped from big cities and settled down in rural areas at the seaside and the mountains, very often in holiday homes designed to be used for only several weeks per year, with limited maintenance work over the years. However, in order to spend more time here, many Turks started to renovate their holiday homes during the -pandemic. As a result, the overall decline was not as big as previously forecast.

No major changes have been recorded for 2020 regarding the product mix. The overall market remains dominated by one-head mixers, which are mostly used for washbasins as they are the most used CSW product. One-head mixers are very often represented in the replacement park and are closely correlated to washbasin replacements, as a washbasin replacement gives the consumer the opportunity to switch the old models. Within all taps and mixers, washbasin mixers are the most

common models.

Washbasin taps and mixers continued to be followed by shower taps and mixers in 2020, as the trend away from baths towards showering continued in Turkey, even though the shower tray market declined. Here again, most of the taps and mixers used for showers remained one-head mixers.

Bath taps and mixers were challenged again in 2020. Firstly, due to the trend away from baths, demand for this type of mixer declined, as expected. Another challenge for these products has been the economic crisis, which continued in 2020 and affected the overall bath market negatively. In terms of ratio, two-head mixers have the highest share in bath applications. The two-head mixer market declined as expected, as they are gradually replaced by one-head mixers. Two-head mixers are becoming more and more a niche product and more often installed in the high-end segment. One-head mixers continue to see the highest demand, whereas thermostatic sales have the smallest share in the bath mixers market.

Bidet taps and mixers remain a niche market. Sales of this type of product declined in 2020, due to weakening demand from the new housing segment. Pillar and self-closing tap and mixer sales continued to decline as previously forecast, as these products present old-fashioned and very basic solutions. Self-closing models continued to be replaced by electric mixers. Pillar taps are very often installed in terraces and gardens, as a simple water source.

Electronic mixer sales stopped their growing trend in 2018 and declined in 2019, which continued in 2020, due to shrinking international investment in the non-residential and public sectors. Flush valves, both WC and urinal models, have also suffered from declining non-housing investments in the country.

Prices and Market Values

Manufacturers of taps and mixers, as well as shower panel products, usually provide dealers with a margin of approximately 45% of the list price; this figure can increase to up to 55% if dealers sign one-year contracts with manufacturers in advance.

Turkey can still be considered a developing market, with the majority of sales in volume taking place in the lower and economy price segments; although this is significantly less in value. The core products within the market are washbasins, in combination with one-head mixers. These products follow the trend of the overall market, with the majority of sales being in the economy and lower segments. Another reason for the very high demand of washbasin mixers in the lower segments is the fact that most of the pillars are installed for washbasins, which are sold only in the lower segments.

Bath, kitchen and shower mixers are also mostly sold in the lower price segments, as the Turkish market is very price-driven. The upper price segment accounts for a larger share of bidet mixers than it does for other mixers, as this type of mixer is

often not essential and very often installed in a household with more financial possibilities.

The very old-fashioned pillar taps are sold only in the low price segments, whereas thermostatic mixers are traded mostly in the upper and luxury segments. Thermostatic mixers are not actually luxury products, but are classified in the upper segment as they tend to be more expensive. This is also the case for electronic and self-closing taps and mixers. Shower panels are also sold in the high-end segment, as these products are not essential for households and are bought as luxury products.

Regarding the MSP, all taps and mixers recorded increases in Turkish lira in 2020, due to growing inflation. Imported taps and mixers occupy only a small share of the Turkish market, as most products are supplied by manufacturing sites within Turkey, partly owned by foreign companies. However, importing energy and raw materials is the most costly expenditure for Turkish manufacturing sites, which are usually paid for in expensive US dollars. Hence, producers have been forced to increase their MSPs for all products. However, the MSP increase on average was smaller than the overall inflation, due to the declining sales and very competitive market.

New Housing

An average new dwelling has a main bathroom and a small toilet room. The main bathroom contains typically a washbasin with a mixer, and another mixer with a shower or bath. In the small toilet it is usual to find another washbasin mixer, which can be sometimes a simple pillar model. In the kitchen there is also a mixer installed. If the household budget allows, shower panels are also often sold in the new housing segment. A washbasin, especially with a one-head mixer, shows in the new build segment very clearly the highest demand, followed by shower mixers.

The new housing market is a relatively important market segment for taps and mixers in Turkey, and significantly bigger than in most other European countries of the same size. In 2020, the decline of new housing activity in Turkey led to a total sales decrease of the overall taps and mixers market.

No major changes are expected in 2021, with low new housing activity causing declining sales volumes in this segment. However, from 2022 onwards, new construction is projected to increase again and support the overall tap and mixer market sales during the forecast period. Moreover, sales of new housing, as a proportion of overall mixer sales, are expected to grow as well.

Housing RMI

In Turkey, BRG estimates that the overwhelming majority of taps and mixers sold to projects are directed to replacement in housing applications. Nevertheless, an exact amount is difficult to estimate, as manufacturers are often not aware of which

projects distributors are supplying. However, it is assumed that the share of both sanitary and kitchen taps and mixers going to the replacement sector is higher than that of other bathroom products. One of the main reasons for this is that Turkish tap and mixer quality is lower than in Europe, thus requiring more frequent replacement. The average lifespan of a mixer in Europe is over 20 years, but the maximum is 12 years in Turkey.

Regarding product mix, the renovation segment is dominated by washbasin mixers, as they have the largest park in existing dwellings, with shower mixers second. Shower mixers are not always like-for-like replacements, but very often replacements of bath mixers due to the trend away from baths towards showering. Shower panels have a very low share in this segment, because of the short history in the Turkish market.

Overall sales in the RMI segment declined in 2020, due to the economic crisis. The domestic currency suddenly became very weak, which increased inflation and declined the purchasing power of Turks. However, the decline was weaker than previously forecast, as many Turks renovated their homes during the lockdowns. The renovation market intensified in the first quarter of 2021, as the COVID-19 pandemic was still out of control and Turks expected further lockdowns in the coming months. As a result, from 2021 onwards, overall RMI sales are predicted to recover.

Non-Housing

Taps and mixers are an essential element to any occupied building. Regarding product mix, washbasin mixers have the highest sales in the non-housing segment, followed again by shower mixers. Looking at different product types, thermostatic, self-closing and electronic taps and mixers are popular in hotels, terminals, restaurants and schools, which have the largest shares. In the non-housing segment, almost all thermostatic mixers are installed in five-star hotels. Similarly to other end-use segments and bath products, the non-housing segment decreased significantly in 2020, due to the economic recession and slowing-down of investments in the public sector.

Supply Structure

In Turkey, the market can be described as having three parts, namely the 'premium', 'medium' and 'economy/standard' price segments. The premium segment is mainly supplied by foreign brands such as HANSGROHE, GROHE, and IDEAL STANDARD. The medium segment is very often supplied by the two biggest manufacturers in Turkey, ECZACIBASI and ELGINKAN.

Most of the sales are in the economy segment, which is supplied by up to 40 manufacturers. While these manufacturers are often local rather than nationwide, the segment is also supplied by several imported brands, mainly from China. DIY stores have started to import directly from foreign manufacturers, and these foreign firms have taken significant market share from local suppliers; however, domestic

manufacturers still remain much more powerful in the sanitary taps and mixers market.

A gradual increase of Turkish manufacturers supplying complete solutions including tiles, CSW, flushing WCs, and taps and mixers can be seen in this market. Manufacturers from other industries have also started to sell taps and mixers. Their market share is still marginal, but expected to increase strongly in the coming years.

Even though imports of taps and mixers play only a limited role in Turkey, the upper market is very clearly dominated by the German GROHE and HANSGROHE, which are the pioneers in this market.

Domestic suppliers, which are actually only importers (mainly from China) selling taps and mixers under their own Turkish brands, were all at risk and some recorded significant losses when the Turkish Lira became significantly weaker against foreign currencies. Moreover, due to the 2019 financial crisis in Turkey, the government greatly increased the import tax on building products, especially for taps and mixers. KALE is the most well-known company to suffer significant losses on imports in 2020. As a result, KALE's competitiveness decreased very strongly as the overall market became even more price-driven after the decline in purchasing power of Turks. Therefore, KALE experienced very significant sales decline in 2019, and then again in 2020.

On the other hand, due to the weak Turkish currency, the export of Turkish products became more profitable. Hence, in 2020 many market players in Turkey increased their production capacity to meet the demand for exports.

In January 2016, the Spanish ROCA acquired for €46 million the Turkish taps and mixers manufacturer NSK. With this transaction ROCA should reinforce its faucet production capabilities, which will amount to up to eight million pieces per annum. The transaction also means ROCA's return to the Turkish market after its departure in 2009. Established in 1965, NSK has approximately 300 employees in its production plant in the industrial zone of Eskisehir, 350 km away from Istanbul.

Forecasts

The negative trend from 2020 stopped in Q1 2021. As a result of the low number of building permits, the number of housing completions is still predicted to decline again in 2020, but the demand for taps and mixers is highly supported by the strong replacement activity in Turkey. Due to the worldwide COVID-19 pandemic, along with lockdown intervals, many Turks spend their time either at home or away from big cities – in rural areas. Hence, they have intensified their renovation activities in homes, which affects the overall taps and mixer market very positively. As a result, no major changes are expected in the coming quarters and the overall taps and mixers market should record some growth as a whole in 2021.

Heading to 2025, overall sales of taps and mixers should fully recover as the market is set to resume steady growth from 2021 onwards, being bolstered by an increase in construction output. New build activities should again be at a very high level, boosted by private investments and the Urban Transformation Plan, as well as the housing development administration of Turkey, TOKI. On top of that, the shorter lifespans of taps and mixers in Turkey versus Europe, plus positive private consumption and the additional renovations due to the COVID-19 lockdowns, should also positively affect RMI sales from 2021 onwards.

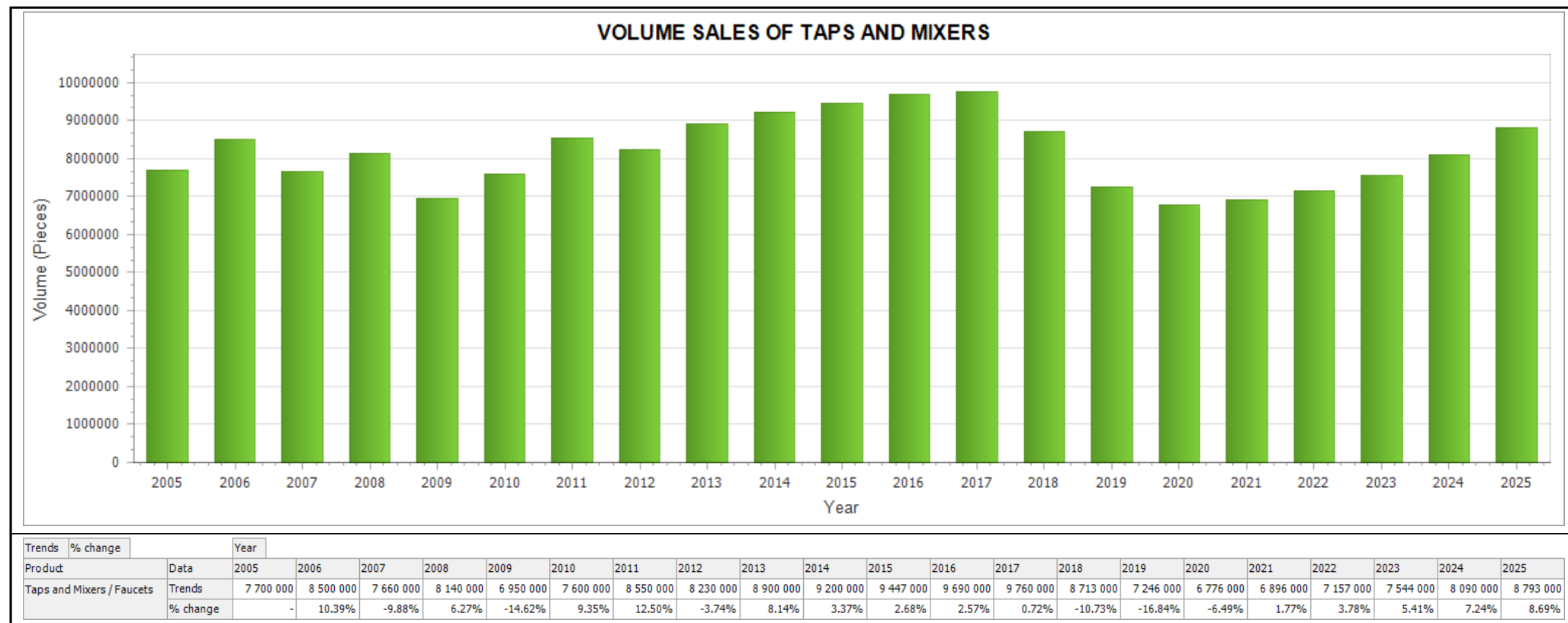
In terms of product mix, BRG does not expect major changes. The overall market is predicted to be dominated by washbasins followed by shower mixers, which are mostly installed with one-head models. Both mixer sales are likely to increase in volume terms, thanks to the predicted growth in new housing and replacement activities. Moreover, shower mixer sales should profit from the shift away from baths towards showering. Kitchen mixer sales should follow the trend of the washbasin mixers, and grow during the forecast period.

On the other hand, the market for bath mixers should decline further over the forecast period, due to the shift away from baths in Turkish bathrooms. Bidet mixers are predicted to decline further at first, due to the trend towards smaller bathrooms, but then should slightly grow again thanks to the expected replacements in this area.

Regarding types of mixers, the trend towards electronic and concealed models should continue. Aesthetics and product design, in combination with the MSP, are predicted to remain key concerns for manufacturers. One-head mixers should retain their popularity. New designs are also likely to be increasingly available at lower price points, as the internet channel continues to gain share, placing continued pressure on MSP margins. Distribution is expected to shift dramatically towards DIY and internet channels in the coming five years, through both private and installer purchases. The increase in internet distribution is likely to enable consumers to obtain higher-quality products and lower prices, which will put pressure on traditional distribution channels such as independent merchants and DIY.

3.2 Historical Trends and Forecasts : Taps and Mixers

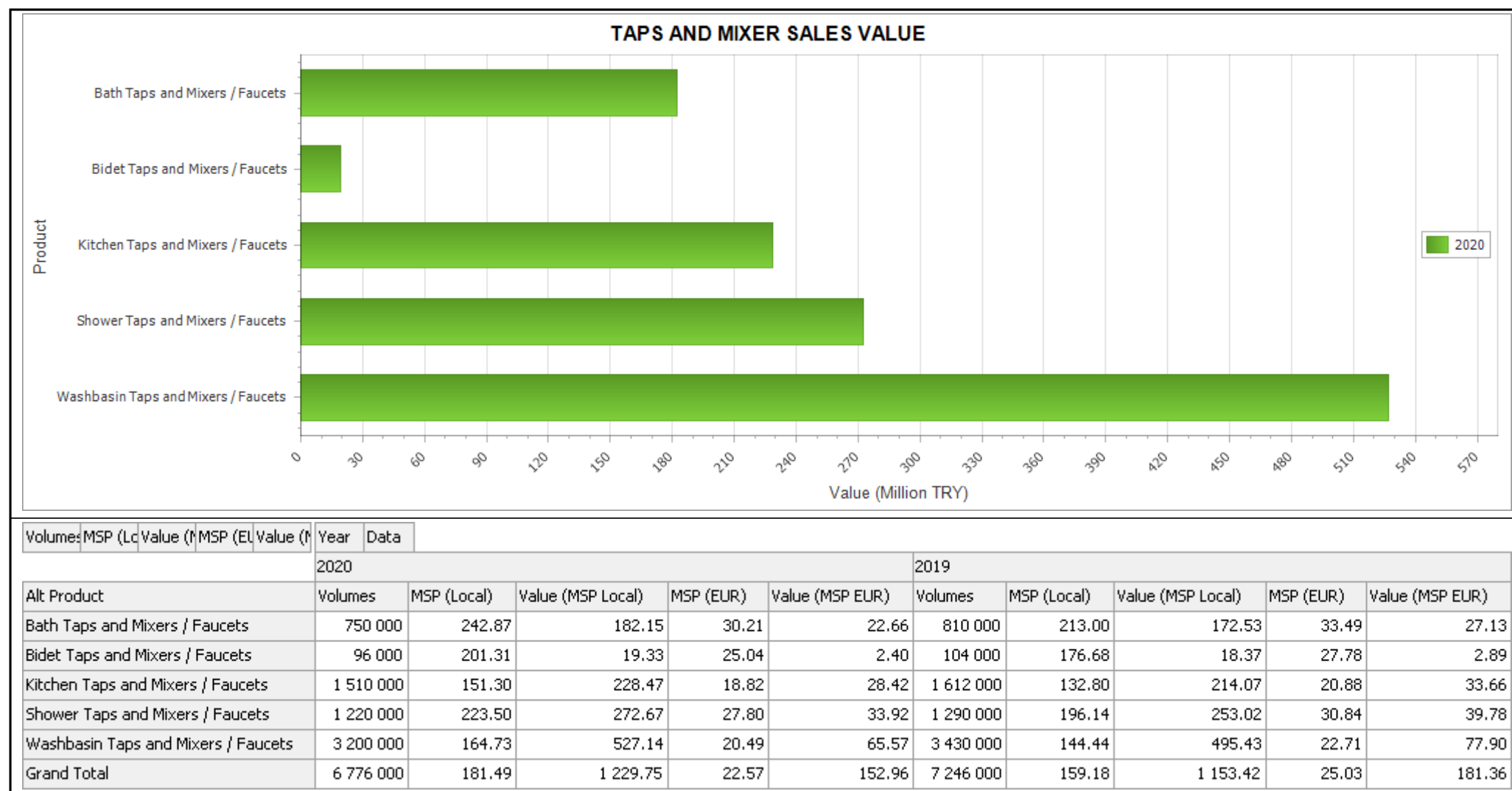
TR - SUMMARY OF TRENDS IN THE TAP AND MIXER MARKET : HISTORICAL AND FORECASTS



Source: BRG based on industry interviews. N.B. Includes Both Kitchen and Sanitary Taps & Mixers

3.3.1 Prices and Market Values : Taps and Mixers

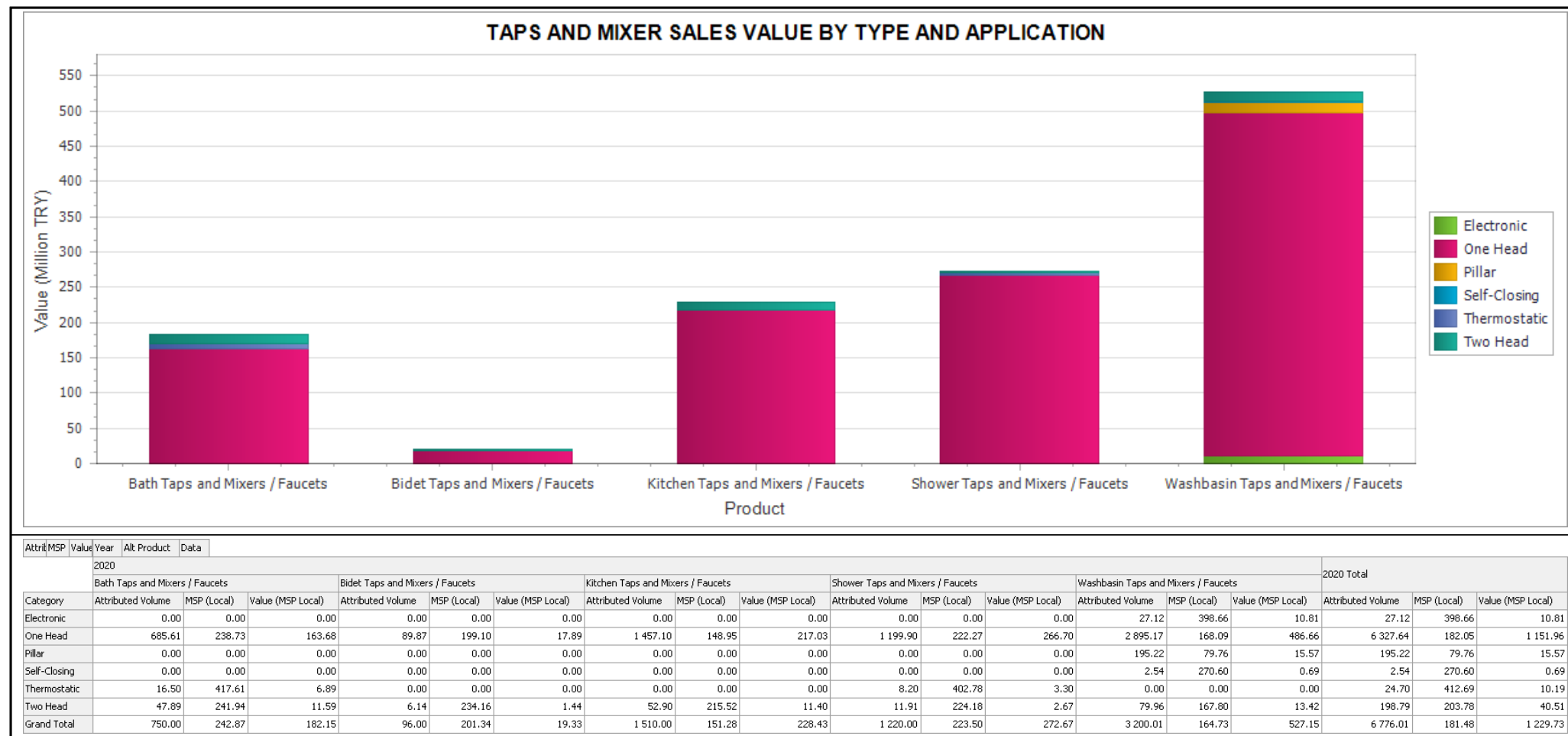
TR - ESTIMATED VALUES OF THE TAP AND MIXER MARKET



Source: BRG based on industry interviews. Local Prices Shown in TRY. Values Shown in Millions (at MSP = Manufacturers' Selling Price).

3.3.2 Prices and Market Values by Type : Taps & Mixers

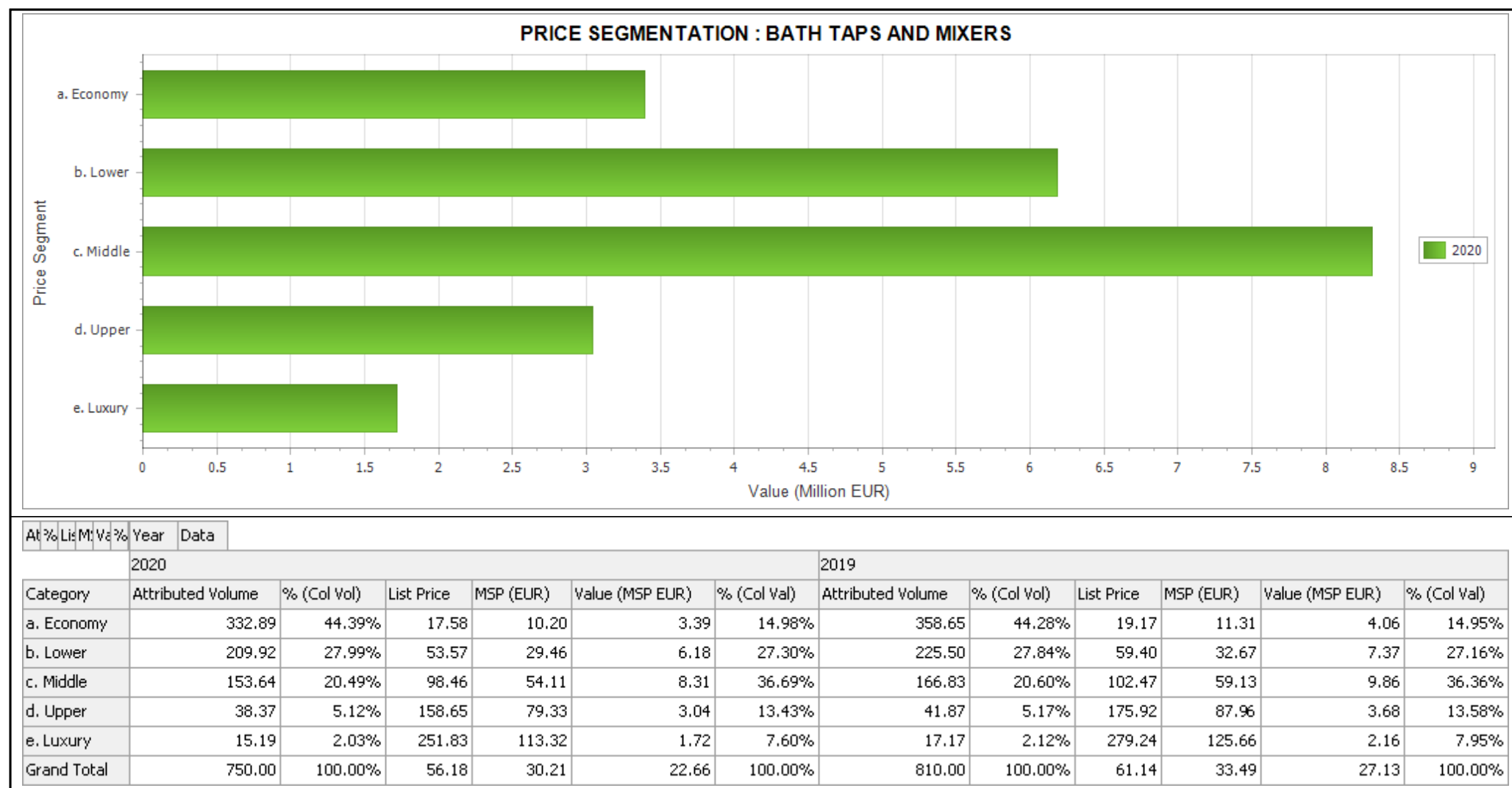
TR - ESTIMATED VALUES OF THE TAP AND MIXER MARKET BY TYPE AND APPLICATION



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Local Prices Shown in TRY. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price).

3.3.3 Price Segmentation : Bath Taps and Mixers

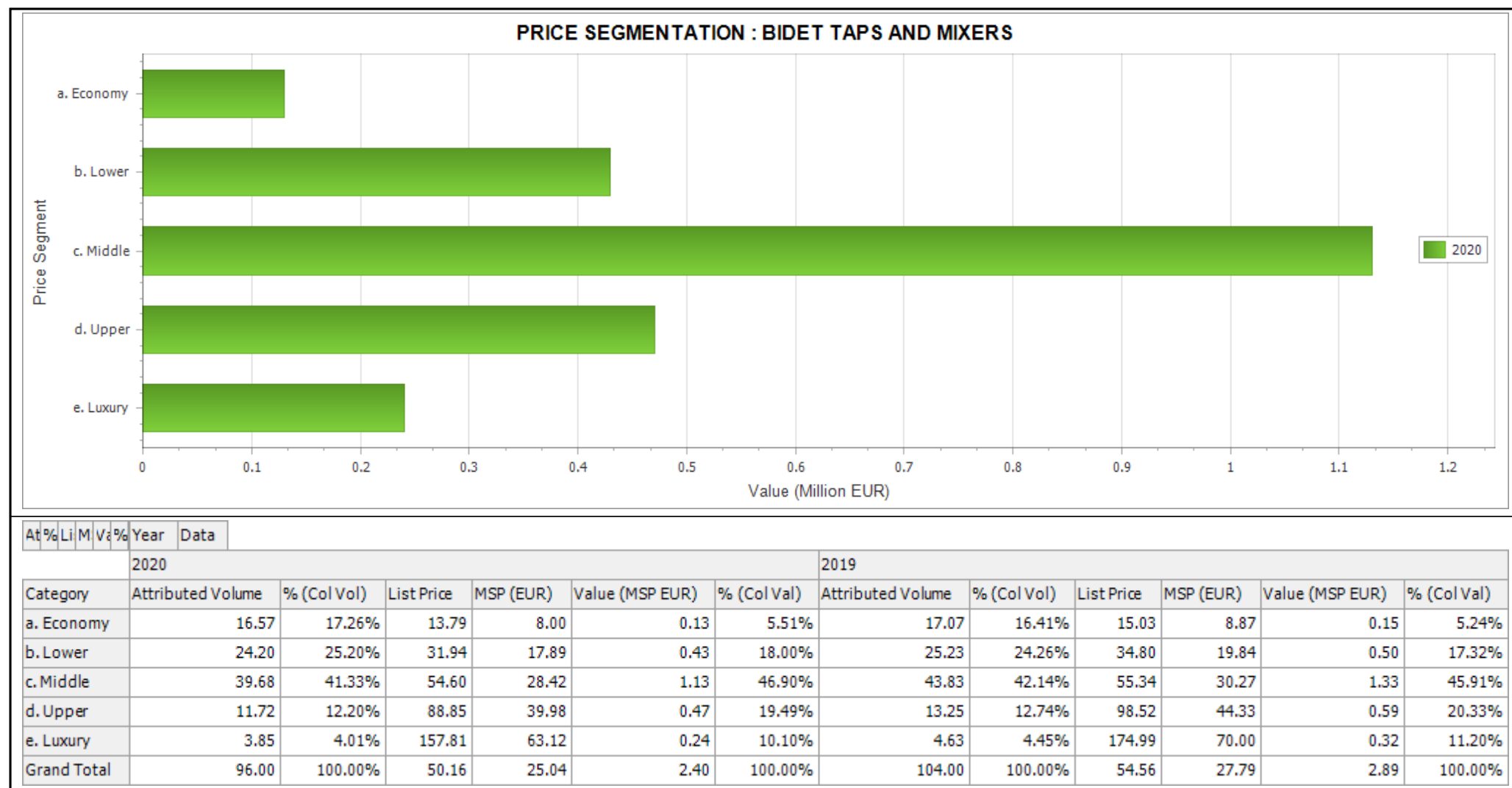
TR - ESTIMATED PRICE SEGMENTATION FOR BATH TAPS AND MIXERS



Source: BRG based on industry interviews. Total may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy <€20, Lower €20-€40, Middle €40-€65, Upper €65-€100 and Luxury >€100

3.3.4 Price Segmentation : Bidet Taps and Mixers

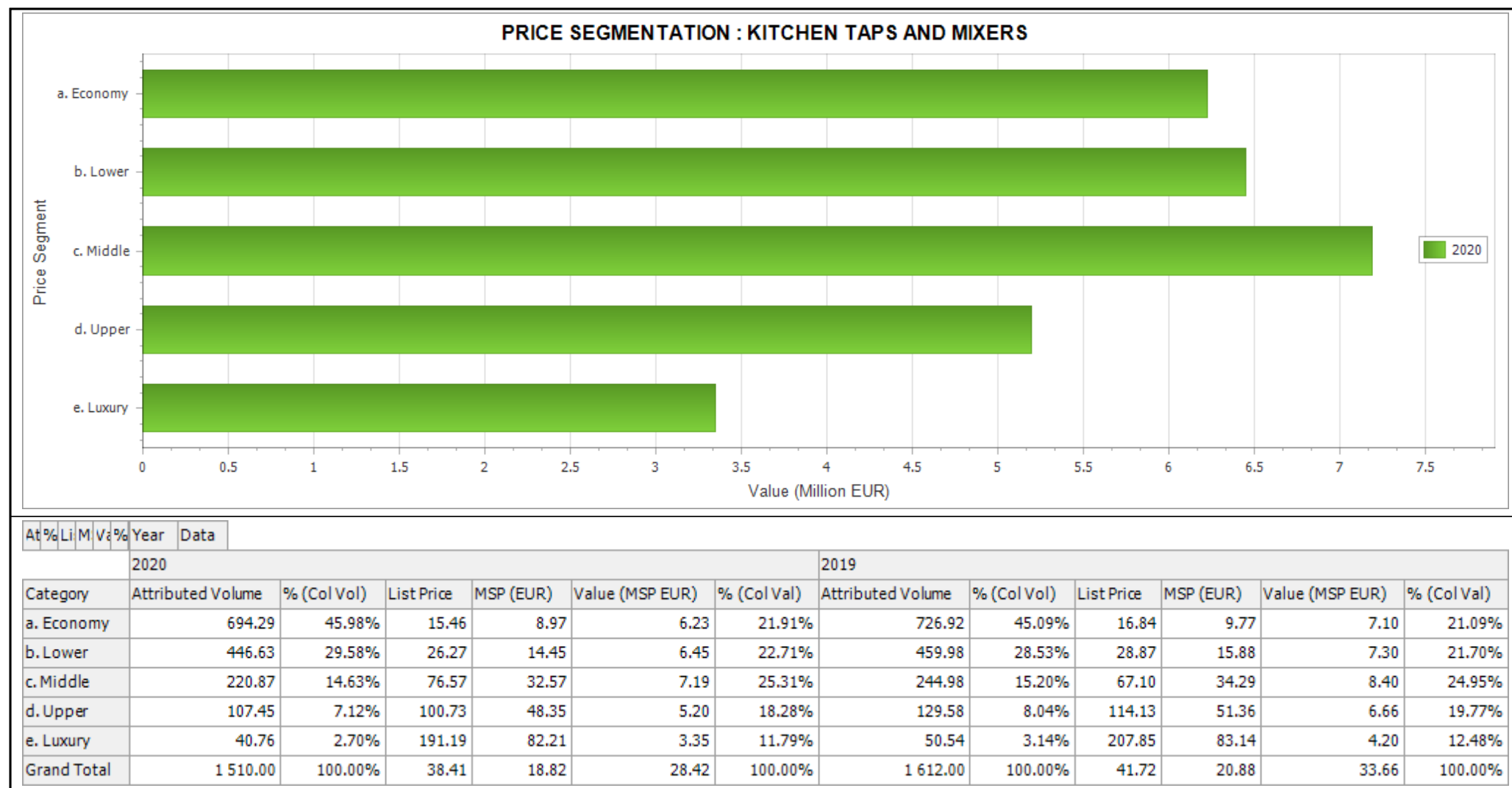
TR - ESTIMATED PRICE SEGMENTATION FOR BIDET TAPS AND MIXERS



Source: BRG based on industry interviews. Total may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy <€15, Lower €15-€25, Middle €25-€40, Upper €40-€60 and Luxury >€60

3.3.5 Price Segmentation : Kitchen Taps and Mixers

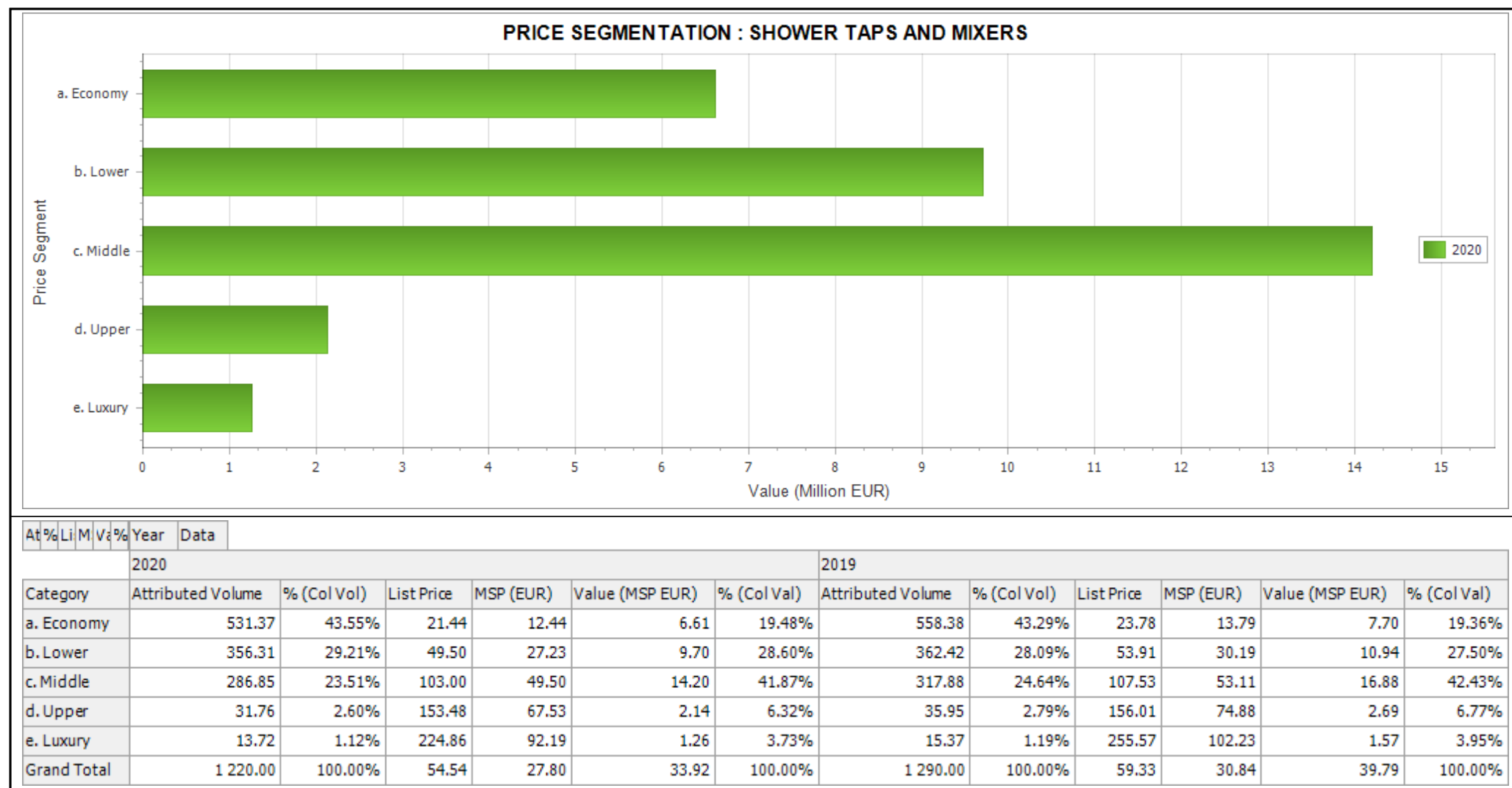
TR - ESTIMATED PRICE SEGMENTATION FOR KITCHEN TAPS AND MIXERS



Source: BRG based on industry interviews. Total may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy <€15, Lower €15-€30, Middle €30-€50, Upper €50-€80 and Luxury >€80

3.3.6 Price Segmentation : Shower Taps and Mixers

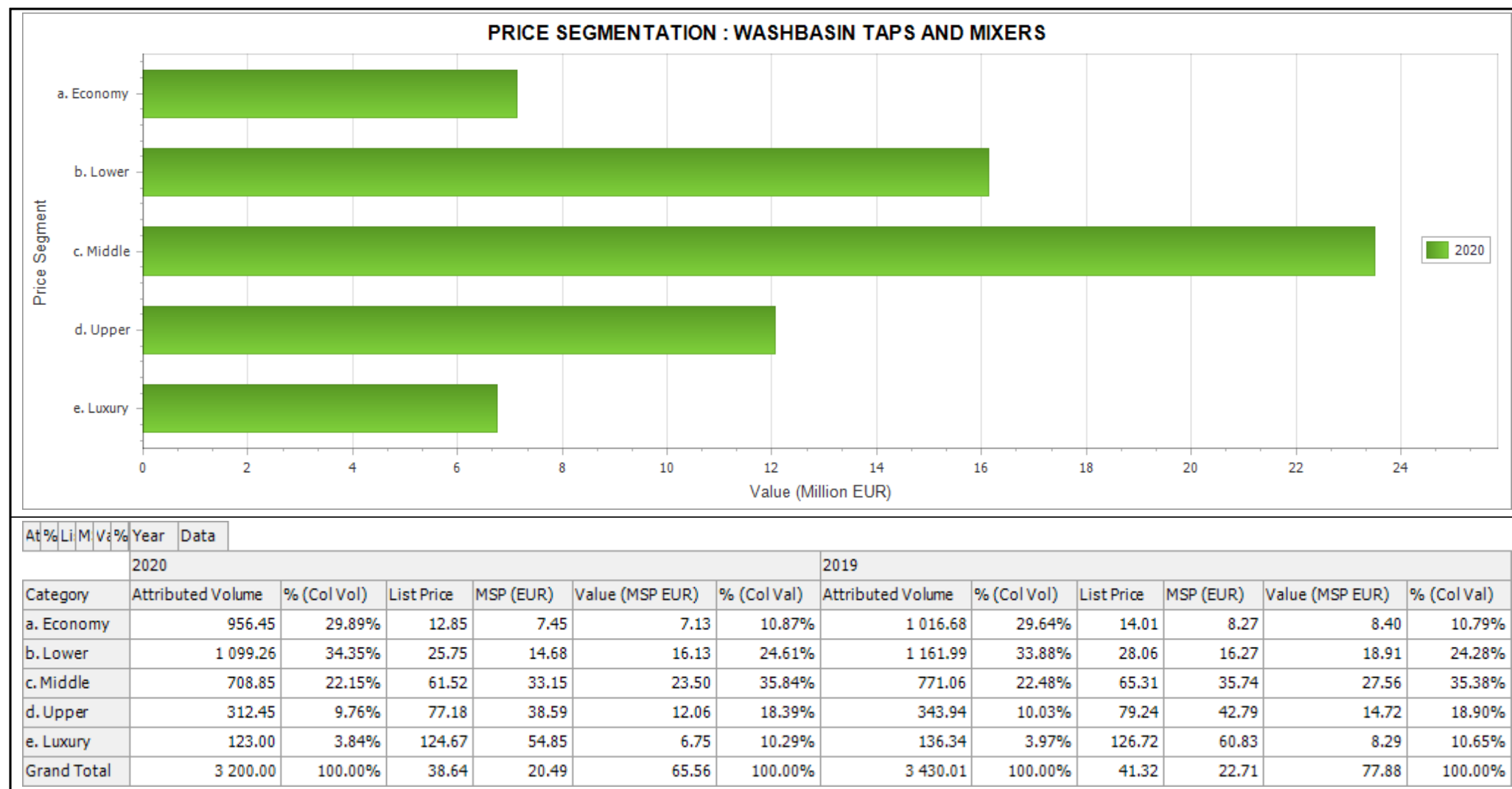
TR - ESTIMATED PRICE SEGMENTATION FOR SHOWER TAPS AND MIXERS



Source: BRG based on industry interviews. Total may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy <€20, Lower €20-€40, Middle €40-€60, Upper €60-€100 and Luxury >€100

3.3.7 Price Segmentation : Washbasin Taps and Mixers

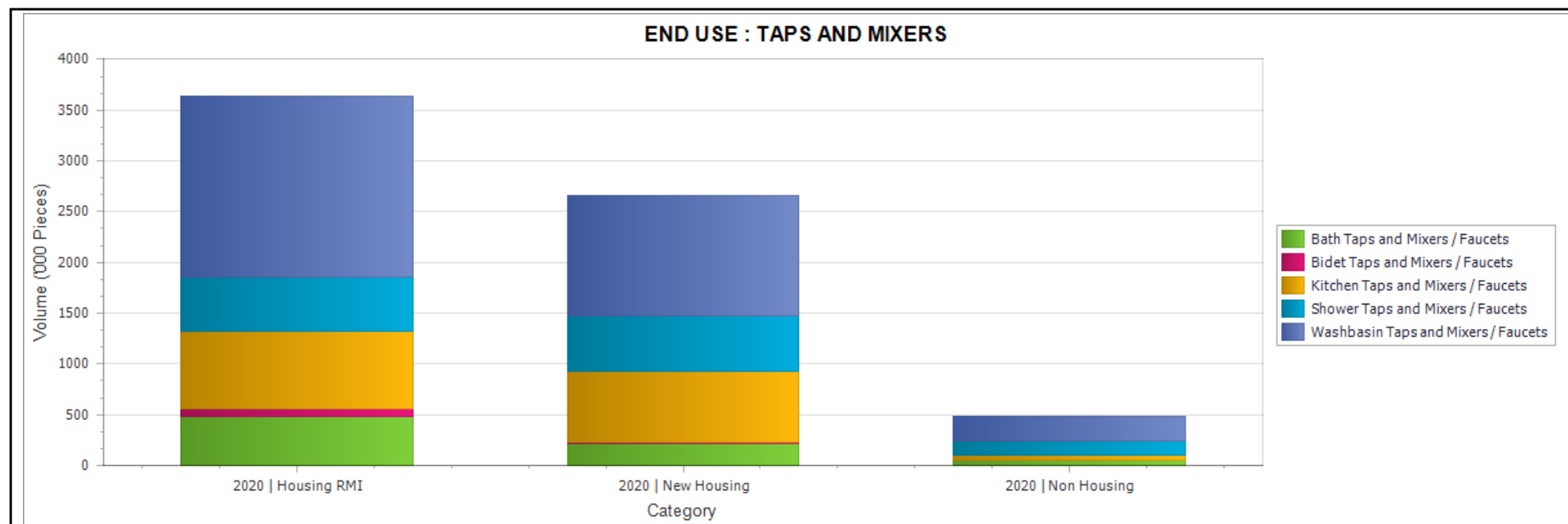
TR - ESTIMATED PRICE SEGMENTATION FOR WASHBASIN TAPS AND MIXERS



Source: BRG based on industry interviews. Total may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy <€15, Lower €15-€25, Middle €25-€40, Upper €40-€60 and Luxury >€60

3.4.1 End Use : Base Year

TR - SALES OF TAPS AND MIXERS BY TYPE AND END USE SEGMENT ('000 PIECES)

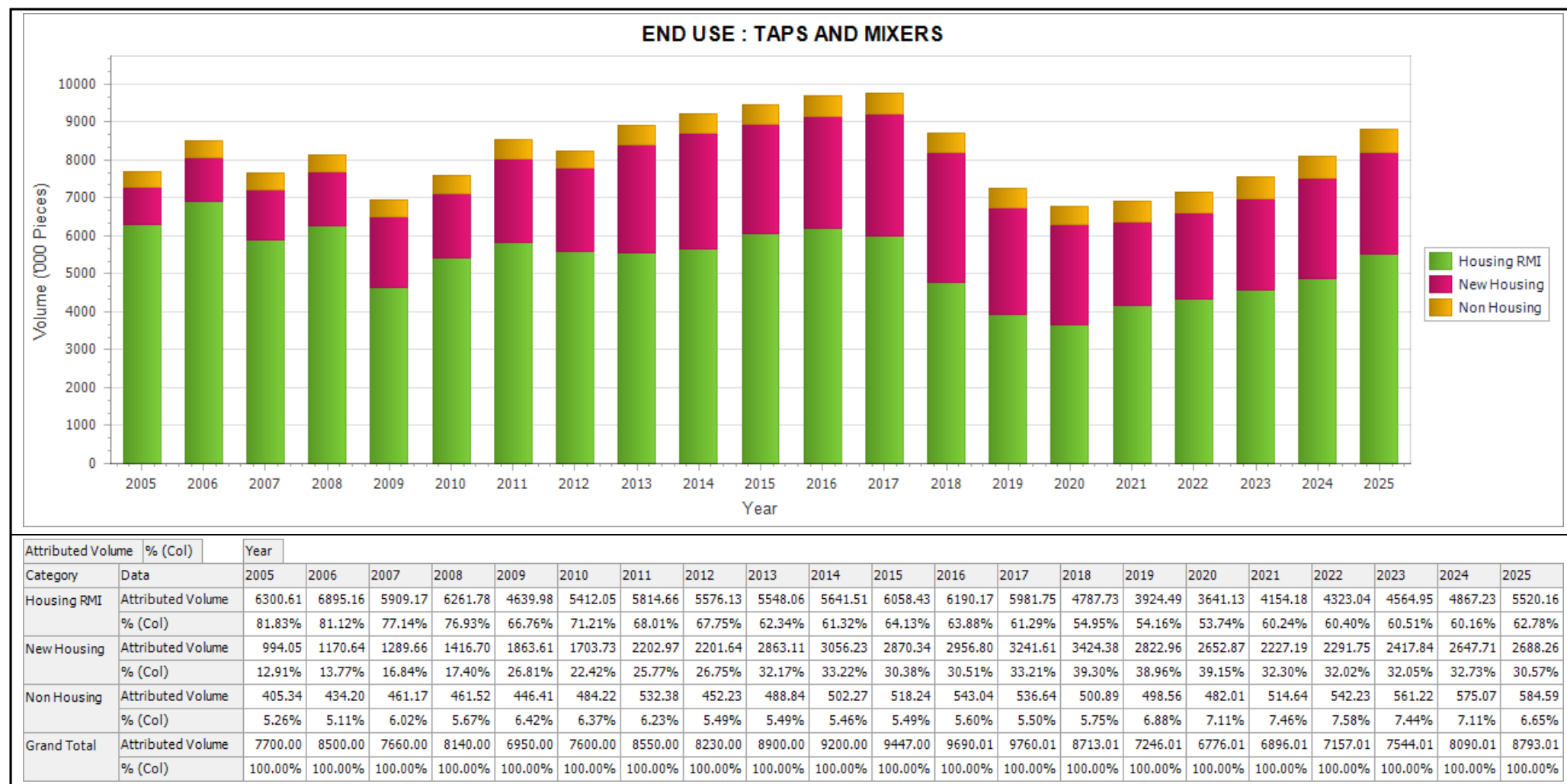


Attributed Volume	% (Col)	% (Row)	Year	Category	Data								
2020													
Housing RMI				New Housing			Non Housing			2020 Total			
Alt Product	Attributed Volume	% (Col)	% (Row)	Attributed Volume	% (Col)	% (Row)	Attributed Volume	% (Col)	% (Row)	Attributed Volume	% (Col)	% (Row)	
Bath Taps and Mixers / Faucets	483.20	13.27%	64.43%	213.90	8.06%	28.52%	52.91	10.98%	7.05%	750.00	11.07%	100.00%	
Bidet Taps and Mixers / Faucets	73.37	2.02%	76.43%	19.06	0.72%	19.85%	3.57	0.74%	3.72%	96.00	1.42%	100.00%	
Kitchen Taps and Mixers / Faucets	764.48	21.00%	50.63%	700.34	26.40%	46.38%	45.18	9.37%	2.99%	1510.00	22.28%	100.00%	
Shower Taps and Mixers / Faucets	537.49	14.76%	44.06%	546.00	20.58%	44.75%	136.51	28.32%	11.19%	1220.00	18.00%	100.00%	
Washbasin Taps and Mixers / Faucets	1782.59	48.96%	55.71%	1173.57	44.24%	36.67%	243.85	50.59%	7.62%	3200.00	47.23%	100.00%	
Grand Total	3641.13	100.00%	53.74%	2652.87	100.00%	39.15%	482.01	100.00%	7.11%	6776.01	100.00%	100.00%	

Source: BRG based on industry interviews. Totals may show slight variance due to rounding. N.B. Includes Kitchen and Sanitary Taps & Mixers

3.4.2 End Use : Historical and Forecasts

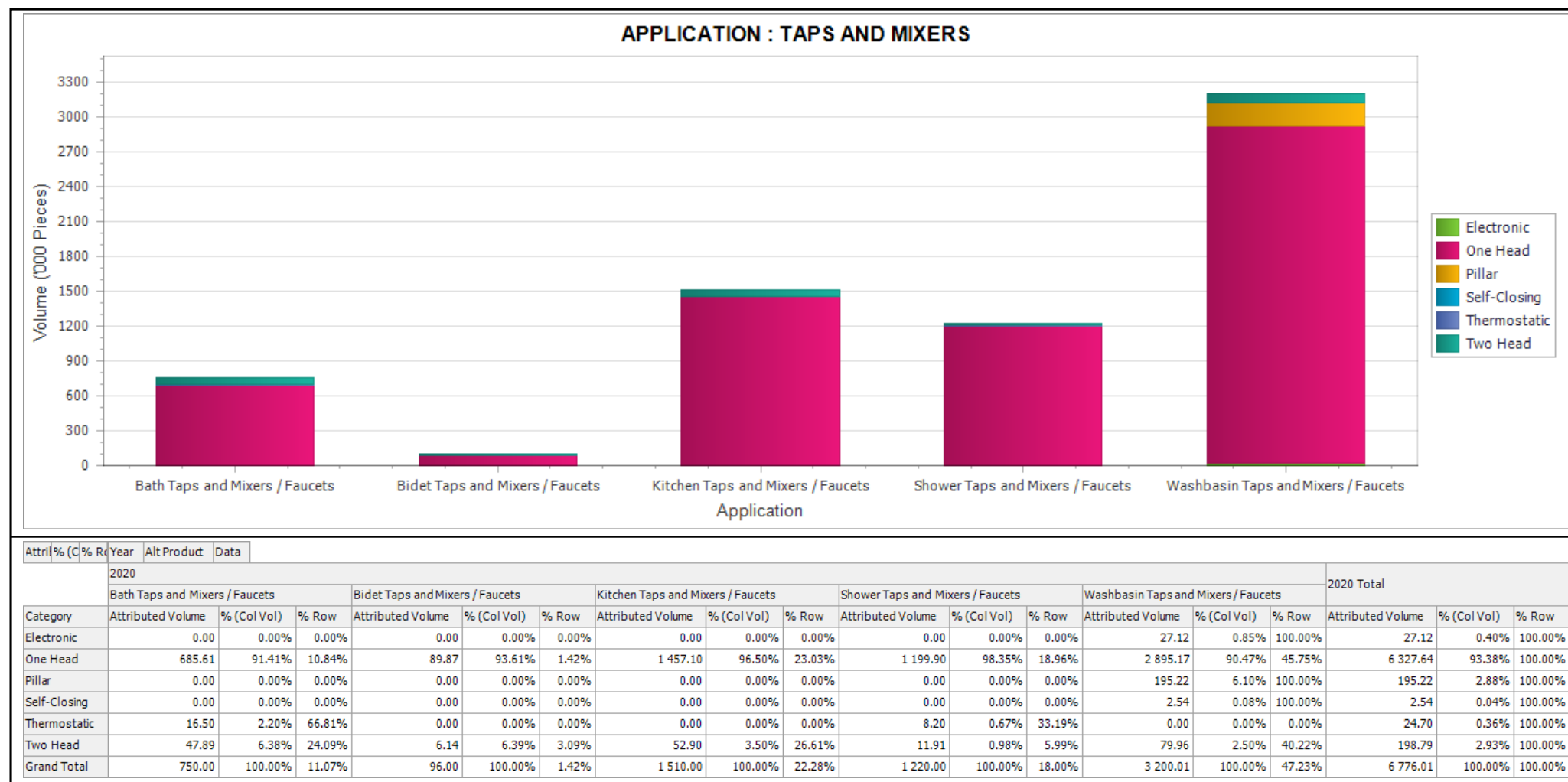
TR - SALES OF TAPS AND MIXERS BY END USE SEGMENT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. N.B. Includes Both Kitchen and Sanitary Taps & Mixers

3.4.3 Sales by Type and Application : Taps and Mixers

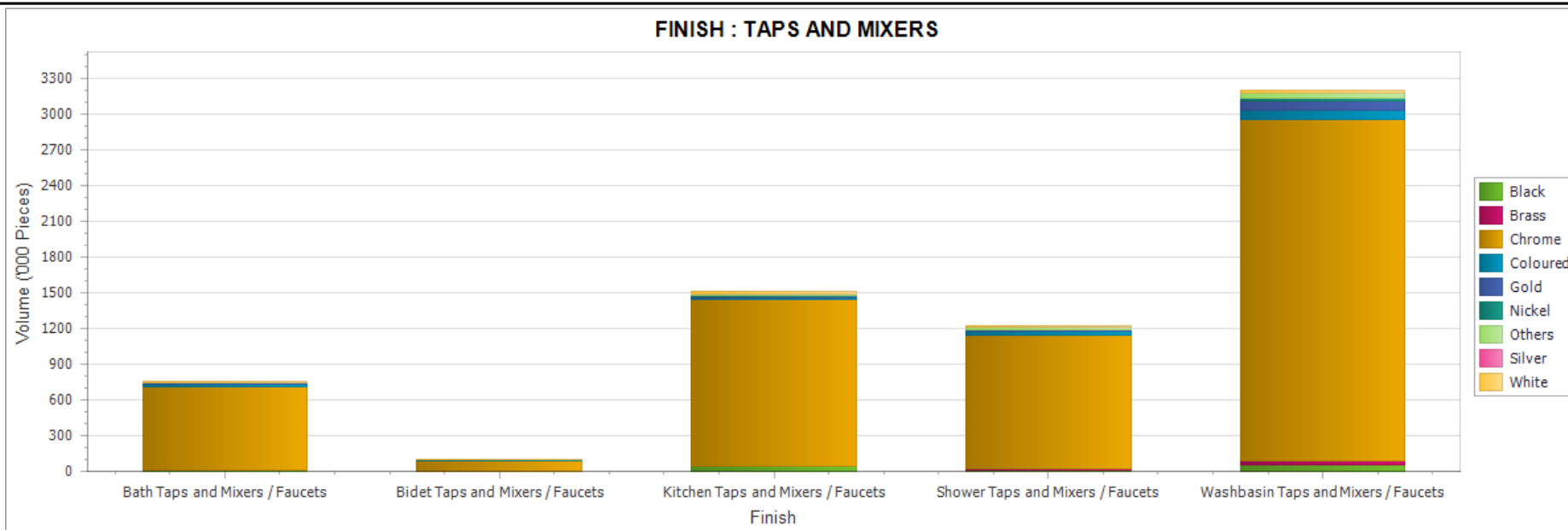
TR - TAP AND MIXER SALES BY TYPE AND APPLICATION ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

3.4.4 Sales by Type of Finish : Taps and Mixers

TR - TAP AND MIXER SALES BY TYPE OF FINISH ('000 PIECES)

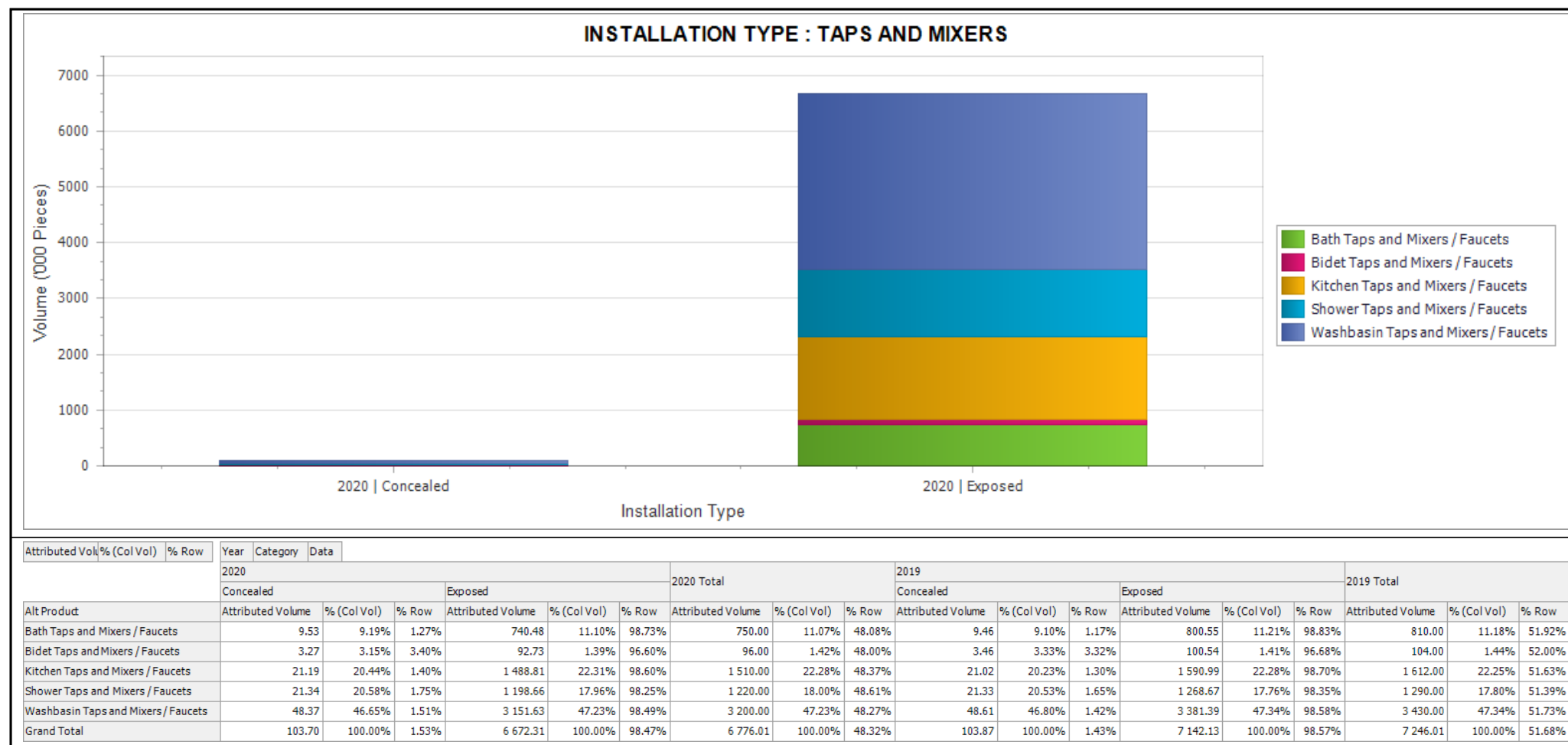


Attri	% (C	% R	Year	Alt Product	Data																			
2020																								
Bath Taps and Mixers / Faucets						Bidet Taps and Mixers / Faucets			Kitchen Taps and Mixers / Faucets			Shower Taps and Mixers / Faucets			Washbasin Taps and Mixers / Faucets			2020 Total						
Category	Attributed Volume	% (Col Vol)	% Row			Attributed Volume	% (Col Vol)	% Row			Attributed Volume	% (Col Vol)	% Row			Attributed Volume	% (Col Vol)	% Row	Attributed Volume	% (Col Vol)	% Row			
Blakd	10.50	1.40%	9.25%			0.48	0.50%	0.42%			40.77	2.70%	35.93%			7.32	0.60%	6.45%	54.40	1.70%	47.94%	113.47	1.67%	100.00%
Brass	5.25	0.70%	8.34%			0.00	0.00%	0.00%			9.06	0.60%	14.40%			13.42	1.10%	21.33%	35.20	1.10%	55.93%	62.93	0.93%	100.00%
Chrome	699.75	93.30%	11.34%			93.89	97.80%	1.52%			1 390.71	92.10%	22.53%			1 123.62	92.10%	18.21%	2 864.00	89.50%	46.40%	6 171.97	91.09%	100.00%
Coloured	18.75	2.50%	12.96%			0.00	0.00%	0.00%			16.61	1.10%	11.48%			29.28	2.40%	20.24%	80.00	2.50%	55.31%	144.64	2.13%	100.00%
Gold	4.50	0.60%	4.82%			0.00	0.00%	0.00%			7.55	0.50%	8.08%			10.98	0.90%	11.75%	70.40	2.20%	75.35%	93.43	1.38%	100.00%
Nickel	4.50	0.60%	8.42%			0.58	0.60%	1.08%			9.06	0.60%	16.95%			7.32	0.60%	13.69%	32.00	1.00%	59.86%	53.46	0.79%	100.00%
Others	0.00	0.00%	0.00%			0.00	0.00%	0.00%			18.12	1.20%	24.62%			17.08	1.40%	23.21%	38.40	1.20%	52.17%	73.60	1.09%	100.00%
Silver	0.00	0.00%	0.00%			0.48	0.50%	100.00%			0.00	0.00%	0.00%			0.00	0.00%	0.00%	0.00	0.00%	0.00%	0.48	0.01%	100.00%
White	6.75	0.90%	10.88%			0.58	0.60%	0.93%			18.12	1.20%	29.21%			10.98	0.90%	17.70%	25.60	0.80%	41.27%	62.03	0.92%	100.00%
Grand Total	750.00	100.00%	11.07%			96.00	100.00%	1.42%			1 510.00	100.00%	22.28%			1 220.00	100.00%	18.00%	3 200.00	100.00%	47.23%	6 776.01	100.00%	100.00%

Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

3.4.5 Sales by Type of Installation : Taps and Mixers

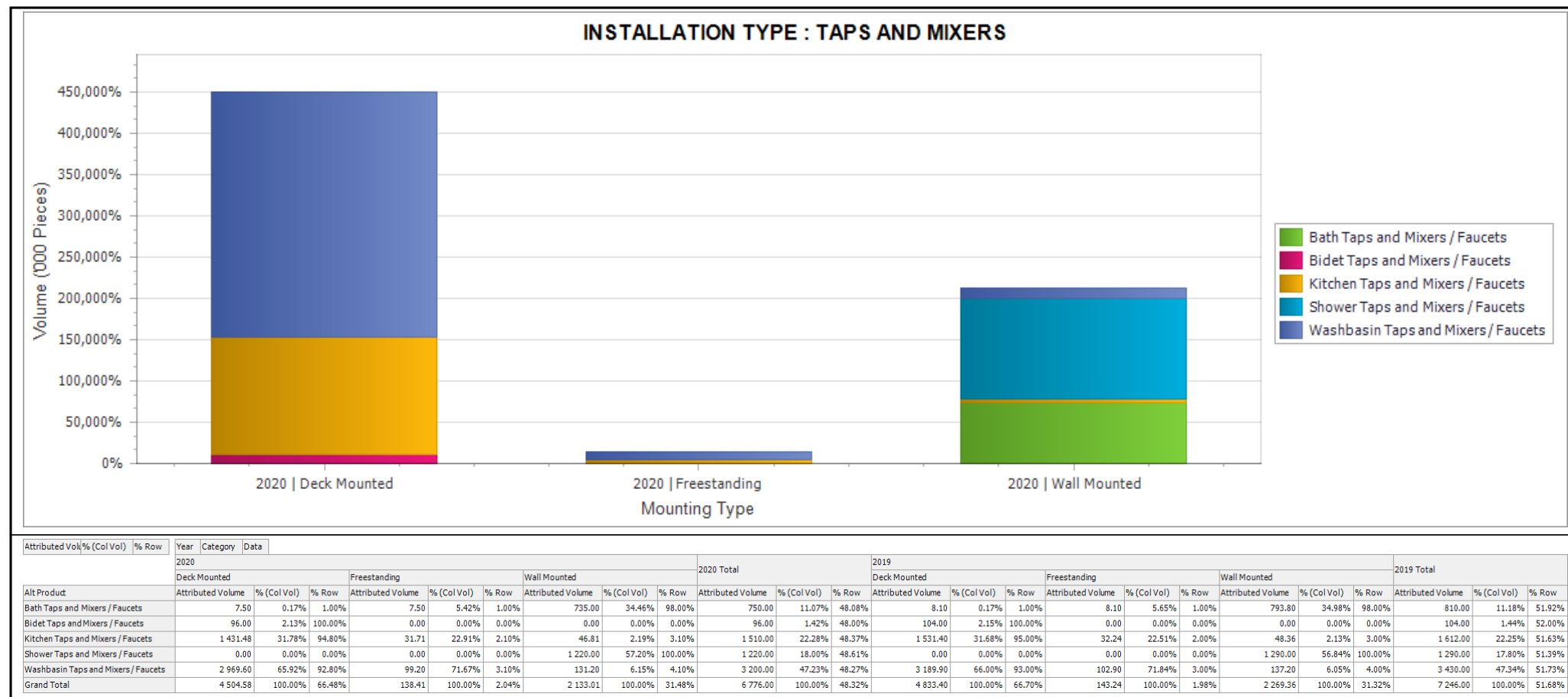
TR - TAP AND MIXER SALES BY INSTALLATION TYPE ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

3.4.6 Sales by Type of Mounting : Taps and Mixers

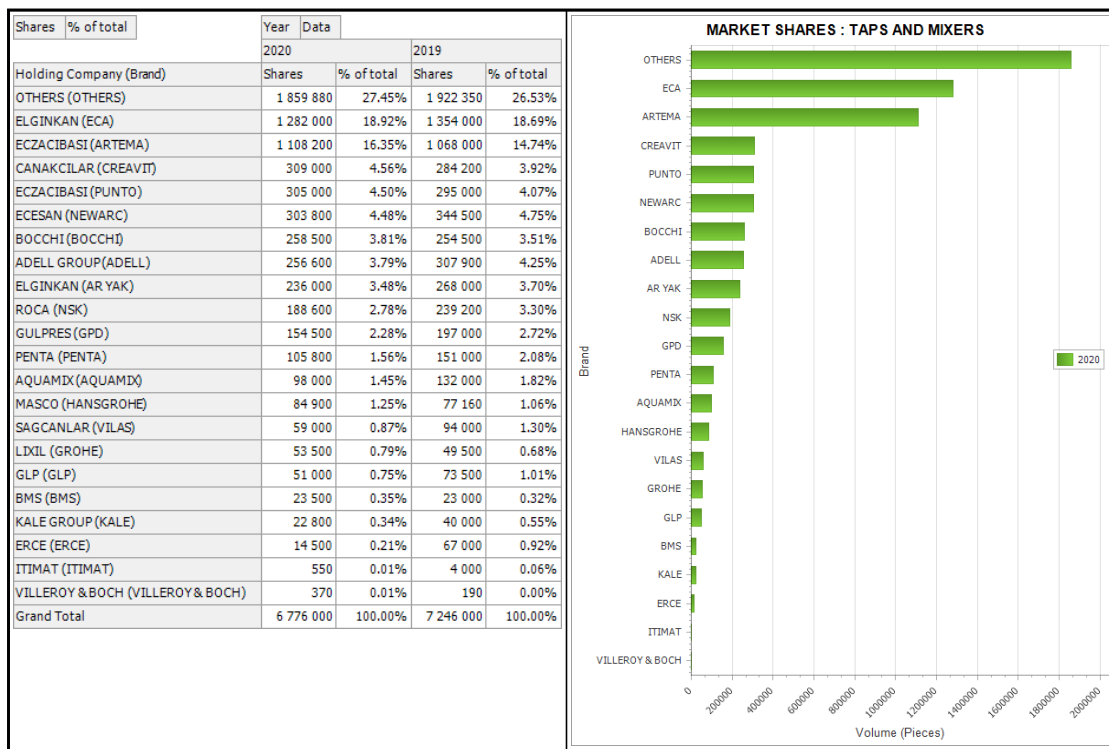
TR - TAP AND MIXER SALES BY TYPE OF MOUNTING ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

3.5 Market Shares : Taps and Mixers

TR - ESTIMATED MARKET SHARES FOR TAPS AND MIXERS



Source: BRG based on industry interviews. N.B. Includes Both Kitchen and Sanitary Taps & Mixers.

3.6.1 TR - IMPORTS OF TAPS & MIXERS 2020

Source Country	2020				2019			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
China	8 126	30.6%	71 709	13.4%	7 171	29.3%	62 476	11.5%
Italy	3 532	13.3%	87 044	16.2%	5 029	20.6%	126 105	23.2%
Germany	2 901	10.9%	104 742	19.5%	2 416	9.9%	84 050	15.5%
Poland	2 035	7.7%	24 916	4.6%	1 735	7.1%	20 210	3.7%
Spain	1 292	4.9%	24 183	4.5%	1 155	4.7%	20 008	3.7%
Czech Republic	950	3.6%	27 062	5.0%	925	3.8%	26 619	4.9%
France	774	2.9%	24 712	4.6%	675	2.8%	23 057	4.2%
Korea, Republic of	763	2.9%	15 822	2.9%	328	1.3%	4 230	0.8%
Netherlands	681	2.6%	11 241	2.1%	625	2.6%	16 872	3.1%
Iran, Islamic Republic of	605	2.3%	781	0.1%	16	0.1%	6	0.0%
Bulgaria	461	1.7%	3 086	0.6%	48	0.2%	447	0.1%
OTHERS	4 458	16.8%	141 511	26.4%	4 317	17.7%	159 510	29.3%
Total	26 578	100.0%	536 809	100.0%	24 440	100.0%	543 590	100.0%

Source : Trademap

3.6.2 TR - EXPORTS OF TAPS & MIXERS 2020

Destination	2020				2019			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
Iraq	4 717	11.5%	31 694	9.4%	3 918	10.0%	31 324	8.8%
Egypt	2 545	6.2%	18 403	5.5%	2 359	6.0%	18 758	5.2%
Russian Federation	2 512	6.1%	19 979	5.9%	1 784	4.6%	12 856	3.6%
Azerbaijan	1 825	4.5%	13 638	4.0%	1 730	4.4%	10 887	3.0%
Germany	1 756	4.3%	20 601	6.1%	2 347	6.0%	23 147	6.5%
Morocco	1 271	3.1%	8 973	2.7%	855	2.2%	8 381	2.3%
Saudi Arabia	1 260	3.1%	8 282	2.5%	1 205	3.1%	12 661	3.5%
Uzbekistan	1 187	2.9%	9 429	2.8%	1 132	2.9%	8 926	2.5%
Libya, State of	1 169	2.9%	7 484	2.2%	1 252	3.2%	9 853	2.8%
Sweden	1 037	2.5%	2 227	0.7%	171	0.4%	1 242	0.3%
Israel	954	2.3%	5 962	1.8%	770	2.0%	5 836	1.6%
OTHERS	20 666	50.5%	190 709	56.5%	21 486	55.1%	213 772	59.8%
Total	40 899	100.0%	337 381	100.0%	39 009	100.0%	357 643	100.0%

Source : Trademap

3.7.1 Route to Market: Taps & Mixers / Faucets
TR - ESTIMATED DISTRIBUTION FOR TAPS & MIXERS / FAUCETS: 2020

Manufacturer / Importer	<table><tr><td colspan="5">Manufacturers / Importers</td></tr><tr><td colspan="5">100.0%</td></tr></table>					Manufacturers / Importers					100.0%																																															
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Wholesale Level	<table><tr><td colspan="5">69.5%</td></tr><tr><td colspan="5">Wholesalers / Merchants</td></tr><tr><td colspan="5">69.5%</td></tr></table>					69.5%					Wholesalers / Merchants					69.5%																																										
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Retail Level	<table><tr><td>Manufacturer</td><td>Wholesaler</td></tr><tr><td>5.6%</td><td>3.5%</td></tr><tr><td colspan="2">Internet Retailers</td></tr><tr><td colspan="2">9.1%</td></tr></table>		Manufacturer	Wholesaler	5.6%	3.5%	Internet Retailers		9.1%		<table><tr><td>Manufacturer</td><td>Wholesaler</td></tr><tr><td>14.4%</td><td>13.6%</td></tr><tr><td colspan="2">DIY Retailers</td></tr><tr><td colspan="2">28.0%</td></tr></table>		Manufacturer	Wholesaler	14.4%	13.6%	DIY Retailers		28.0%		<table><tr><td>Manufacturer</td><td>Wholesaler</td></tr><tr><td>1.2%</td><td>21.0%</td></tr><tr><td colspan="2">Specialist Retailers</td></tr><tr><td colspan="2">22.2%</td></tr></table>		Manufacturer	Wholesaler	1.2%	21.0%	Specialist Retailers		22.2%																													
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Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer																																																					
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	<table><tr><td colspan="3">New Build</td><td colspan="3">Residential RMI</td></tr><tr><td colspan="3">39.2%</td><td colspan="3">53.7%</td></tr><tr><td>Project</td><td colspan="2">Non-Project</td><td>Project</td><td colspan="2">Non-Project</td></tr><tr><td>9.0%</td><td colspan="2">30.2%</td><td>9.6%</td><td colspan="2">44.1%</td></tr></table>					New Build			Residential RMI			39.2%			53.7%			Project	Non-Project		Project	Non-Project		9.0%	30.2%		9.6%	44.1%		<table><tr><td colspan="3">New Build</td><td colspan="3">Non-Residential RMI</td></tr><tr><td colspan="3">2.0%</td><td colspan="3">5.1%</td></tr><tr><td>Project</td><td colspan="2">Non-Project</td><td>Project</td><td colspan="2">Non-Project</td></tr><tr><td>1.4%</td><td colspan="2">0.6%</td><td>1.4%</td><td colspan="2">3.7%</td></tr></table>					New Build			Non-Residential RMI			2.0%			5.1%			Project	Non-Project		Project	Non-Project		1.4%	0.6%		1.4%	3.7%	
New Build			Residential RMI																																																							
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1.4%	0.6%		1.4%	3.7%																																																						

3.7.2 Route to Market: Taps & Mixers / Faucets
TR - ESTIMATED DISTRIBUTION FOR TAPS & MIXERS / FAUCETS: 2025

Manufacturer / Importer	Manufacturers / Importers					
	100.0%					
Wholesale Level	69.0%					
	Wholesalers / Merchants					
	69.0%					
Retail Level	Manufacturer		Wholesaler			
	6.8%		3.7%			
	Internet Retailers					
	10.5%					
	Manufacturer		Wholesaler			
	14.6%		13.7%			
	DIY Retailers					
	28.3%					
	Manufacturer		Wholesaler			
	1.2%		21.0%			
Specialist Retailers						
22.2%						
					Manufacturer	
					Wholesaler	
					Internet	
					DIY	
					Specialist	
					8.4%	
					27.8%	
					3.5%	
					12.7%	
					15.1%	
					Installers	
					67.5%	
End Use Level	Manufacturer		Wholesaler		Internet	
	0.0%		1.0%		6.5%	
				</		

Source: BRG Based on Industry Interviews. N.B. Include both Kitchen and Sanitary Taps & Mixers / Faucets - Totals may vary due to roundings

3.7.3 Route to Market - Consolidated View : Taps and Mixers / Faucets

TR - ESTIMATED DISTRIBUTION FOR TAPS AND MIXERS / FAUCETS : 2020

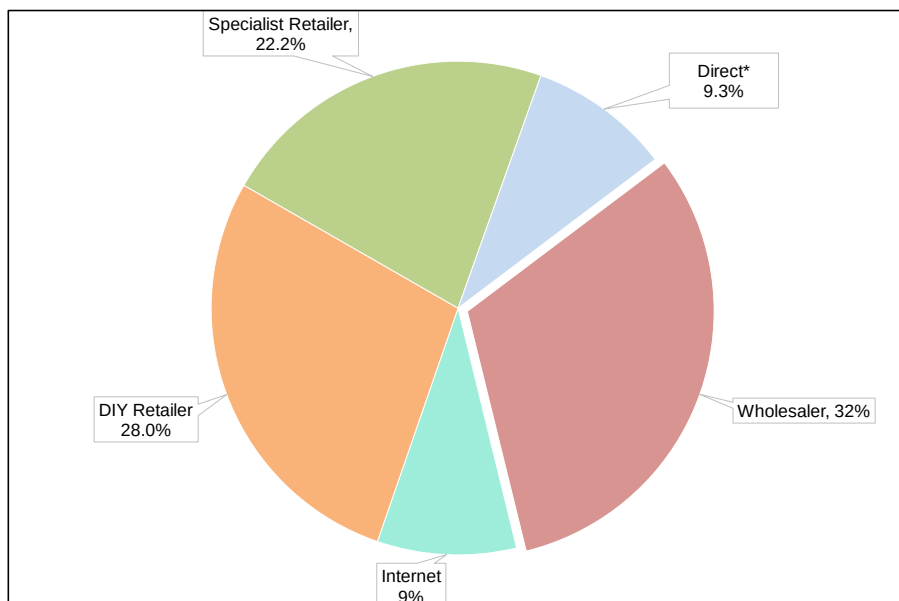
Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
31.4%	9.1%	28.0%	22.2%	9.3%	18.6%	2.8%	74.3%	4.3%
100%					21.4%		78.6%	

3.7.4 Route to Market - Consolidated View : Taps and Mixers / Faucets

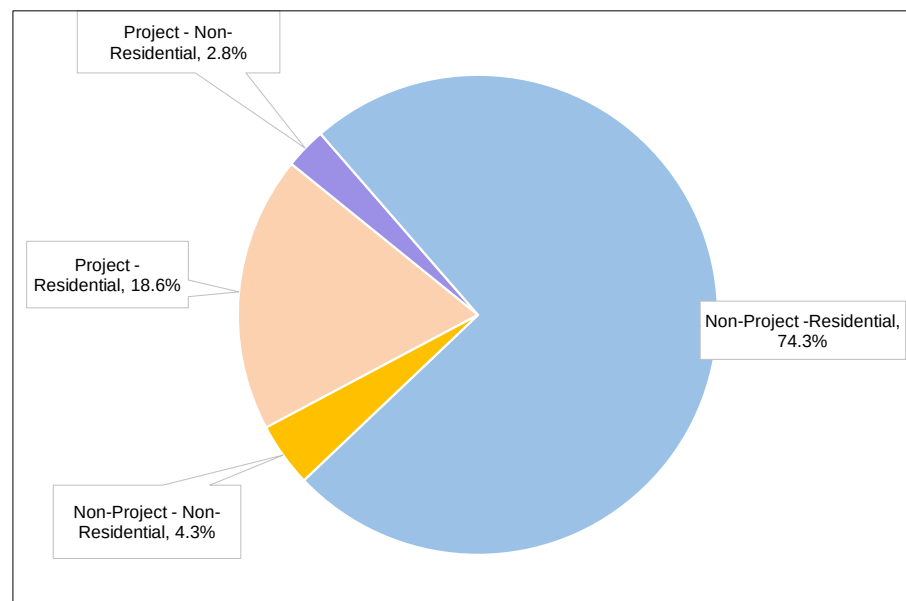
TR - ESTIMATED DISTRIBUTION FOR TAPS AND MIXERS / FAUCETS : 2025

Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
30.6%	10.5%	28.3%	22.2%	8.4%	18.0%	2.7%	75.4%	3.9%
100%					20.7%		79.3%	

TR - SALES OF TAPS AND MIXERS / FAUCETS BY DISTRIBUTION CHANNEL: 2020



TR - SALES OF TAPS AND MIXERS / FAUCETS BY POINT OF DECISION AND END-USE SECTOR: 2020



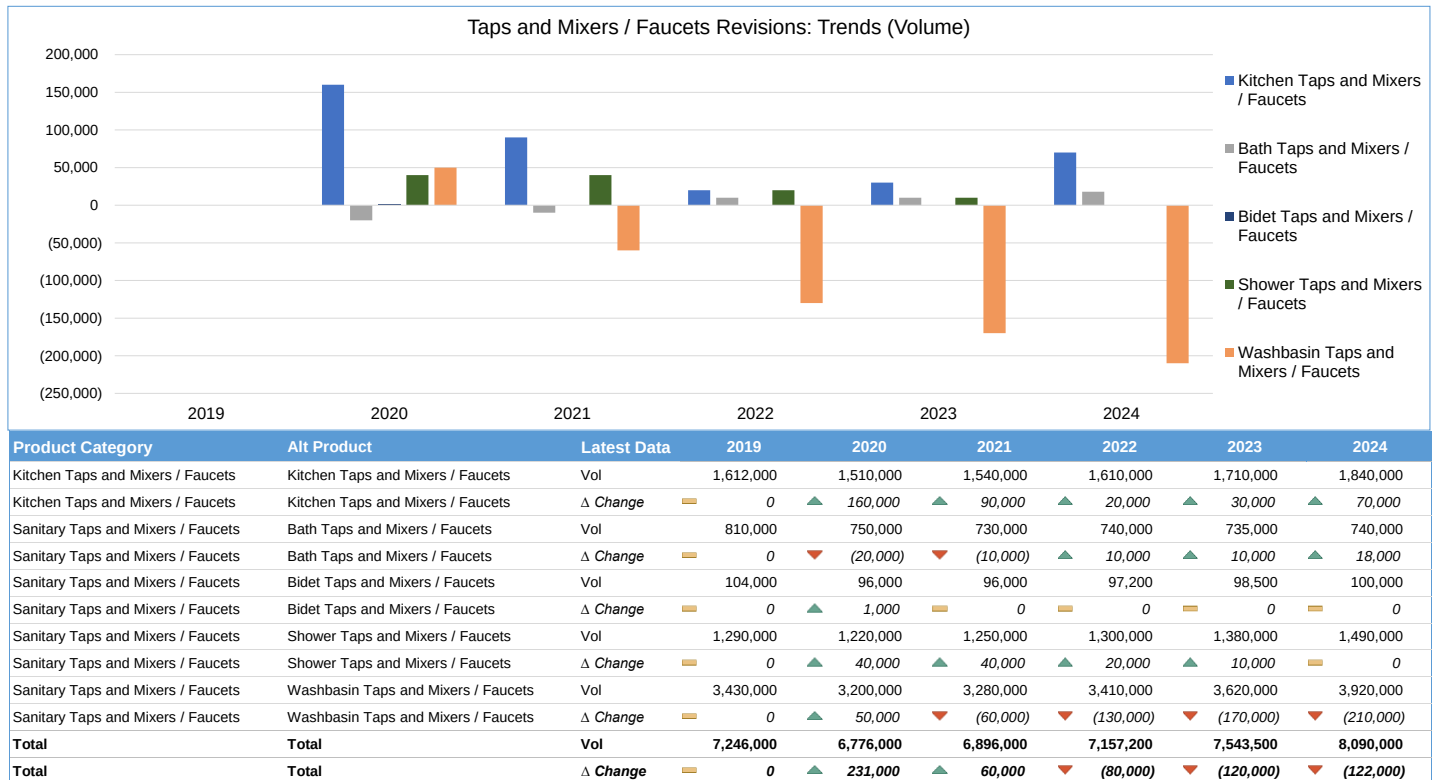
Source: BRG Based on Industry Interviews.

*Direct includes sales going directly from manufacturers to installers, contractors and private customers.

3.8 Revisions in the Turkish Taps and Mixers / Faucets Market (April 2021)

TR - REVISIONS IN THE TURKISH TAPS AND MIXERS / FAUCETS MARKET

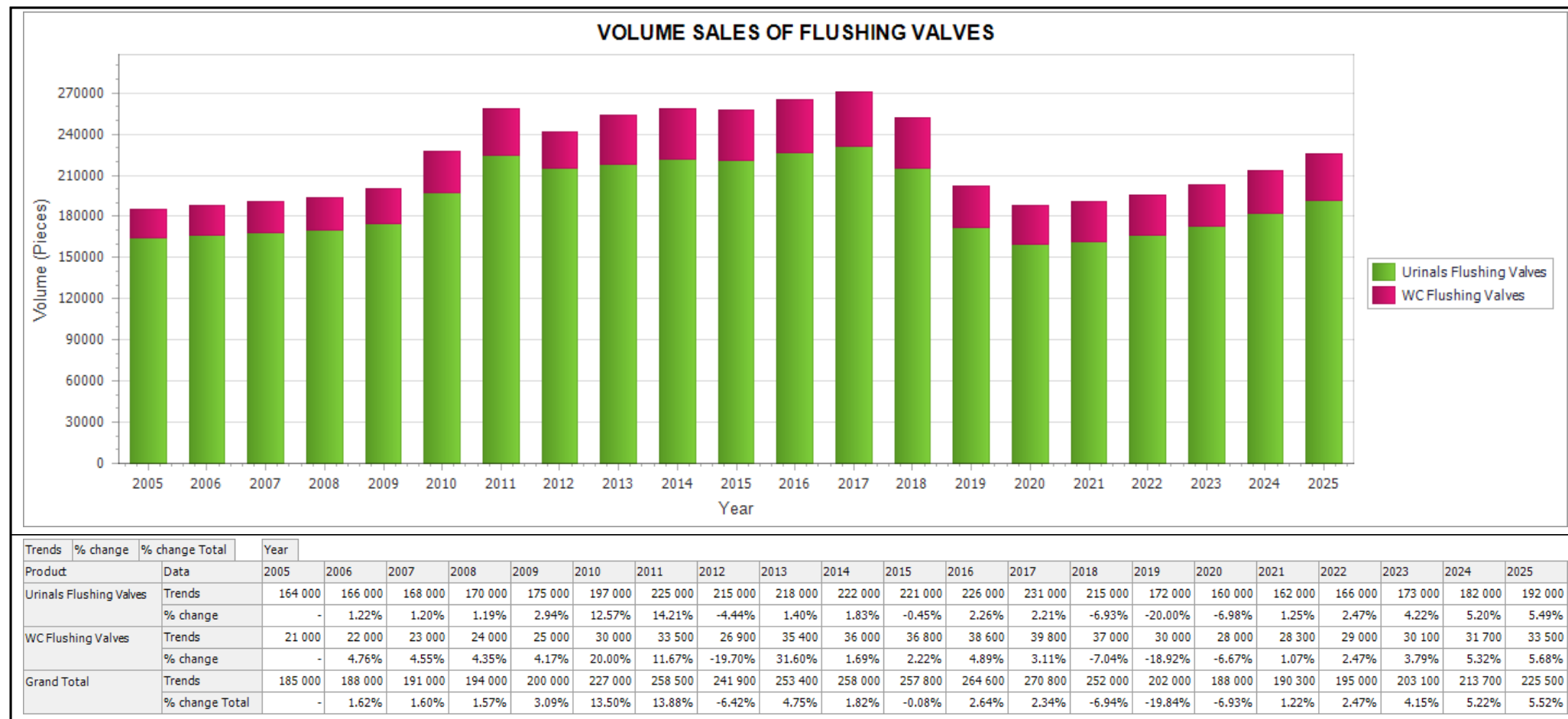
Data Comparison: October 2020



No revisions have been made to historical volumes and values for this market.

3.9.1 Historical Trends and Forecasts : Flushing Valves

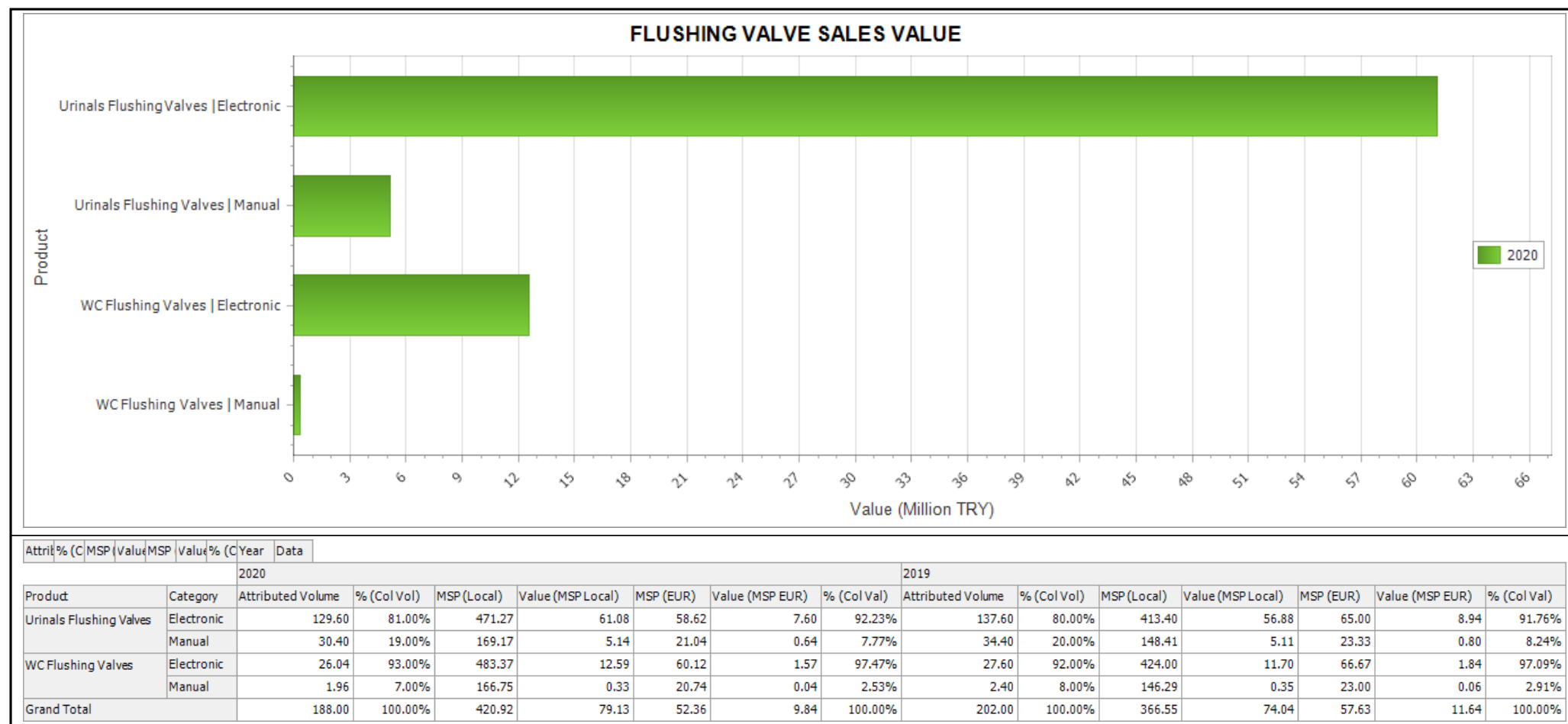
TR - SUMMARY OF TRENDS IN THE FLUSHING VALVE MARKET : HISTORICAL AND FORECASTS



Source: BRG based on industry interviews.

3.9.2 Prices and Market Values : Flushing Valves

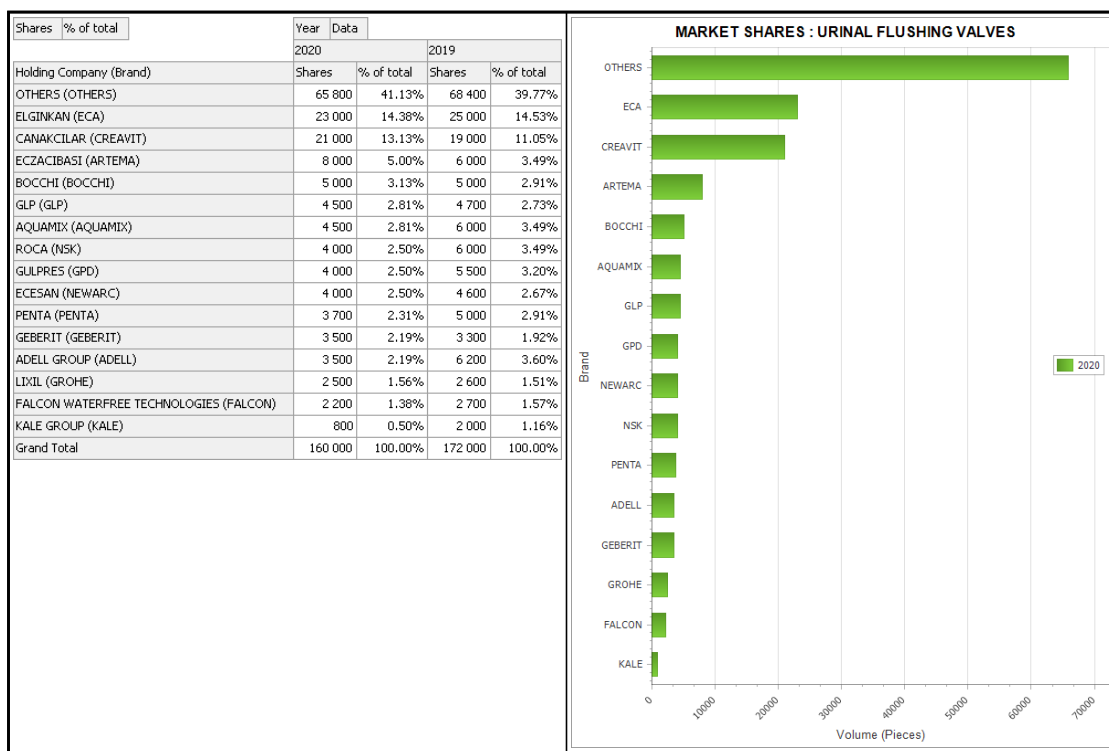
TR - ESTIMATED VALUES OF THE FLUSHING VALVE MARKET



Source: BRG based on industry interviews. Total may show slight variance compared to market data due to rounding. Local Prices Shown in TRY. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price).

3.9.3 Market Shares : Urinal Flushing Valves

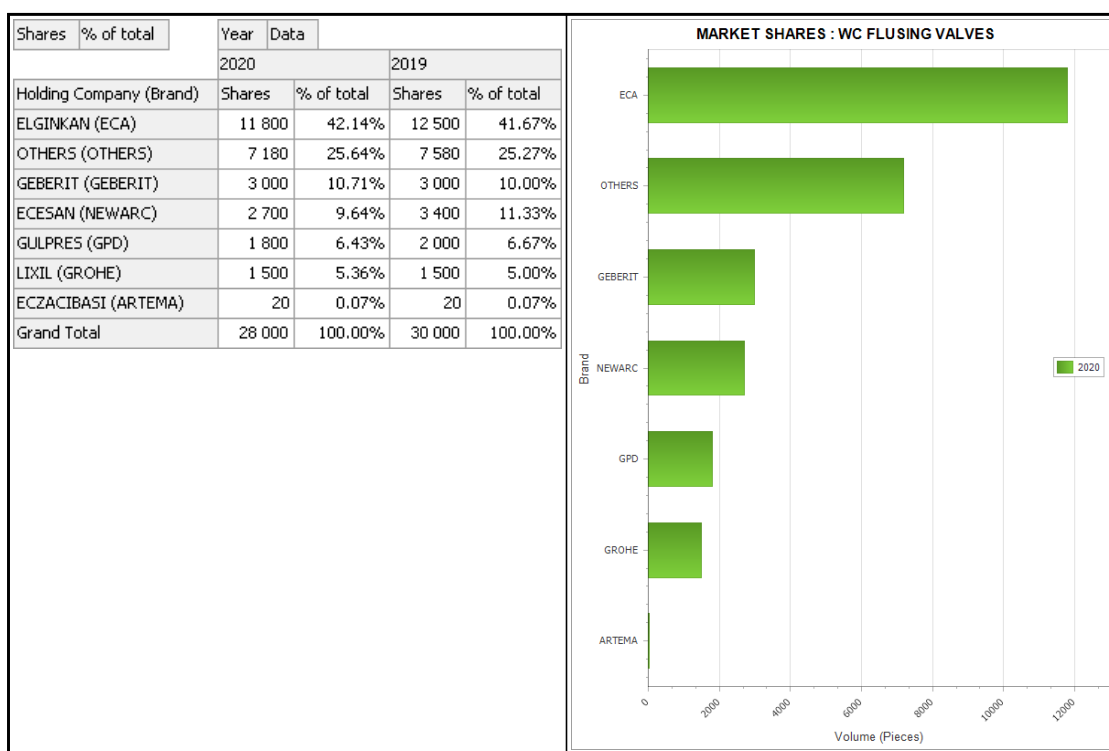
TR - ESTIMATED MARKET SHARES FOR URINAL FLUSHING VALVES



Source: BRG based on industry interviews.

3.9.4 Market Shares : WC Flushing Valves

TR - ESTIMATED MARKET SHARES FOR WC FLUSHING VALVES

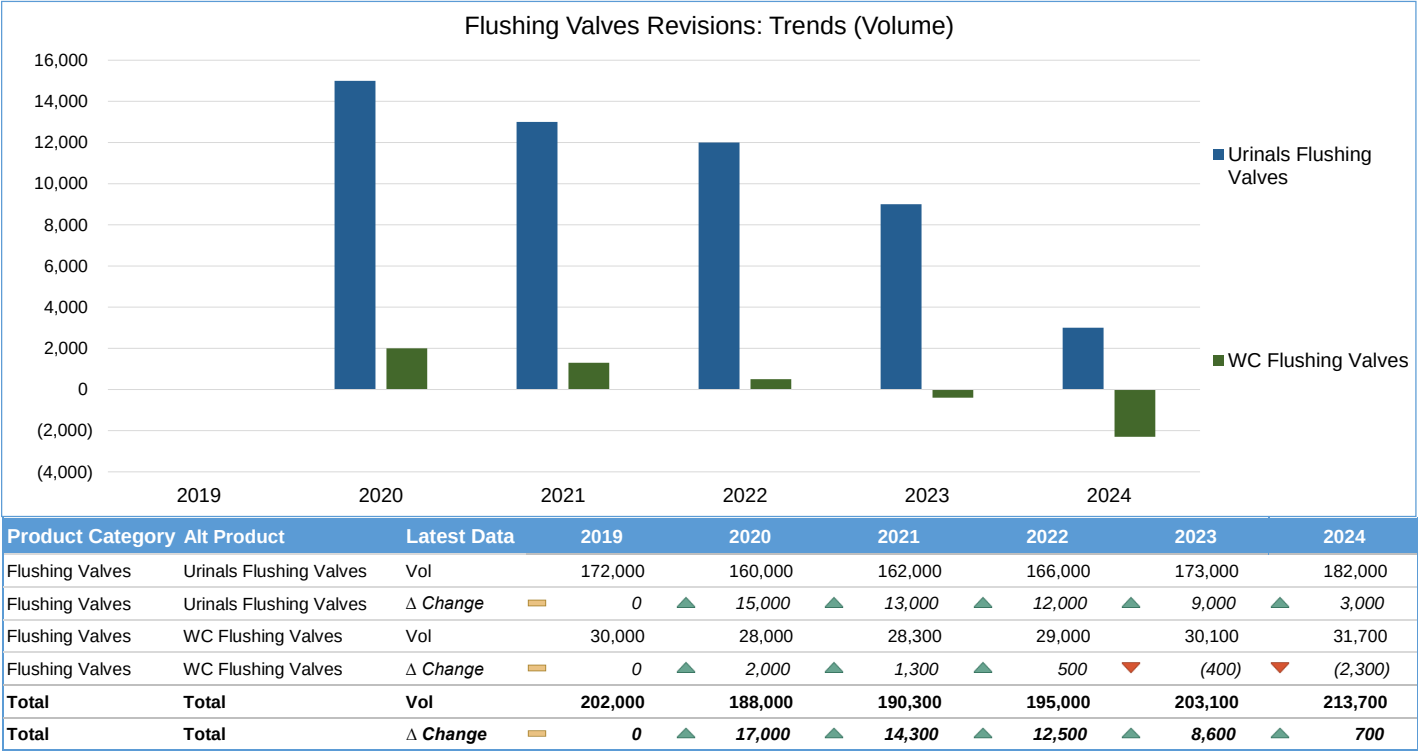


Source: BRG based on industry interviews.

3.10 Revisions in the Turkish Flushing Valves Market (April 2021)

TR - REVISIONS IN THE TURKISH FLUSHING VALVES MARKET

Data Comparison: October 2020



No revisions have been made to historical volumes and values for this market.

TURKEY

SHOWER WALL

THE EUROPEAN BATHROOM & KITCHEN PRODUCT MARKETS



Market Characteristics and Product Mix

The size of the average household has been decreasing for years. As a result, new dwellings are becoming smaller – designed for singles, couples or families with one or two children. As a result, Turkish bathrooms are also losing size to become more suitable for smaller households. Hence, most of the new dwellings contain increasingly only a shower, rather than a bath.

On top of that, the overall Turkish shower wall market has benefitted over the years from the trend towards showering, similarly to other European countries. Showering is becoming significantly more popular than bathing in Turkey. In the new housing market, most apartments are fitted with a shower because of the convenience and space-saving benefits offered by showers rather than baths.

This trend has had negative consequences for sales of baths and, therefore, bath screens. Shower enclosure sales exceeded sales of bath screens for the first time in 2002, and have continued to outperform the bath screen market ever since.

Bath screens entered the Turkish market first, and most of them are installed in the RMI segment. The penetration of bath screens increased significantly from 2000 onwards. Until then, curtains were the typical 'walls' installed with baths. Since big private contractors are offering more dwellings in a very competitive market, bath screens have become essential for baths in new dwellings. Small contractors very often still do not install bath screens, and many new house buyers use basic curtains, which will be replaced with bath screens after some time. However, although bath screens are the oldest shower wall type, there is still potential for market growth, driven mostly by new housing activity.

Shower wall sales were linked strongly to those of shower trays in the past. However, wet shower rooms, using a shower drainage system with very often a tile or stone floor, have recently become very popular in Turkey, just as they have in Western Europe. As a result, shower trays and shower walls are not developing in parallel anymore. Shower trays are decreasing due to the increasing popularity of walk-in showers, but shower walls are growing thanks to the trend towards showers away from baths. Hence, shower wall manufacturers are increasingly introducing an enclosure to suit this new modern theme.

Shower enclosures with one door continue to be the most popular, followed by corner entry models. Sales of five-angle shower enclosures remained low in 2019 again. Shower enclosures are the most popular shower wall type in Turkey, and this is expected to stay the case in the coming years. The thickness of the glass has also recently become a factor of interest to the consumer, with thicknesses ranging from the budget three-millimetre-thick glass to the luxury eight-millimetre-thick models.

Conventional shower cubicles can be considered a niche product in Turkey; sales are low and have remained stable for over 10 years. The vast majority are made out of glass, similarly to shower enclosures.

Within all hydrotherapy products, the biggest increases for many years have been recorded for hydrotherapy shower cubicles. This growth has been driven by Turkey's economic performance and private consumption growth in the past, as well as the increasing desire to save time. Shower cubicles in this market are becoming increasingly sophisticated, with more and more luxury features available such as radios, phones, a larger number of jets, steam shower programmes, and chromotherapy and aromatherapy options. This trend towards more luxury products is a result of the fact that manufacturers are adapting their offers to a fragmented market, flooded by cheap imports.

Recent Market Trends

Overall sales of shower wall products decreased again in 2020, following the decline in 2019. New construction activity remained the most important driver in this market, but due to the weakening of the Turkish currency coupled with increasing inflation and the decreasing purchasing power of the Turkish population, demand for new housing declined strongly. Hence, many contractors could not sell their projects and generate their needed finances. As a result, a significant number of homes could not be completed, and the overall shower wall market volume declined again in 2020.

On top of that, the trends of the shower wall market experienced additionally unexpected developments in 2020, due to the worldwide COVID-19 pandemic. As a result, the overall shower wall market had been forecast to be very negative for 2020, with very strong decline rates. This projection was supported at the beginning of the year, when the COVID-19 pandemic started in the first quarter. The first lockdowns led to some restrictions regarding conventional distribution channels, such as specialists and installers.

However, after the COVID-19 lockdown many Turks escaped from big cities and settled down in rural areas at the seaside and the mountains – very often in holiday homes designed to be used for only several weeks per year, with limited maintenance work over the years. In order to spend more time here, many Turks started, after the ease of the restrictions, to renovate their holiday homes. As a result, overall shower wall sales have been revised upwards for 2020, and the overall decline was not as big as previously forecast.

In terms of product mix, BRG did not see any major changes in 2020, as shower enclosures continued to dominate the shower wall market. The trend towards glass products continued. Sliding and pivoting doors are the most popular door types, followed by side wall doors, while folding doors are the least preferred option. As the Turkish population is relatively young, easy-access and elderly-designed products are not required very often yet, but due to aesthetic reasons, tiled and stoned floors without shower trays were very popular until a few years ago – but

this trend slowed down as Turks had bad experiences, such as leaking water, due to using low-quality materials. Therefore, the share of walk-in showers continued to decline further in 2020.

However, shower enclosure sales decreased further in 2020, mainly due to declining housing completions, but at a lower rate when compared to shower trays due to the significant share of walk-in showers in Turkey.

Sales of bath screens in 2020 declined at a higher rate than shower enclosures, due to the shift towards showering rather than bathing. This negative trend was pushed even more by the above-mentioned economic recession and the crisis in new housing construction industry. Bath sales developed very negatively, although the decline of bath screens was not as strong as baths because not all baths have a screen, but the ratio continues to increase. Product awareness is increasing and the trend away from curtains towards bath screens is still ongoing.

Similarly to other shower wall products, conventional shower cubicle market volume has also declined, still operating as a niche market. In 2020, shower walls and bath screens with ornaments were quite trendy, being boosted by the current hype for ottoman-style products. In 2020, the negative demand trend of hydrotherapy cubicles continued, which was negatively affected by the economic crisis.

Prices and Market Values

In Turkey, the average MSP of shower walls and hydrotherapy cubicles is low when compared to MSPs in other European countries, mainly due to the fact that the market remains very competitive, labour costs are low and there are a high number of domestic manufacturers.

Turkey can still be considered a developing market, with the majority of volume sales taking place in the lower and economy price segments, although this generates significantly less revenue. The core products within the market are shower enclosures/doors, which follow the trend of the overall market and see the majority of sales in the economy and lower segments. Some small shares have been recorded in the middle and upper segments. The same price trends are observed for the remaining shower wall products, with a shifting away from luxury segments towards the lower segments, due to the economic crisis in the country in 2020.

Imports of shower walls and hydrotherapy cubicles play only a very limited role in Turkey, as most products are supplied by domestic manufacturers. However, importing energy and raw materials are the most costly expenditures for Turkish manufacturers, which has forced producers to increase their MSPs for all products. The reason for this is the weakening of the Turkish currency in 2020, which strongly increased inflation. However, the average MSP growth was smaller than the inflation, due to the market being very competitive and declining.

New Housing

Generally, shower enclosures and bath screens are not always installed in new housing at the time of handover, as buyers more often install them when they move into the new home. These installations are still considered as new housing demand by BRG. So-called social housing projects managed by the government, such as TOKI, offer cheap solutions for Turks that often do not include any shower walls in the dwellings. The buyers here also do not see a proper shower wall as an essential product, and continue to use curtains. New dwellings built by private well-known contractors typically include shower walls, due to the very competitive contractor market in Turkey.

Like previously mentioned, the new build market developed negatively in 2020, and therefore demand from new build dwellings suffered accordingly. However, the new housing segment remains the most important source for the overall shower wall and hydrotherapy cubicles markets, although the penetration of wet shower rooms is increasing very strongly in the new build segment in Turkey.

Shower enclosures remain the most commonly chosen product in the new housing segment. Turks gradually prefer shower walls in new housing, and sales of curtains, which were very popular in the past, are slowly declining in this end-use segment. Another reason for the shower wall popularity is the fact that more and more dwellings are equipped only with showers, due to the trend away from baths. Nevertheless, shower enclosure sales decreased in 2020, due to the declining construction activity along with increasing inflation.

The poor performance of the new housing segment also affected negatively sales of other shower wall products in 2020, with bath screens suffering additionally due to the shift away from baths towards showering. However, from 2022 onwards the total market volume of the new housing segment is projected to recover and increase again.

Housing RMI

Shower walls and hydrotherapy cubicles in Turkey are very cheap, and the quality is quite low. A lifespan of around seven years, compared to around 15 for shower trays, has meant that replacement is around twice as often as that for shower trays.

In low-budget new housing projects, customers do not install shower wall products immediately after moving to a new house, but do so after several months. Therefore, considerable sales in the RMI segment are to first-time users who have recently moved to a new house. Moreover, people are more likely to install shower wall products as they are considered fashionable. Thus, major demand for this segment comes from renovation.

With ongoing lockdown periods over 2020, Turks started to spend more time at home and renovate their homes, which was not the case before, due to a lack of

time for many people. At the same, many Turks escaped from big cities to their homes in rural areas. Here they invested in their housing, from essential and/or old facilities to new and modern bathrooms. As a result, housing RMI became a very important driver for the overall bathroom market. Overall sales in this segment declined as a whole in 2020, but this decline was less dramatic than previously forecast.

Non-Housing

Shower walls installed in non-housing remained the smallest market segment in 2020. Most shower walls go into hospitals and hydrotherapy cubicles in hotels. Sales of all shower walls in the non-housing segment decreased only very slightly due to the renovations during the lockdown, especially in the privately held non-residential area. However, installations in this area are predicted to increase in the coming years, thanks to the projected economic recovery.

Supply Structure

The supply structure of the shower wall and hydrotherapy market is always challenging to estimate due to the high fragmentation of the market, especially the big number of small local players that operate only on a regional basis. Hence, those companies are included in the large share of 'others'.

Most shower screens, enclosures and hydrotherapy cubicles are supplied by Turkish manufacturers. There are very low levels of imports recorded, and they mainly come from Italy, Germany and, increasingly, China. The relatively low share of European imports is due to these products being usually high priced and high quality, thus targeting the upper segments of the market. This is the same for most other bathroom products. Most of the market players supply all the different shower walls, as well as baths and shower trays.

SERICO, previously named HYDROCABIN, grew very strongly up until a couple of years ago. The company experienced some financial problems in 2017 and sold its shares to TEKFEN, which also owns one of the biggest DIY chains in Turkey, TEKZEN. HYDROCABIN used to primarily sell its products via Turkey's biggest DIY chain, KOCTAS. After the cooperation with KOCTAS's competitor TEKZEN, KOCTAS stopped selling HYDROCABIN products and so its distribution channels decreased significantly in 2017. However, in 2018, KOCTAS added HYDROCABIN back to its portfolio. In 2020, HYDROCABIN restructured its business and created the SERICO brand, which could be a very large supplier in the coming years.

DOCCE, EREL, and CORIN all recorded some financial challenges in 2020, due to the competitive and decreasing market. STINA and DUSAKABINCIM are selling a significant number of products via the internet.

Forecasts

In the first four months of 2021, bath screen sales continued to decline. The ongoing challenge of this market, which is shifting away from baths towards showering, has intensified and continued to negatively affect overall sales. On top of this, the number of new housing completions, which is the most important market driver for bath screen sales, is expected to decline further in 2021.

However, the ongoing COVID-19 pandemic, along with the growing replacements, continued in the first quarter of 2021 and affected the total shower wall sales positively. As a result, the decline rate of the bath screen sales slowed down between January and April 2021. On the other hand, the shift away from baths towards showering continued, which supports the shower wall sales. Driven by the strong replacement activity, shower enclosure sales stopped declining and started to grow again in the first quarter of 2021. Hydrotherapy shower cubicles also benefitted from this trend, whereas conventional shower cubicle sales, operating in niche markets, declined further.

The mid-term outlook for the Turkish shower wall market is very similar to the first quarter of 2021. The substitution trend from baths to showers should still be prevalent and is unlikely to slow down, given that the factors for the trend are not set to reverse anytime soon, such as smaller living spaces, busier lives and an ageing population. Hence, sales of shower enclosures are expected to account for the largest share of the shower wall market, being followed by bath screens – whereas conventional shower cubicles should stay a niche market. Generally, shower wall demand is expected to be higher than that for bath screens, as the shift away from bathing towards showering is predicted to continue in the coming years.

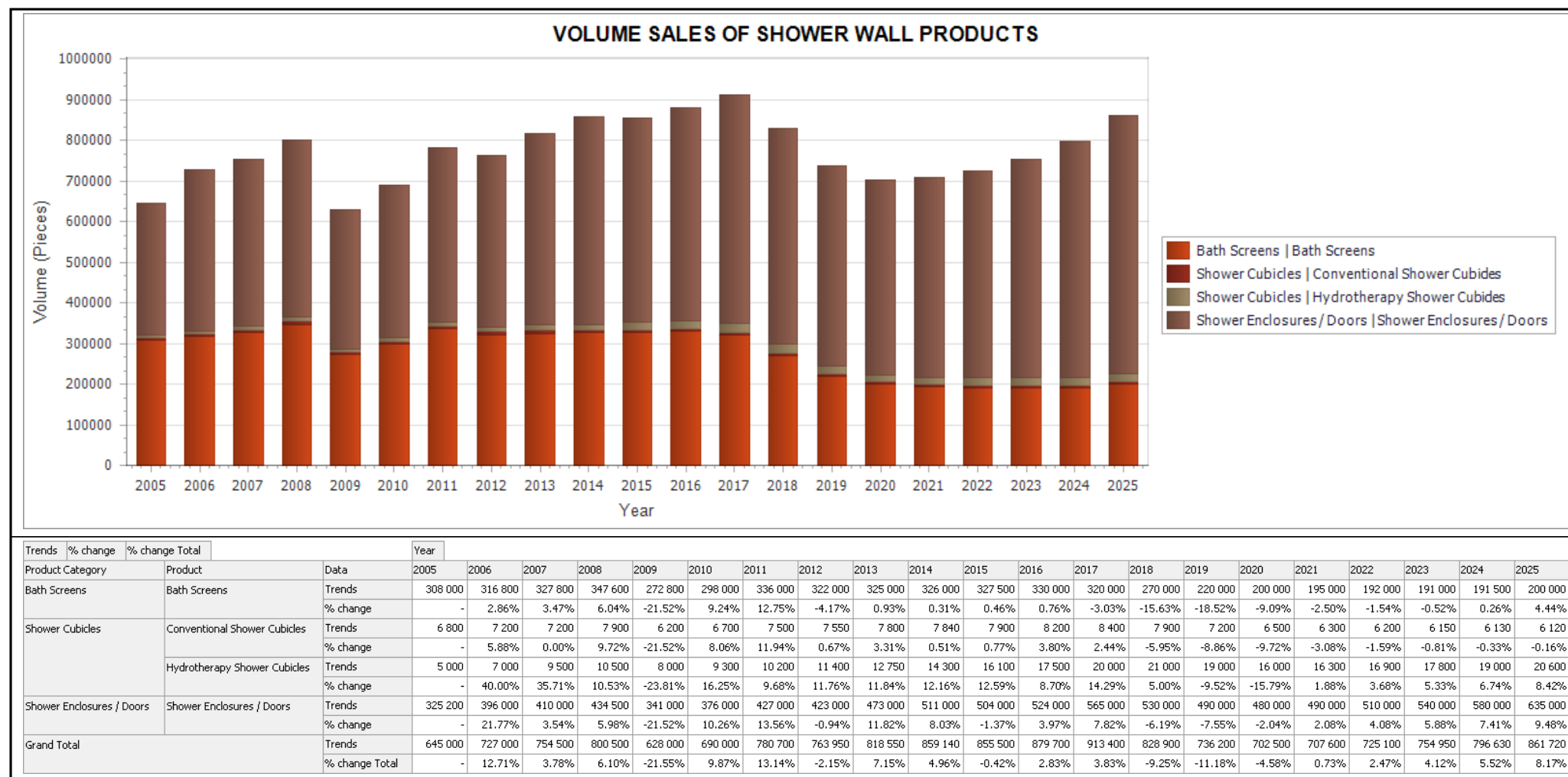
The medium-term outlook regarding the economic environment seems to be positive, which should positively influence the shower wall market. Private investments in new housing, as well as the Urban Transformation Plan and the TOKI projects, should foster overall sales of shower walls, and steady growth is predicted through the forecast period. The importance of the housing RMI segment is likely to grow further over the forecast period as the first shower walls, which were installed in the last decade, already require gradual replacement. Like other low-quality bathroom products, shower walls in Turkey have shorter lifespans and need to be replaced more frequently. Hence, the RMI segment is becoming a more and more important driver for the shower wall market.

Political instability could affect the economy and construction output negatively, which is likely to lead to a decrease in the overall shower wall market. The large new housing share leaves the sector very sensitive to construction activity. In addition, the market is very sensitive to the political and economic situation. A stable political environment leads to consumer confidence and higher spending (which translates into replacement, renovation and construction, leading to more installations); however, during periods of political/economic instability, people tend

to spend less, causing a decrease in shower enclosure sales. The purchasing power of the general population remains lower than in most Western European countries. As a result, not everyone can afford shower enclosure installations, particularly people in rural areas. Therefore, economic growth and the increasing purchasing power of households should positively affect sales.

4.2 Historical Trends and Forecasts : Shower Wall

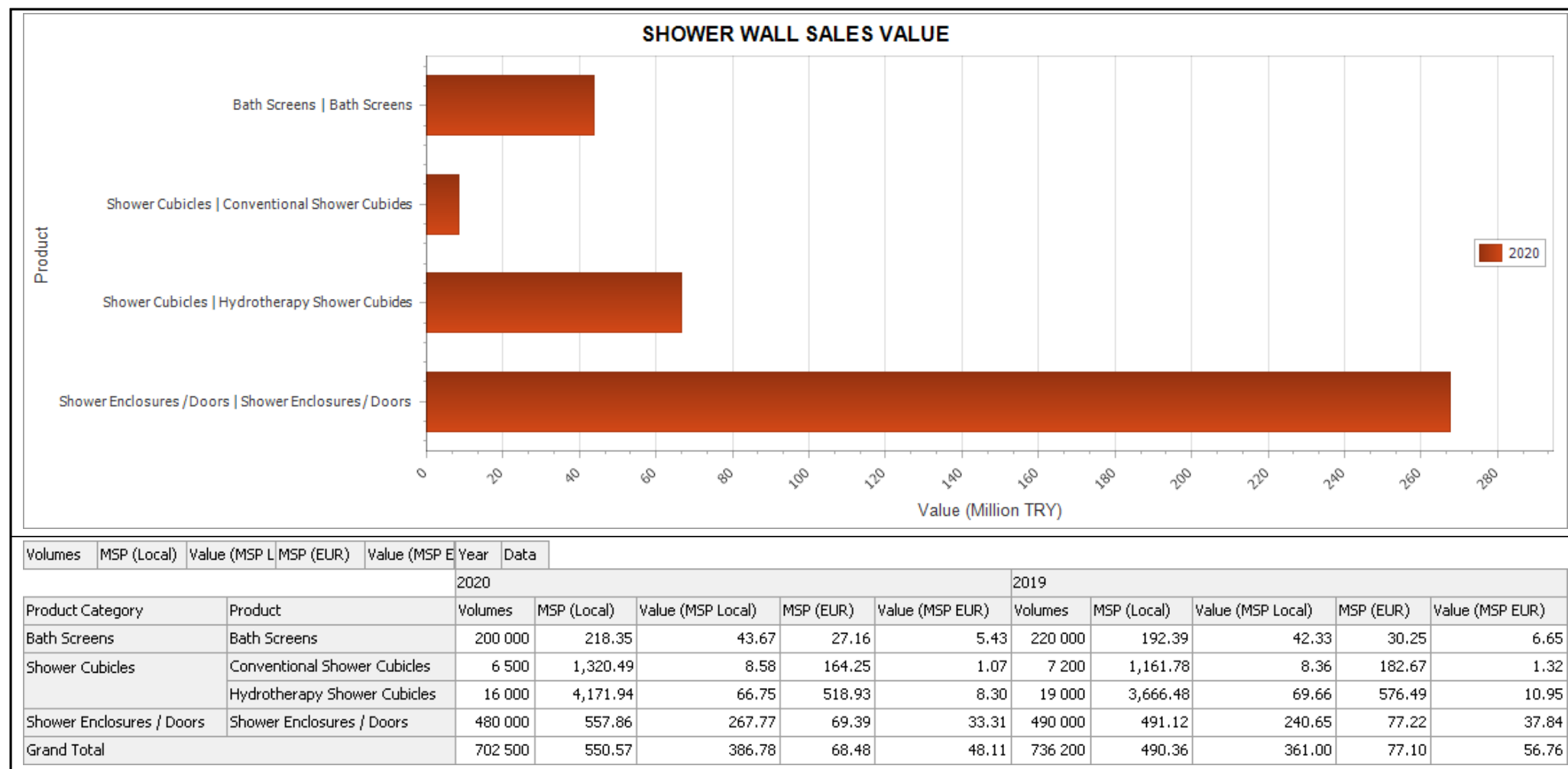
TR - SUMMARY OF TRENDS IN THE SHOWER WALL MARKET : HISTORICAL AND FORECASTS



Source: BRG based on industry interviews.

4.3.1 Prices and Market Values : Shower Wall

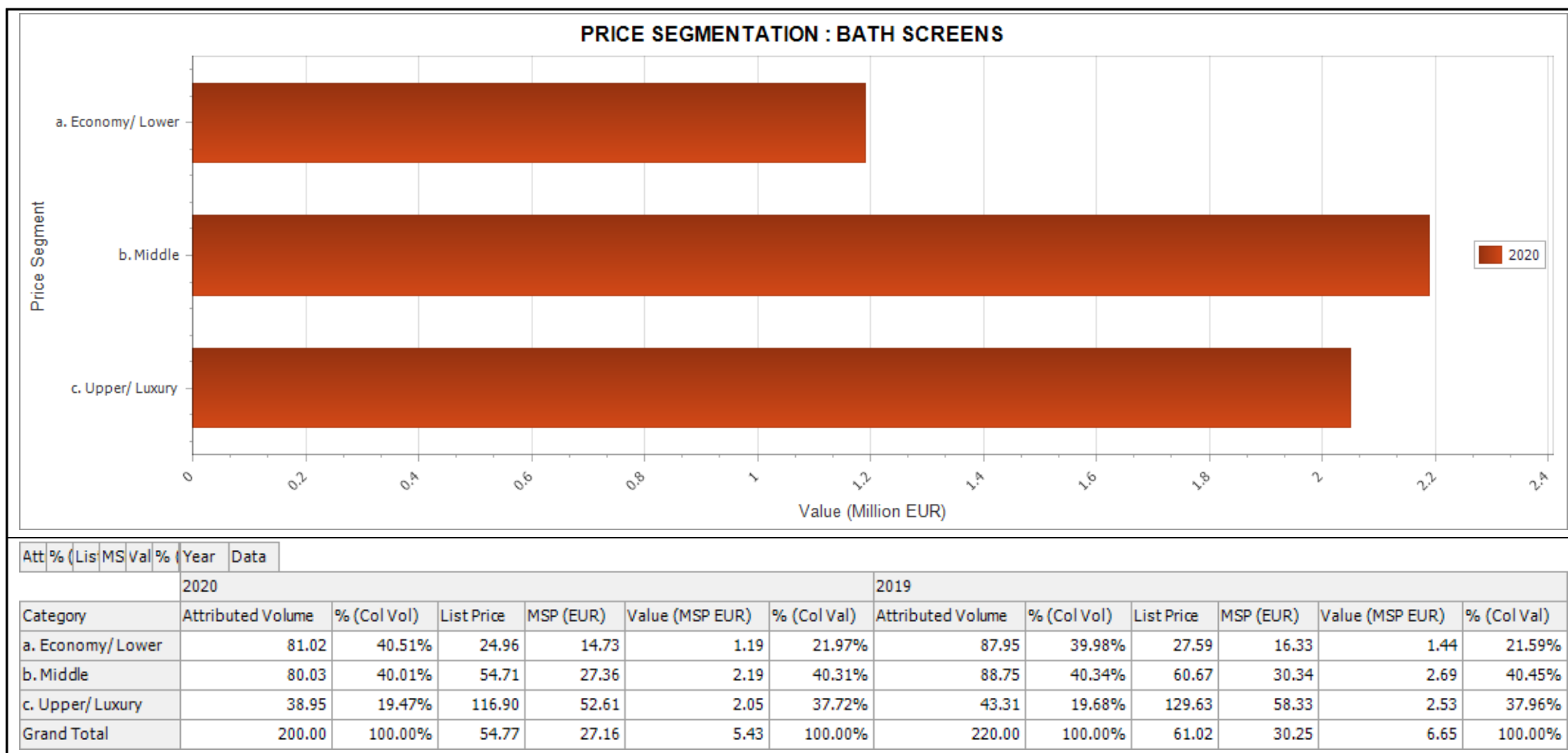
TR - ESTIMATED VALUES OF THE SHOWER WALL MARKET



Source: BRG based on industry interviews. Local Prices Shown in TRY. Values Shown in Millions (at MSP = Manufacturers' Selling Price)

4.3.2 Price Segmentation : Bath Screens

TR - ESTIMATED PRICE SEGMENTATION FOR BATH SCREENS

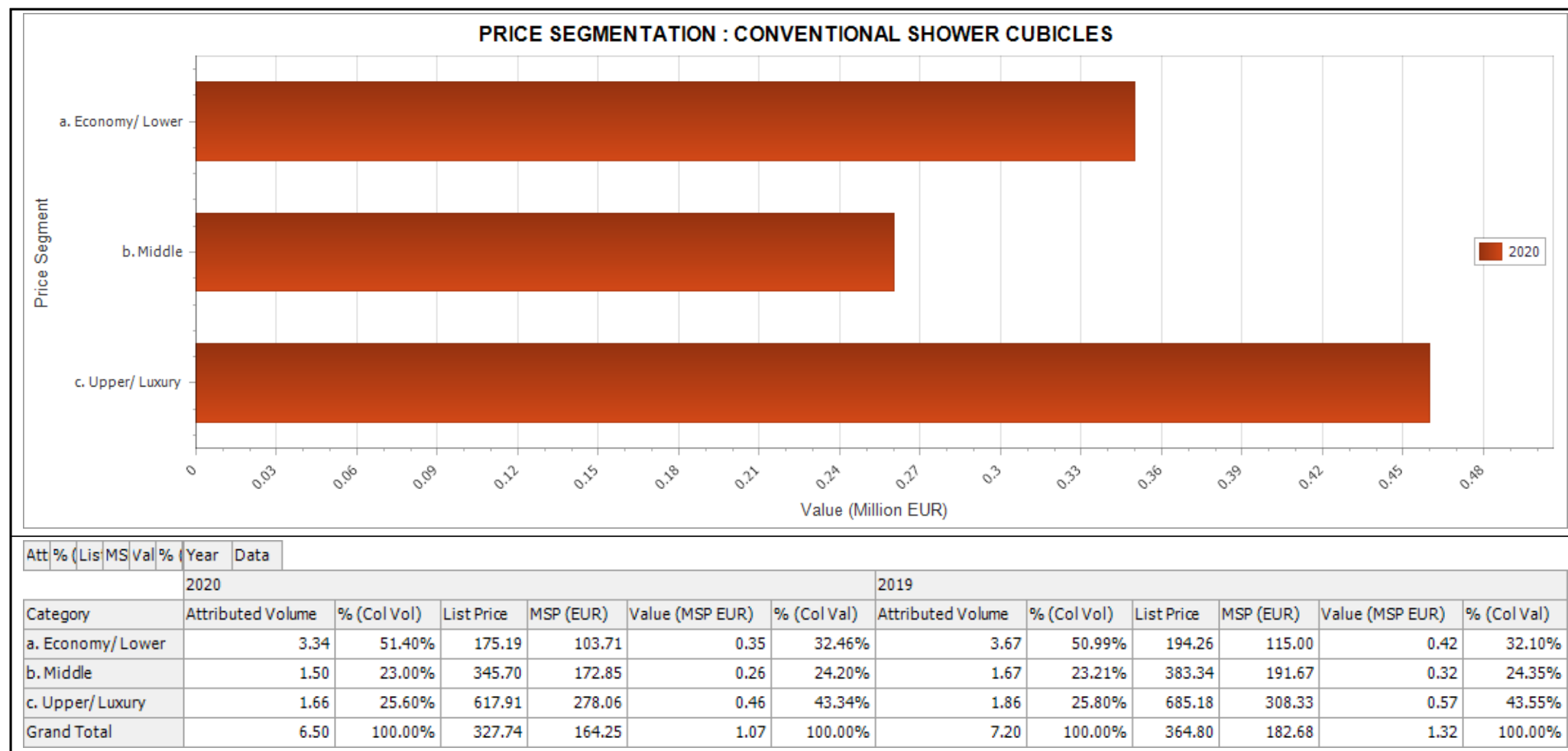


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€20, Middle €20-€40, Upper/ Luxury >€40

4.3.3 Price Segmentation : Conventional Shower Cubicles

TR - ESTIMATED PRICE SEGMENTATION FOR CONVENTIONAL SHOWER CUBICLES

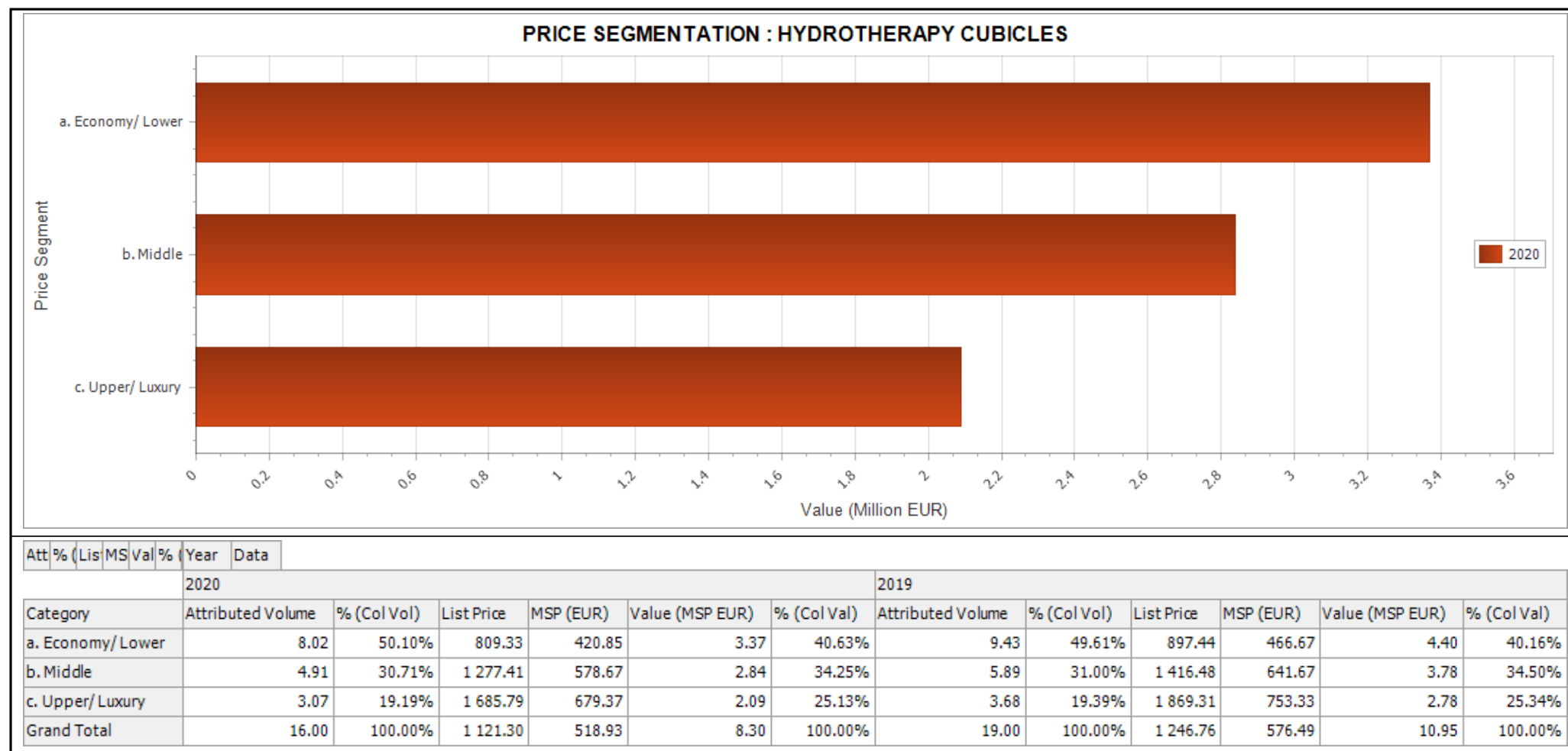


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€125, Middle €125-€250, Upper/ Luxury >€250

4.3.4 Price Segmentation : Hydrotherapy Shower Cubicles

TR - ESTIMATED PRICE SEGMENTATION FOR HYDROTHERAPY CUBICLES

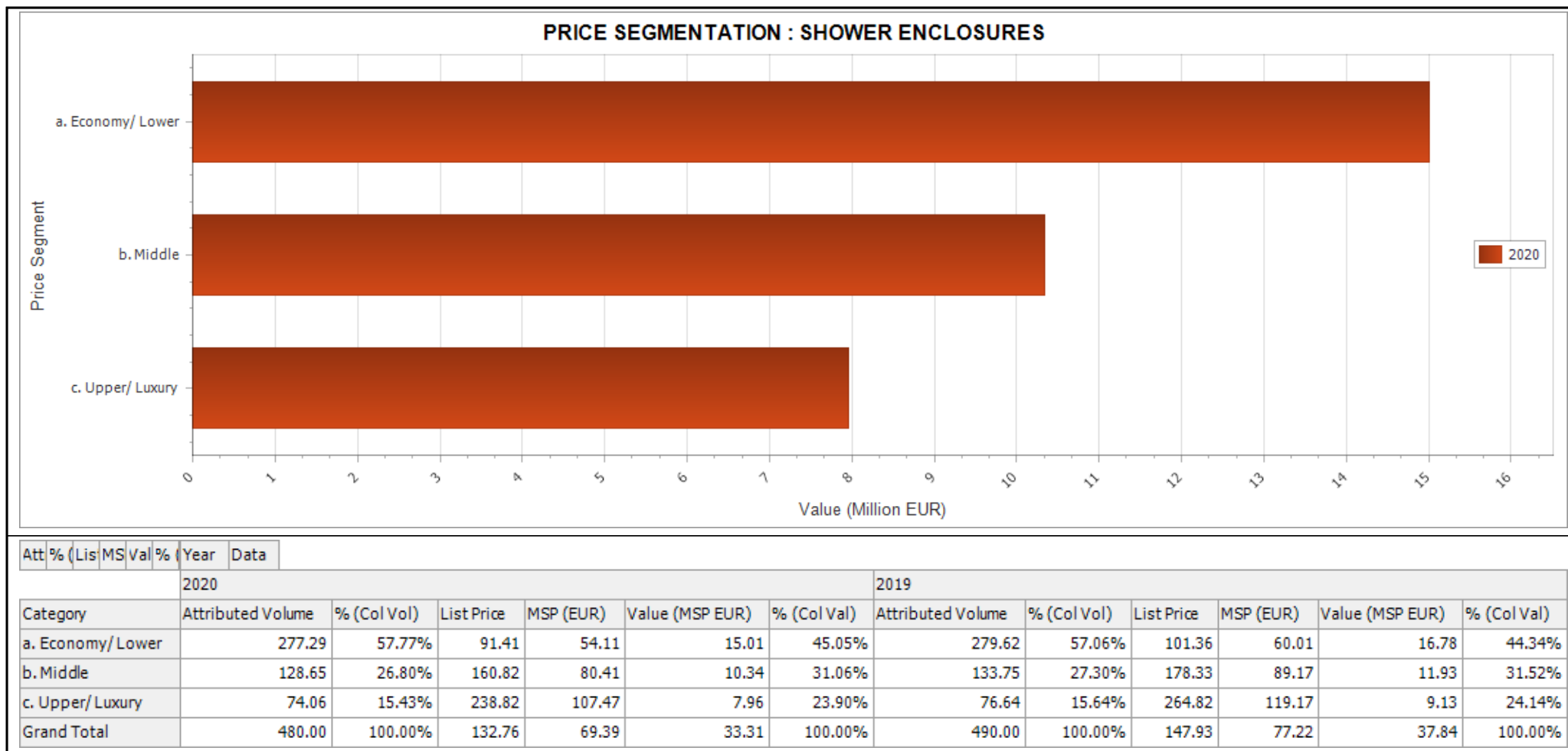


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€500, Middle€500-€650, Upper/ Luxury >€650

4.3.5 Price Segmentation : Shower Enclosures

TR - ESTIMATED PRICE SEGMENTATION FOR SHOWER ENCLOSURES

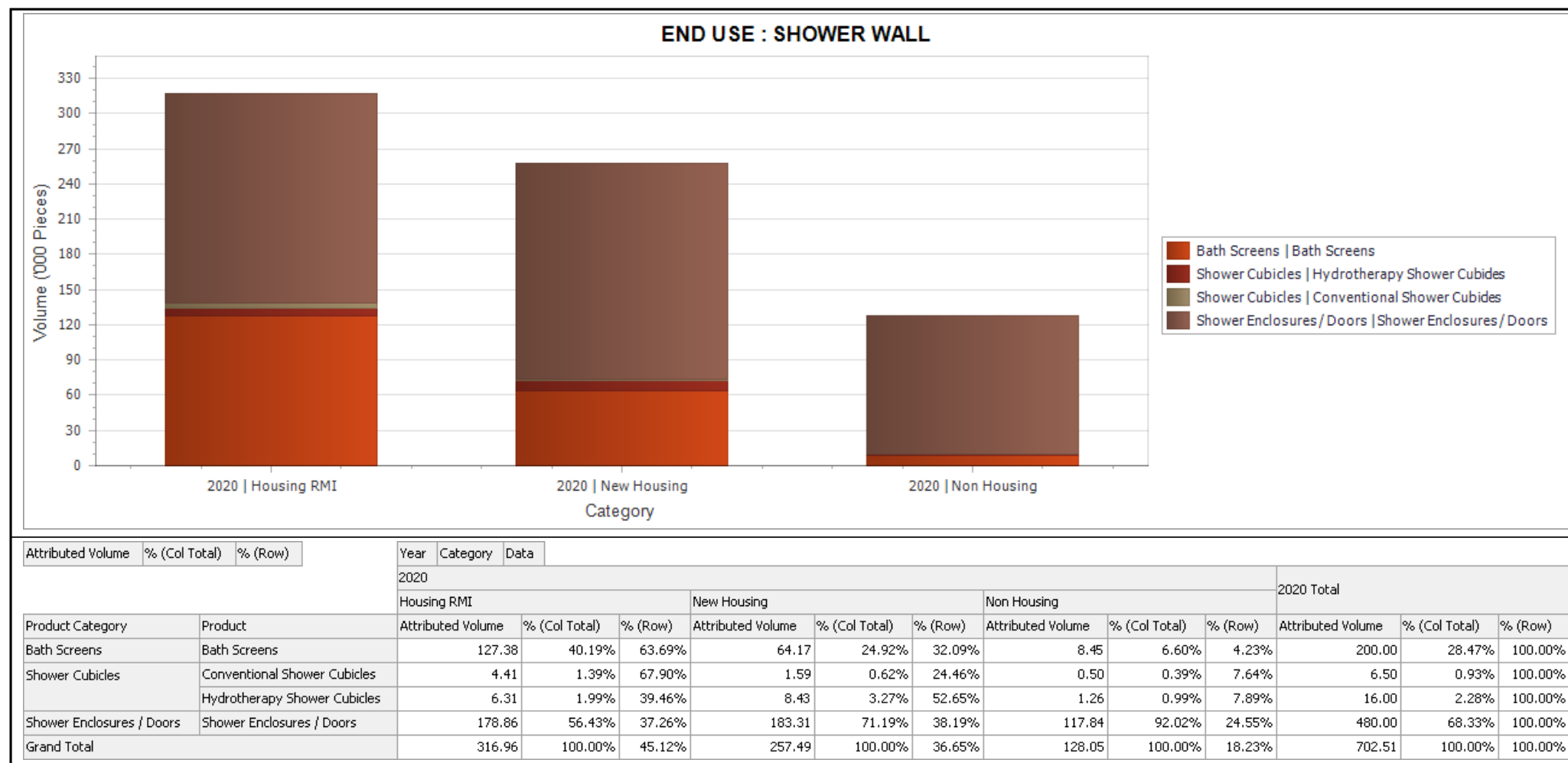


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€70, Middle €70-€100, Upper/ Luxury >€100

4.4.1 End Use : Base Year

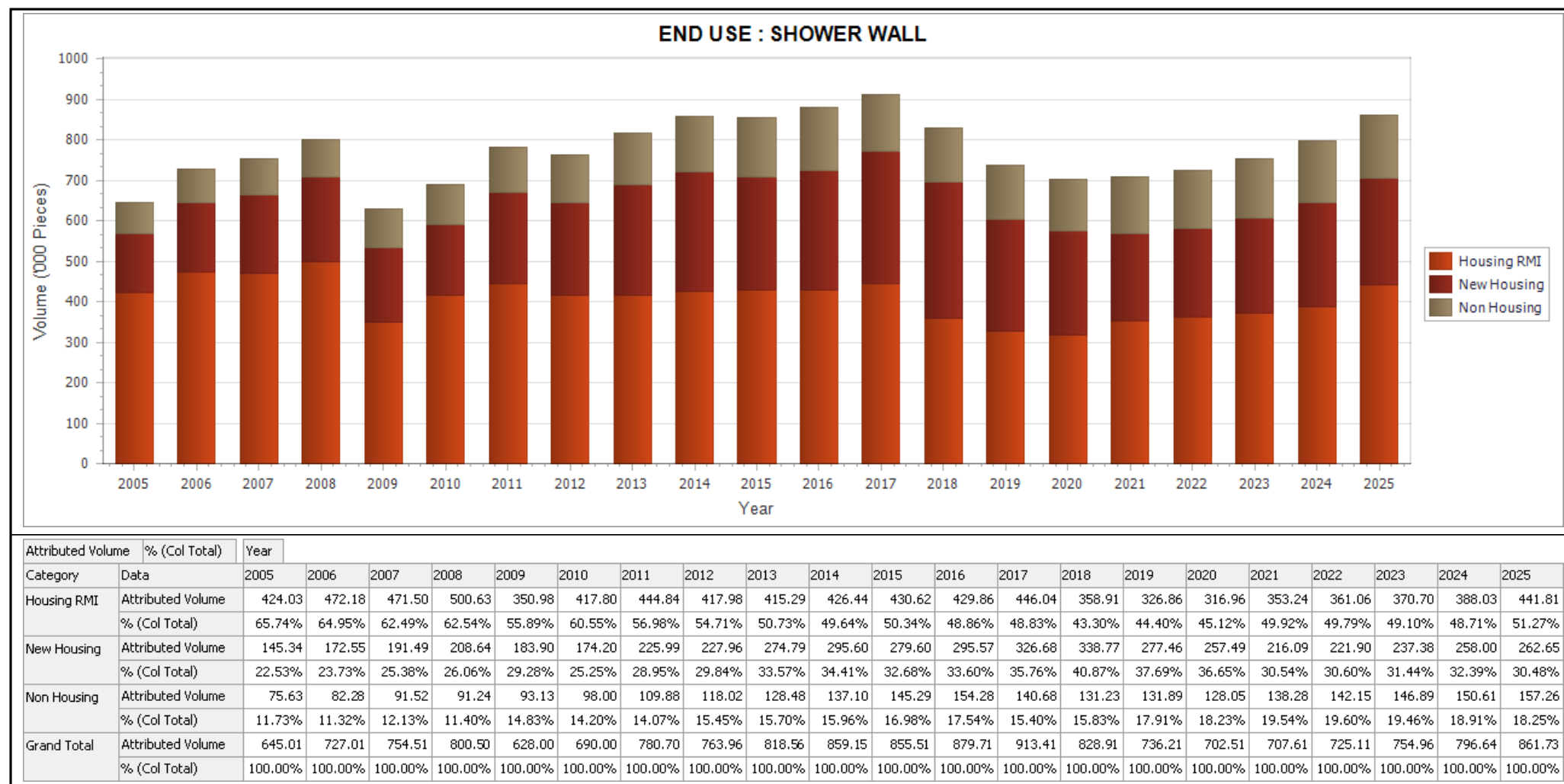
TR - SALES OF SHOWER WALL BY TYPE AND END USE SEGMENT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

4.4.2 End Use : Historical and Forecasts

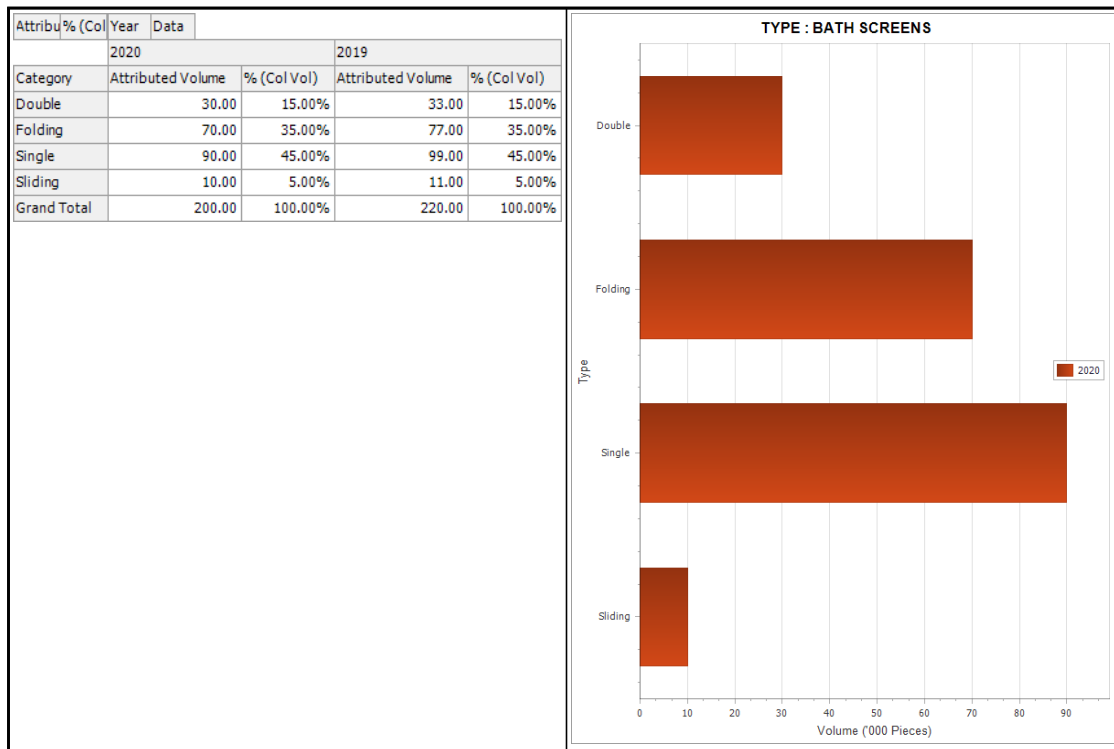
TR - SALES OF SHOWER WALL PRODUCTS BY END USE SEGMENT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

4.4.3 Sales by Type : Bath Screens

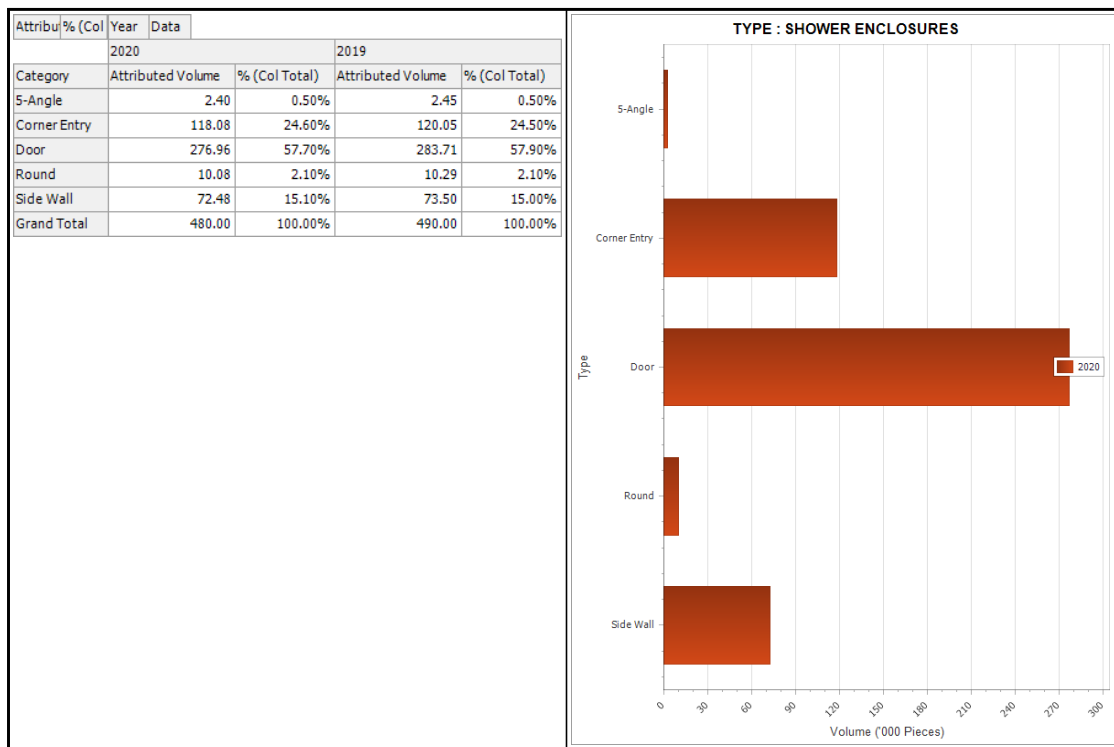
TR - BATH SCREEN SALES BY TYPE ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

4.4.4 Sales by Type : Shower Enclosures

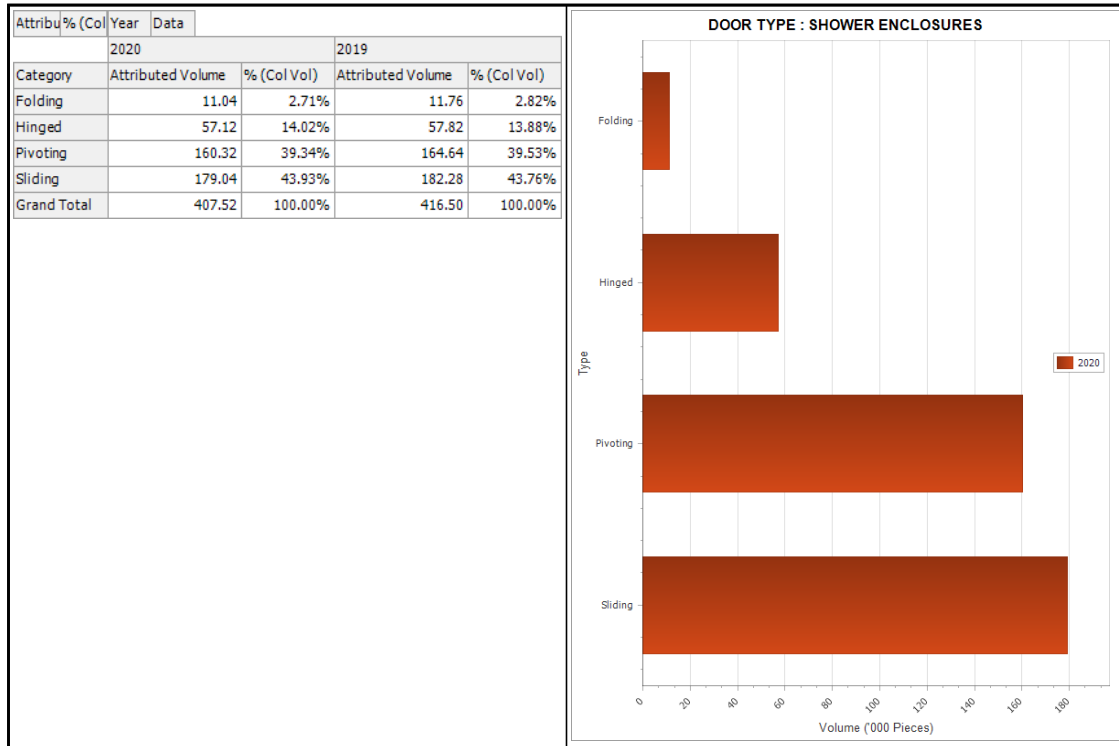
TR - SHOWER ENCLOSURE SALES BY TYPE ('000 PIECES)



Source: BRG based on industry interviews

4.4.5 Sales by Type of Door : Shower Enclosures

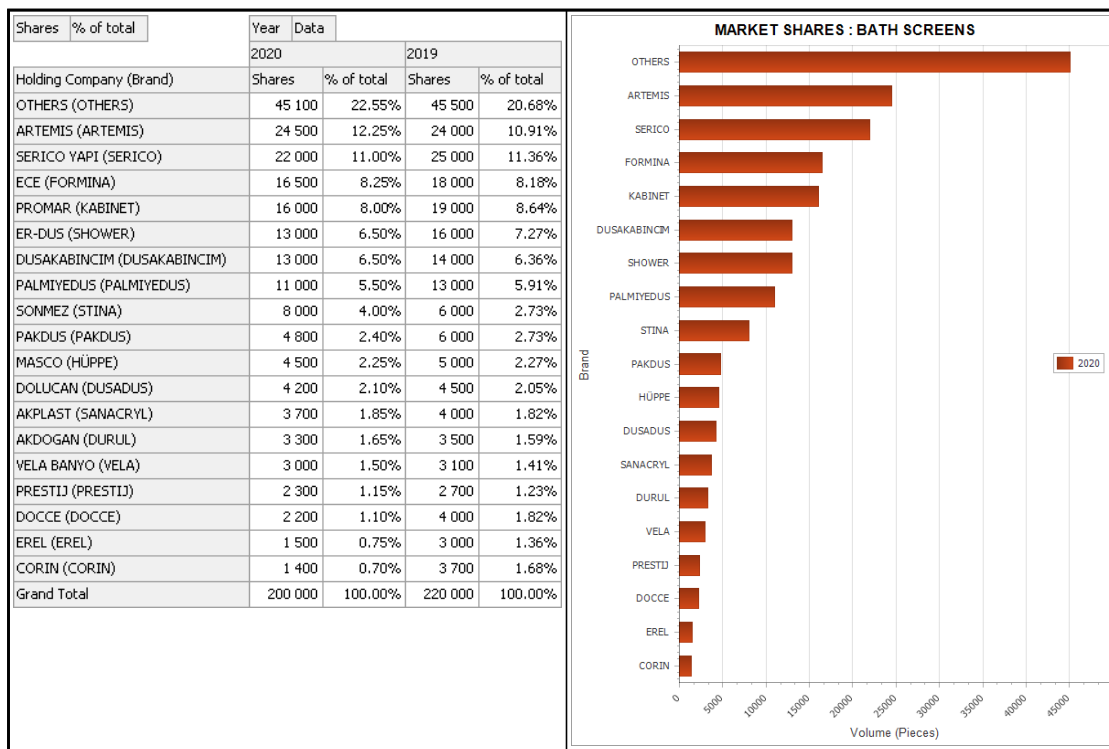
TR - SHOWER ENCLOSURE SALES BY DOOR TYPE ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

4.5.1 Market Shares : Bath Screens

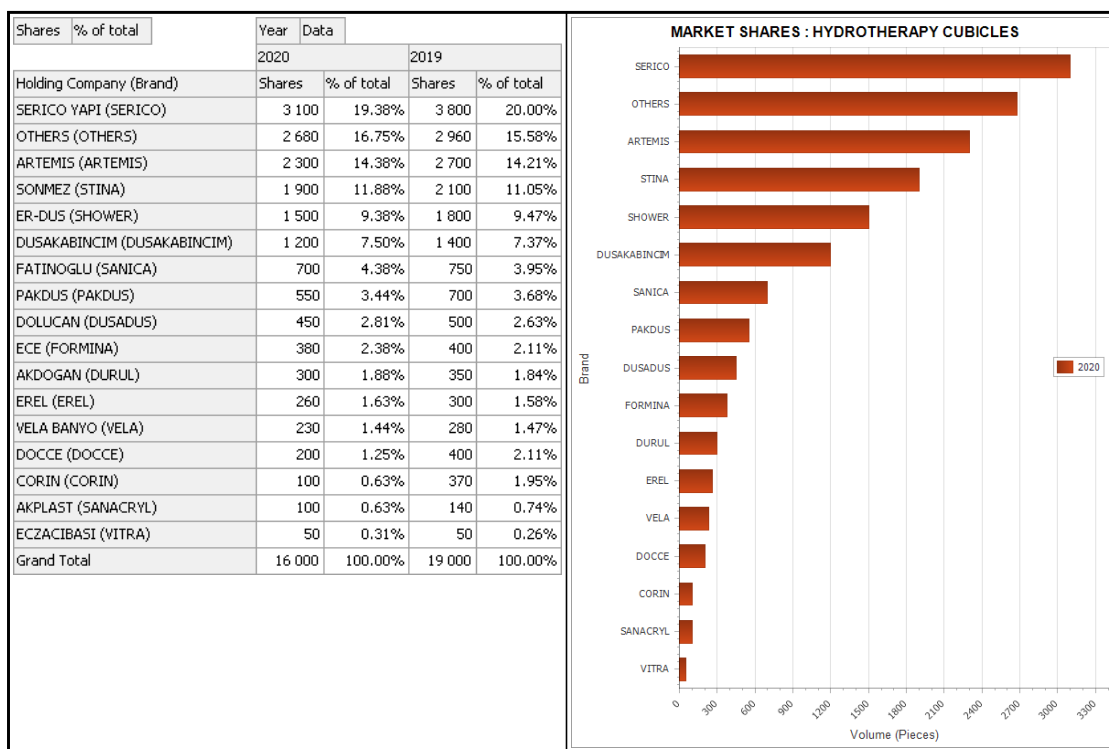
TR - ESTIMATED MARKET SHARES FOR BATH SCREENS



Source: BRG based on industry interviews.

4.5.2 Market Shares : Hydrotherapy Shower Cubicles

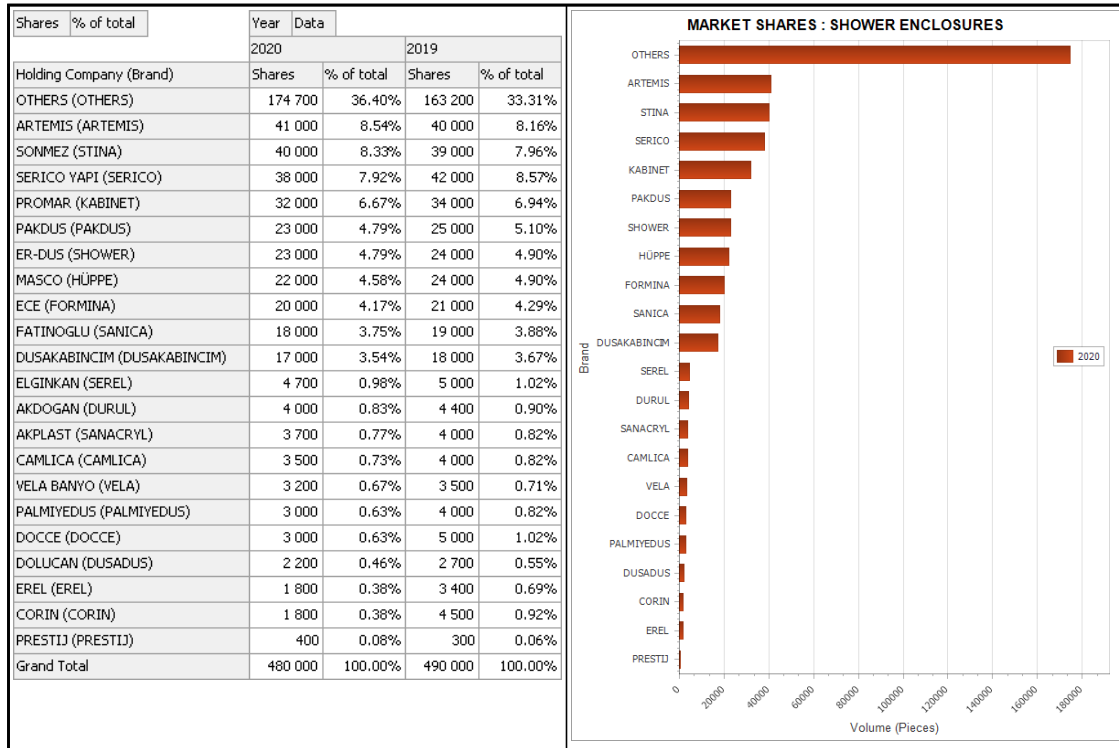
TR - ESTIMATED MARKET SHARES FOR HYDROTHERAPY CUBICLES



Source: BRG based on industry interviews.

4.5.3 Market Shares : Shower Enclosures

TR - ESTIMATED MARKET SHARES FOR SHOWER ENCLOSURES



Source: BRG based on industry interviews.

4.6.1 Route to Market: Shower Wall
TR - ESTIMATED DISTRIBUTION FOR SHOWER WALL: 2020

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	69.5%
	Wholesalers / Merchants
	69.5%

Retail Level	Manufacturer 4.2%	Wholesaler 0.7%	Manufacturer 9.5%	Wholesaler 3.5%	Manufacturer 8.1%	Wholesaler 2.0%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	4.9%		13.0%		10.1%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
8.7%	37.5%	1.6%	11.3%	9.1%
Installers				
68.2%				

End Use Level	Manufacturer 0.0%	Wholesaler 19.8%	Internet 3.3%	DIY 1.7%	Specialist 1.0%	Installer 56.0%	Manufacturer 0.0%	Wholesaler 6.0%	Internet 0.0%	DIY 0.0%	Specialist 0.0%	Installer 12.2%
	Residential						Non-Residential					
	81.8%						18.2%					
	New Build			Residential RMI			New Build			Non-Residential RMI		
	36.7%			45.1%			9.0%			9.2%		
	Project	Non-Project		Project	Non-Project		Project	Non-Project		Project	Non-Project	
	22.7%	14.0%		20.0%	25.1%		7.0%	2.0%		3.1%	6.1%	

4.6.2 Route to Market: Shower Wall
TR - ESTIMATED DISTRIBUTION FOR SHOWER WALL: 2025

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	69.0%
	Wholesalers / Merchants
	69.0%

Retail Level	Manufacturer 4.6%	Wholesaler 0.8%	Manufacturer 9.6%	Wholesaler 3.6%	Manufacturer 8.1%	Wholesaler 2.0%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	5.4%		13.2%		10.1%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
8.7%	37.4%	1.7%	11.4%	9.1%
Installers				
68.3%				

End Use Level	Manufacturer 0.0%	Wholesaler 19.2%	Internet 3.7%	DIY 1.8%	Specialist 1.0%	Installer 56.1%	Manufacturer 0.0%	Wholesaler 6.0%	Internet 0.0%	DIY 0.0%	Specialist 0.0%	Installer 12.2%
	Residential						Non-Residential					
	81.8%						18.2%					
	New Build			Residential RMI			New Build			Non-Residential RMI		
	30.5%			51.3%			9.0%			9.2%		
	Project	Non-Project		Project	Non-Project		Project	Non-Project		Project	Non-Project	
	21.0%	9.5%		21.0%	30.3%		7.0%	2.0%		3.2%	6.0%	

Source: BRG Based on Industry Interviews. - Totals may vary due to roundings

4.6.3 Route to Market - Consolidated View : Shower Wall

TR - ESTIMATED DISTRIBUTION FOR SHOWER WALL : 2020

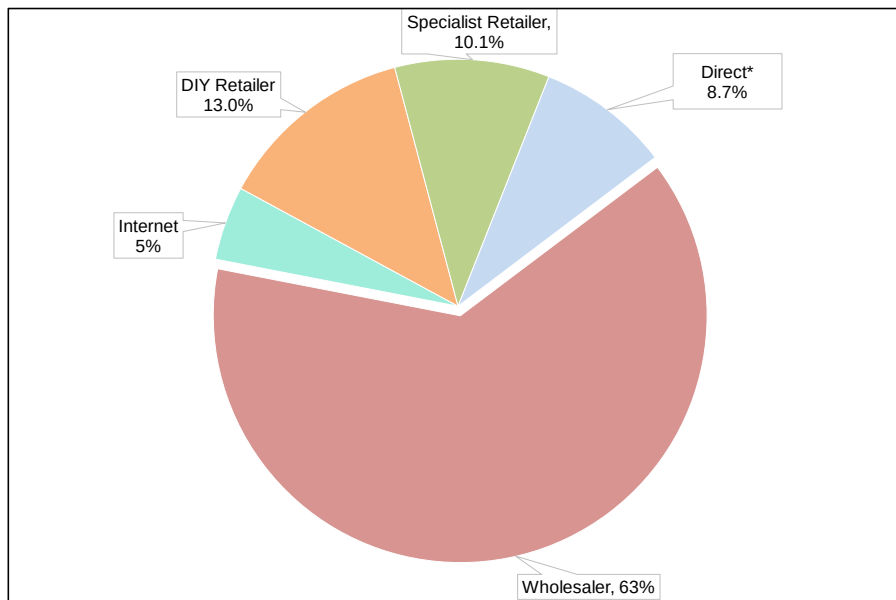
Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
63.3%	4.9%	13.0%	10.1%	8.7%	42.7%	10.1%	39.1%	8.1%
100%					52.8%		47.2%	

4.6.4 Route to Market - Consolidated View : Shower Wall

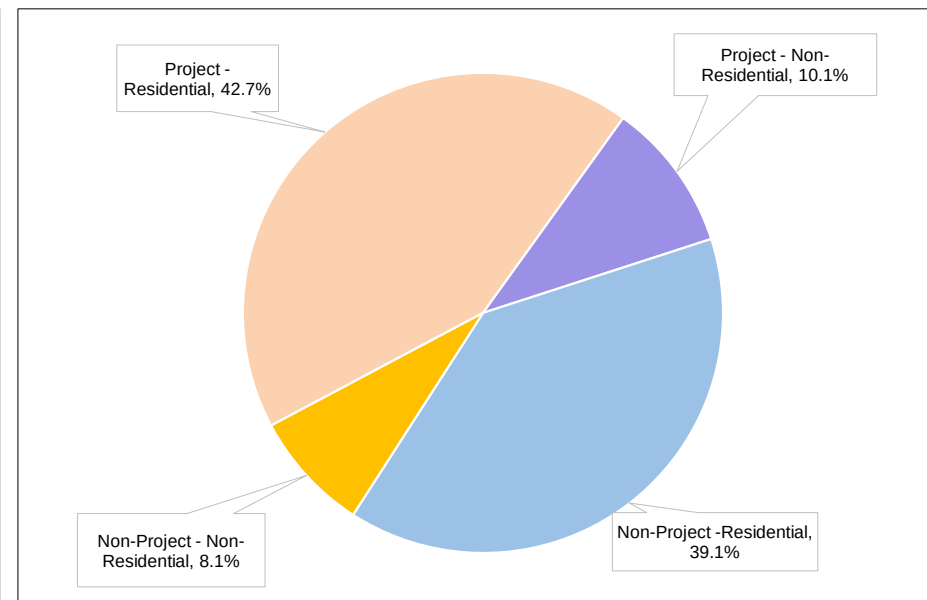
TR - ESTIMATED DISTRIBUTION FOR SHOWER WALL : 2025

Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
62.6%	5.4%	13.2%	10.1%	8.7%	42.0%	10.2%	39.8%	8.0%
100%					52.2%		47.8%	

TR - SALES OF SHOWER WALL BY DISTRIBUTION CHANNEL: 2020



TR - SALES OF SHOWER WALL BY POINT OF DECISION AND END-USE SECTOR: 2020



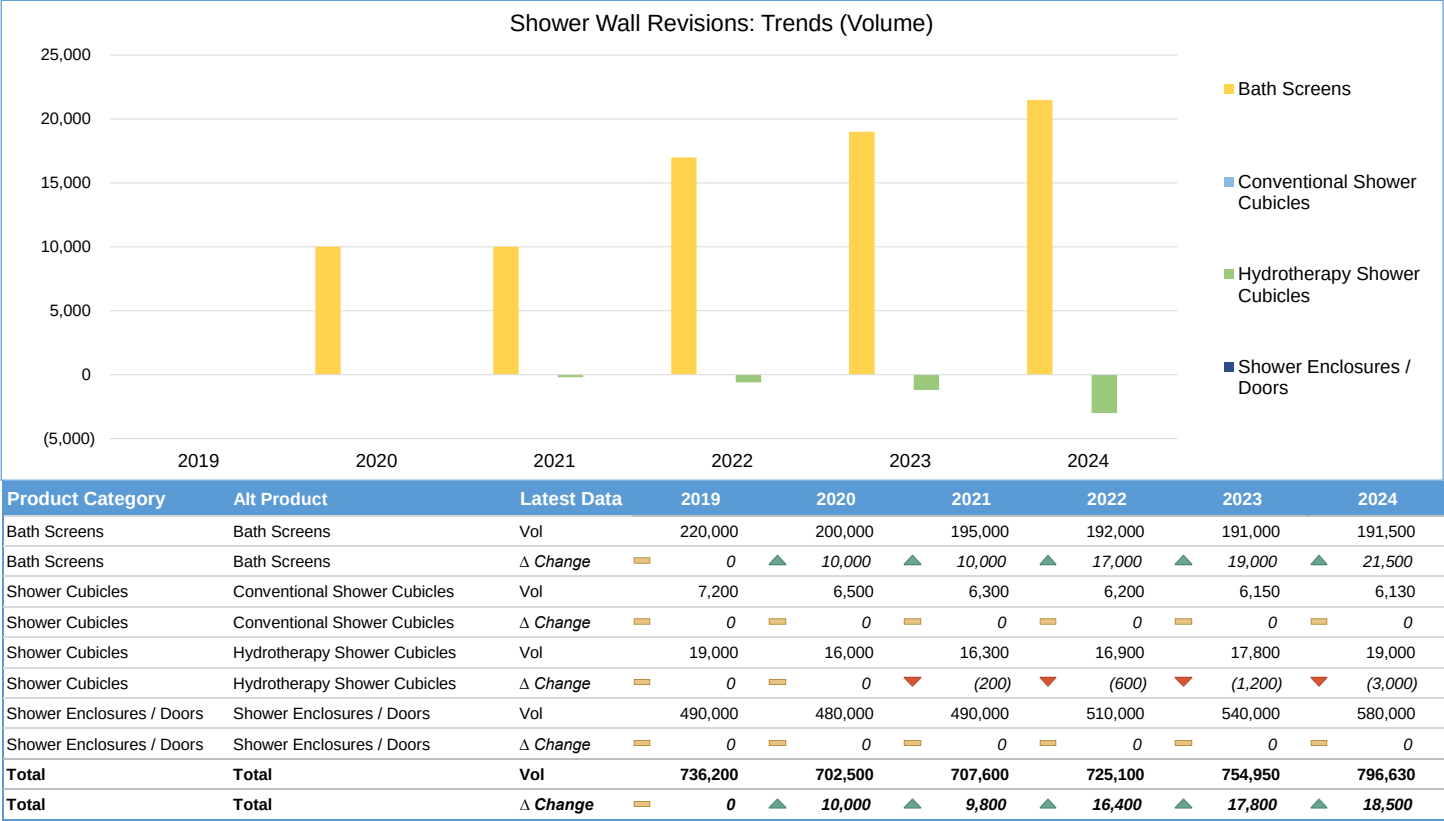
Source: BRG Based on Industry Interviews.

*Direct includes sales going directly from manufacturers to installers, contractors and private customers.

4.7 Revisions in the Turkish Shower Wall Market (April 2021)

TR - REVISIONS IN THE TURKISH SHOWER WALL MARKET

Data Comparison: October 2020



No revisions have been made to historical volumes and values for this market.